Cortex Cloud

Runtime Security
Documentation -

Detailed

1. Get started with Cortex Cloud 

1.1. What is Cortex Cloud?  1.1.1. Key features 

1.2. What is Cortex Cloud Runtime Security? 

1.3. Agentic AI in Cortex Cloud  1.3.1. Cortex Agentic Assistant 

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🟢note

By default, you have access to the Cortex
Assistant, which includes a natural language
interface for entity investigation and provides a
list of recommended responses such as running
a playbook, performing a scan, or collecting
support files.
If you enable the Cortex Agentic Assistant, it
replaces the Cortex Assistant interface entirely.
For more...

💡

🔵tip

The system Help Center Agent provides fast
access to documentation. You can ask natural
language questions, such as "How do I create a
dashboard?" or "Where can I review my data
retention policies?" and the agent retrieves
concise, relevant information from the official
documentation site.

🧭

➡️n…

Settings → Configurations → General → Server
Settings → Agentic Assistant

1.3.1.1. Agentic Assistant use cases 

1.3.1.2. Agentic Assistant security 

1.4. Supported web browsers  📝

🟢note

Cortex Cloud is optimized for best
performance using the HTTPv2 protocol. If you
are using a non-supported browser version or a
network middleware that uses a protocol other
than HTTPv2, you may experience slowness in
loading and response times between the
browser and the Cortex Cloud service.
Consult with your browser or middleware
vendor to see how...

1.5. Use the interface 

1.6. In-product support case creation 

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🟢note

When opening a support case through the
Customer Support Portal, you need to manually
select Cortex Cloud as the product. While
there may be discrepancies between the
categories in this wizard and the Customer
Support Portal process, that's because this
wizard is designed specifically to focus on
options relevant to Cortex Cloud.

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🟢note

Selecting an endpoint from the endpoint table
and retrieving TSF requires full Retrieve
Endpoint Data permissions under Endpoint
Administration.

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🟢note

Console recordings cannot exceed 10 minutes.
The current recording time is displayed at the
top of the window.

1.7. Understand your user persona 

1.8. Understand license plans 

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🟢note

A Cloud Runtime license includes both Posture
Scanning and Runtime Protection on the same
asset. Usage, including any overflow, is tracked
automatically to ensure accurate reporting
across both licenses without duplicate counting.

1.8.1. Data retention 

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🟢note

Case data is retained according to the Last
Updated date.
Issue data is retained according to the
Observation Time. Data collected within these
dates is kept and displayed for 186 days. To
ensure the accuracy of issues, Cortex Cloud
provides a grace period of up to 31 days for
issues displayed in the Issues View, Issues table,
and Cases View.

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🟢note

Requires the Forensics add-on.

1.9. Fair Usage policy for Cortex Cloud 

2. Onboard and configure Cortex
Cloud 

2.1. Plan and prepare 

2.2. Deployment steps and checklist 

2.2.1. Activate Cortex Cloud 

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The Cortex Cloud activation email.
A Customer Support Portal (CSP) account.
You need to set up your CSP account. For more
information, see How to Create Your CSP User
Account.
When you create a CSP account, you can set
up two-factor authentication (2FA) to log into
the CSP by using an Email, Okta Verify, or
Google Authenticator (non-FedRAMP
accounts)....

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🟢note

Activation can take about an hour and does not
require you to remain on the activation page.
Cortex Cloud sends a notification to your email
when the process is complete.

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🟢note

You can only change the subdomain once, and
it cannot be undone.
After deleting the subdomain, you can reuse it
after 7 days.

2.2.1.1. Cortex Cloud supported regions 

2.2.1.2. Enable access to required PANW
resources



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🟢note

<tenant-name> refers to the selected
subdomain of your Cortex Cloud tenant, and
<region> is the region in which your tenant is
deployed. For more information, see Cortex
Cloud supported regions.

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🟢note

Refer to Regional Docker registry mapping for
your specific tenant location and
corresponding Docker registry URL.

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➡️n…

SSL decryption: If you use SSL decryption and
experience difficulty connecting the Cortex
XDR agent to the server, we recommend that
you add the FQDNs required for access to your
SSL Decryption Exclusion list in Device →
Certificate Management → SSL Decryption
Exclusion

2.2.2. Upgrade from Prisma Cloud to Cortex
Cloud 

2.2.2.1. About the Upgrade Helper  📝

🟢note

Your Prisma Cloud tenant remains available and
fully operational during the agreed upgrade
period.

2.2.2.2. Link Cortex Cloud to Prisma Cloud  🧭

➡️n…

Settings → Configurations → Upgrade Helper

2.2.2.3. Copy content 

2.2.2.3.1. Copy Global configurations 

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🟢note

Keep the following caveats in mind:
Scope-Based Access Control (SBAC)
configurations, such as resources or account
lists are not copied. You can manually assign
scope-based access to the relevant users or
groups.

When migrating permission groups and roles,
the total count of items successfully copied
may be lower than the initial number selected.
This...

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➡️n…

Settings → Configurations → Access
Management

2.2.2.3.2. Copy CSPM configurations  🧭

➡️n…

Posture Management → Rules & Policies →
Policies → Cloud Security

Investigation & Response → Automation →
Automation Rules

Posture Management → Compliance →
Standards

2.2.2.3.3. Copy CWP configurations  🧭

➡️n…

Posture Management → Rules & Policies →
Cloud Workload

2.2.2.3.4. Copy Cortex Cloud Application
Security configurations  👀
🟠i…

If labels are imported into Cortex Cloud less
than three hours after they were last added or
modified in Prisma Cloud, they may not be
included in the import.

2.2.2.4. Migrate Cortex CLI  ☑️

🟣pr…

Before you begin, ensure you have the following:
Cortex Cloud API key: An active API key for
your Cortex Cloud tenant with associated CLI
role permissions. Refer to Manage API keys for
more information
Install the Cortex CLI. You can find the
installation instructions here

2.2.3. Set up users and roles 

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🟢note

Some features are license-dependent.
Accordingly, users may not see a specific
feature if the feature is not supported by the
license type or if they do not have access
based on their assigned role or scope.

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🟢note

The user who activated the Cortex product is
assigned the Account Admin role. You cannot
create additional Account Admin roles in the
Cortex Cloud tenant. If you do not want the
user to have Account Admin permission, you
must remove the Account Admin role in Cortex
Gateway.

2.2.3.1. User group management 

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🟢note

You can create user groups in the tenant or
Cortex Gateway. User groups created in Cortex
Gateway cannot be mapped to SAML groups.
Only user groups created in the tenant support
SAML group mapping and scoring. We
recommend creating user groups in the Cortex
tenant because:
User groups are available for all tenants, and
you may want different user...

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🟢note

If users have been created in the CSP, but you
want them to access the tenant through SSO
only, skip this field and add only SAML group
mapping after SSO is set up, otherwise, users
can access the tenant through both the CSP
and SSO.
If you have not yet created any users, skip this
field and add them later. See Set up
authentication .

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🟢note

When using Azure AD for SSO, the SAML group
mapping needs to be provided using the group
object ID (GUID) and not the group name.

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🟠i…

By default, Enable Scope Based Access
Control is disabled in Settings →
Configurations → General → Server Settings,
and granular scoping is not enforced. Before
enabling SBAC, we recommend that an
administrator or a user with Access
Management permissions first ensures that the
users, user groups, and API Keys defined in
Cortex Cloud are granted...

2.2.3.2. Assign user roles and groups 

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🟢note

If an existing user in the Cortex Gateway no
longer has a role or a user group assigned, the
user is revoked. Any roles, user groups, or
egress configurations created by that user are
shown as created by Revoked user instead of
the user's email address.

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🟢note

Visibility of Security domain Issues that refer to
assets with agents is controlled by the
Endpoints scoping configuration.

👀

🟠i…

Setting Cortex Query Language (XQL) dataset
access permissions for a user role can only be
performed from Cortex Cloud Access
Management. For more information, see
Manage user roles.

👀

🟠i…

Before configuring, ensure that you review
Understand scoping in the Manage user scope
section.

👀

🟠i…

By default, Enable Scope Based Access
Control is disabled in Settings →
Configurations → General → Server Settings,
and granular scoping is not enforced. Before
enabling SBAC, we recommend that an
administrator or a user with Access
Management permissions first ensures that the
users, user groups, and API Keys defined in
Cortex Cloud are granted...

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🔵tip

To apply the same settings to multiple users,
select them, and then right-click and select Edit
Users Permissions.

2.2.4. Manage API keys 

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🟢note

Visibility of Security domain Issues that refer to
assets with agents is controlled by the
Endpoints scoping configuration.

👀

🟠i…

Before configuring, ensure that you review
Understand scoping in the Manage user scope
section.

👀

🟠i…

By default, Enable Scope Based Access
Control is disabled in Settings →
Configurations → General → Server Settings,
and granular scoping is not enforced. Before
enabling SBAC, we recommend that an
administrator or a user with Access
Management permissions first ensures that the
users, user groups, and API Keys defined in
Cortex Cloud are granted...

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🟠i…

You will not be able to view the API key again
after you complete this step. Ensure that you
copy the API key before closing the notification.

2.2.5. Set up authentication 

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🔵tip

You should have at least one user in the
Customer Support Portal for backup, in case of
any authentication issues with your IdP provider.

2.2.5.1. Authenticate users through the
Customer Support Portal 

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🟢note

You must be assigned the Super User role in the
Customer Support Portal to add users in the
Customer Support Portal.
The first Super User who logs into Cortex
Gateway is automatically assigned the Account
Admin role and has access to the tenant. The
user who activates the Cortex Cloud tenant will
also be assigned the Account Admin role (if
there is no...

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🟢note

After users are added to the Customer Support
Portal and they accept the invitation, you can
manage them in Cortex Gateway or the Cortex
Cloud tenant.

2.2.5.2. Authenticate users using SSO 

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🟢note

To set up SSO authentication in the tenant, you
must be assigned an Instance Administrator or
Account Admin role.
SAML 2.0 users must log in to Cortex Cloud
using the FQDN (full URL) of the tenant. To
allow login directly from the IdP to , you must
set the relay state on the IdP to the FQDN of
the tenant.
If you have multiple tenants, you must set up...

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🟢note

The first SSO cannot be deleted, it can only be
deactivated by toggling SSO Enabled to off.
The Domain parameter is predefined for the
first SSO.
If you add additional SSO providers, you must
provide the email Domain in the SSO Integration
settings for all providers except the first.
Cortex Cloud uses this domain to determine to
which identity provider...

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🟢note

Cortex Cloud requires the IdP to send the
group membership as part of the SAML token.
Some IdPs send values in a format that include
a comma, which is not compatible with Cortex
Cloud. In that case, you must configure your IdP
to send a single value without a comma for
each group membership. For example, if your
IdP sends the Group DN (a...

2.2.5.3. Set up Okta as the Identity Provider
Using SAML 2.0  📝
🟢note

When using SAML 2.0, users are required to
authenticate by logging in directly at the tenant
URL. They cannot log in via Cortex Gateway.

2.2.5.4. Set up Azure AD as the Identity Provider
Using SAML 2.0 

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🟢note

The default attributes shown on the main single
sign-on page in Azure AD are not the values
you need. You must click Edit next to Attributes
and Claims to view and copy the actual values.

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🟢note

The attribute claim name must exactly match
the value sent by your IdP. In some cases, this
may be the full attribute name/namespace,
depending on the configuration of our IdP

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🟢note

When using SAML 2.0, users are required to
authenticate by logging in directly at the tenant
URL. They cannot log in via Cortex Gateway.

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🟠i…

When copying the Single Sign-On URL value,
remove idp/saml and leave the trailing /.
For example, if the Single Sign-On URL is
https://clientname.panproduct.region.paloaltone
tworks.com/idp/saml, just copy
https://clientname.panproduct.region.paloaltone
tworks.com/.

2.2.6. Cloud service provider (CSP) onboarding 

2.2.6.1. Ingest cloud assets  📝

🟢note

The cloud accounts being onboarded must be
owned by the customer performing the
onboarding process.

2.2.6.2. Onboard Amazon Web Services 

☑️

🟣pr…

Ensure you have access to AWS Management
Console.
Ensure you have the Required AWS permissions.

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🟢note

Scanning with an outpost may require
additional CSP permissions and may incur
additional CSP costs.

📝

🟢note

When onboarding an AWS organization or
organizational unit (OU), Cortex Cloud creates
IAM resources in every account within that
organization or OU. This occurs even if you
choose to exclude specific accounts from
being scanned. While excluded accounts will
not be scanned and will not appear in the asset
inventory, the IAM resources may still be...

2.2.6.2.1. Manually upload template to AWS 

☑️

🟣pr…

Before you begin, ensure you have:
An AWS account
Access to AWS Management Console
Permission to create a stack and its resources in
AWS CloudFormation

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🟢note

You must ensure that the KMS key region and
the SNS topic region are the exact same as the
AWS region where you are deploying the
CloudFormation stack.

2.2.6.2.2. Configure AWS integration instances
and monitor integration instance health 

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🟢note

If you have not yet onboarded your cloud
integration, see Ingest cloud assets.

📝

🟢note

Currently, automation permission errors or
missing automation permissions do not affect
the Automation health status. You can view any
permission errors or missing permissions in the
the Permissions health status.

2.2.6.3. Onboard Google Cloud Platform 

☑️

🟣pr…

Ensure you have cccess to Google Cloud
Console.
Ensure you have an admin user with the
required admin GCP permissions.
Ensure you have the following APIs in the GCP
project you are onboarding:
Cloud Resource Manager API
Identity and Access Management (IAM) API
Cloud Pub/Sub API (if audit logs are enabled)
If you plan on enabling Automation as...

📝

🟢note

Scanning with an outpost may require
additional CSP permissions and may incur
additional CSP costs.

📝

🟢note

If you want to connect to your GCP Workspace,
you must first complete onboarding with the
option disabled. Once the GCP cloud instance
is created, perform the steps detailed in
Connect Google Workspace with your GCP
cloud instance.

2.2.6.3.1. Manually upload template to GCP 

☑️

🟣pr…

Before you begin, ensure you have:
A GCP account.
Permission to create the required resources in
Google Cloud Deployment Manager.
Installed Terraform on your local machine. You
can download Terraform from the official
Terraform website and follow the installation
instructions for your operating system.
Installed the GCP gcloud CLI tool.
Reviewed the...

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🟢note

The directory you create must be a
subdirectory of the home directory.

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🟠i…

You must not delete or move the Terraform files
from this folder. It will prevent you from being
able to edit your cloud instance in the future.

2.2.6.3.2. Connect Google Workspace with your
GCP cloud instance 

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Ensure you have the Super Admin role in
Google Workspace.

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Ensure you have the organization ID of the
Google Workspace you want to connect:
Log in to your Google Admin Console. and
navigate to Account → Account settings →
Profile. Next to Customer ID is your
organization ID.
Ensure you have the organization ID of the
Google Workspace you want to connect:
Log in to your Google Admin Console. and
navigate...

2.2.6.3.3. Configure GCP integration instances
and monitor integration instance health 

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🟢note

If you have not yet onboarded your cloud
integration, see Ingest cloud assets.

📝

🟢note

Currently, automation permission errors or
missing automation permissions do not affect
the Automation health status. You can view any
permission errors or missing permissions in the
the Permissions health status.

2.2.6.3.4. Monitor GCP resources inside service
perimeters

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🟢note

Each GCP cloud instance is assigned a scope
within GCP. If the scope, whether it be
organization, folder, or project, includes any
projects with a service perimeter, this
procedure must be performed for that cloud
instance to authorize Cortex Cloud to scan the
resources in the perimeter.

2.2.6.4. Onboard Microsoft Azure 

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Ensure you have a Microsoft Azure subscription.
Ensure you have the admin permissions required
to onboard Microsoft Azure or the built-in
Security Administrator role.
Obtain the tenant ID and subscription ID. You
can view these in the Microsoft Azure Portal in
Management groups.

2.2.6.4.1. Finalize Microsoft Azure onboarding by
executing the authentication template 

☑️

🟣pr…

Before you begin, ensure you have:
An Azure subscription.
A user with the required permissions for the
relevant scope (subscription, management
group, tenant). We recommend you create a
dedicated role.
Tenant ID and subscription ID. You can view
these in Microsoft Azure Portal in Management
groups.

Installed Terraform on your local machine. You
can...

☑️

🟣pr…

Before you begin, ensure you have:
An Azure subscription.
A user with the required permissions for the
relevant scope (subscription, management
group, tenant). We recommend you create a
dedicated role.
Tenant ID and subscription ID. You can view
these in Microsoft Azure Portal in Management
groups.
Installed the Azure CLI tool.
Authorization to create...

☑️

🟣pr…

Before you begin, ensure you have:
An Azure subscription.
A user with the required permissions for the
relevant scope (subscription, management
group, tenant). We recommend you create a
dedicated role.
Tenant ID and subscription ID. You can view
these in Microsoft Azure Portal in Management
groups.
Installed the Azure CLI tool.
Authorization to create...

👀

🟠i…

Do not delete or move the Terraform files from
this folder. It will prevent you from being able
to edit your cloud instance in the future.

2.2.6.4.2. Configure Azure integration instances
and monitor integration instance health 

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🟢note

If you have not yet onboarded your cloud
integration, see Ingest cloud assets.

📝

🟢note

Currently, automation permission errors or
missing automation permissions do not affect
the Automation health status. You can view any
permission errors or missing permissions in the
the Permissions health status.

2.2.6.5. Onboard Oracle Cloud Infrastructure 

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Ensure you have access to the Oracle Cloud
Infrastructure console
Permissions for all of the following are required:
Creation of identity groups (for more
information, refer to Managing Groups)
Policies (for more information, refer to How
Policies Work)
Tag namespaces in the root compartment (for
more information, refer to Tags and Tag
Namespace...

2.2.6.5.1. Manually upload template to OCI  ☑️

🟣pr…

Before you begin, ensure you have:
An Oracle Cloud Infrastructure account and
the tenancy OCID.
Permission to deploy a custom template and
create its resources in OCI.
Installed Terraform on your local machine. You
can download Terraform from the official
Terraform website and follow the installation
instructions for your operating system.
Installed...

2.2.6.5.2. Configure the OCI connector for log
collection 

2.2.6.6. Manually connect a cloud instance 

2.2.6.7. Edit your onboarded CSP configuration  👀

🟠i…

When using Terraform authentication
templates, you must execute the updated
Terraform template from the same folder where
the original Terraform template was executed.

2.2.6.8. Outposts 

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🟠i…

Outpost scan is an alternative to the
recommended standard cloud scan. Cloud scan
is recommended because it is fully managed by
Palo Alto Networks and incurs no compute
costs for your organization. Outpost scan is an
advanced deployment model reserved for
specific data residency or architectural
requirements.

2.2.6.8.1. Outpost fundamentals and planning 

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🟢note

Before you create your outpost, verify that
your internet connection is active. An active
internet connection is necessary for the
notification to be sent to Cortex Cloud to
create the new outpost.

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🟠i…

While outposts provide maximum control over
the scanning environment, cloud scan mode is
the recommended default for most
organizations.

2.2.6.8.2. Create an outpost 

☑️

🟣pr…

Before you begin, ensure you have:
An AWS account
Permission to create a stack and its resources in
AWS
Installed Terraform on your local machine. You
can download Terraform from the official
Terraform website and follow the installation
instructions for your operating system.
Installed the AWS CLI tool and configured your
profile with the aws...

☑️

🟣pr…

Before you begin, ensure you have:
A GCP account
Permission to create the required resources in
Google Cloud Deployment Manager
Installed Terraform on your local machine. You
can download Terraform from the official
Terraform website and follow the installation
instructions for your operating system.
Installed the GCP gcloud CLI tool

☑️

🟣pr…

Before you begin, ensure you have:
An active Azure subscription.
Installed the Azure CLI tool.
Permission to deploy a custom template and
create its resources in Microsoft Azure
("Owner" or "Contributor" on the designated
outpost subscription scope, and Active
Directory "Cloud Application Administrator" or
"Application Administrator" privileged...

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🟢note

Verify that your internet connection is active.
An active internet connection is necessary for
notifications to be sent to Cortex Cloud for
creating the new outpost. If you are unable to
establish an internet connection, contact
customer support for a manual workaround.

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🟢note

When creating an outpost for a specific Azure
subscription, the outpost account must be in
the same Azure organization as the monitored
subscriptions.

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🟠i…

While outposts provide maximum control over
the scanning environment, cloud scan mode is
the recommended default for most
organizations. For details, see When to choose
outpost scan.

2.2.6.9. Introduction to Terraform for Cloud

service provider (CSP) onboarding 

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🟠i…

Always run these commands in the same folder
where the original .tf files and .terraform folder
live—this is where the state is stored.

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This .tfstate state file is critical because it is
needed by the terraform destroy command to
clean up created resources. It is critical that
you never delete this file.

2.2.6.10. Container Registry Scanning 

2.2.6.10.1. Overview of container registry
scanning 

2.2.6.10.1.1. Registry Components 

2.2.6.10.1.2. How Container Registry Scanning
Works



2.2.6.10.2. Configure registry scanning for cloud
accounts  ☑️

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Ensure that you have performed the all steps
till Additional Security Capabilities as listed in
the onboarding wizard for the required CSP:
Onboard Amazon Web ServicesOnboard
Amazon Web Services
Onboard Google Cloud PlatformOnboard
Google Cloud Platform
Onboard Microsoft AzureOnboard Microsoft
Azure
Onboard Oracle Cloud InfrastructureOnboard
Oracle...

2.2.6.10.3. Modify the container registry
scanning scope 

2.2.6.10.4. Scan re-evaluation process 

2.2.6.10.5. Connect Docker Hub registry 

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Ensure an Outpost is connected to your tenant.

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Set up and configure Broker VM
Configure High Availability Cluster

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🟢note

If you choose Azure as the Cloud Provider, you
must also select the Tenant Id. The Tenant Id is
required to approve Cortex as an enterprise
application in your Azure tenant.

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🟢note

If you choose Azure as the cloud provider, only
Outposts associated with the selected tenant
ID are displayed.

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🟢note

The list of Broker VMs displays only VMs that
support registry scanning.
The list of high-availability Clusters displays
only clusters that contain at least one VM
supporting registry scanning.
The registry scanning status for each VM
appears in brackets if it was previously
activated for that specific VM.

2.2.6.10.5.1. Manage a Docker Hub connector  📝

🟢note

If you selected Scan with Broker VM mode, you
can't change to a different scan mode (such as
Cloud Discovery or Scan with Outpost) when
you edit the instance.
When editing an instance configured for Scan
with Broker VM, you must re-enter your
authentication credentials, including Username,
Password, and CA certificate.

2.2.6.10.6. Connect Docker V2 compliant
container registry 

☑️

🟣pr…

Ensure an Outpost is connected to your tenant.
Outposts

☑️

🟣pr…

Ensure one of the following is configured:
Set up and configure Broker VM.
Configure High Availability Cluster.

📝

🟢note

If you choose Azure as the Cloud Provider, you
must also select the Tenant Id. The Tenant Id is
required to approve Cortex as an enterprise
application in your Azure tenant.

📝

🟢note

If you choose Azure as the cloud provider, only
Outposts associated with the selected tenant
ID are displayed.

📝

🟢note

The list of Broker VMs displays only VMs that
support registry scanning.
The list of high-availability Clusters displays
only clusters that contain at least one VM
supporting registry scanning.
The registry scanning status for each VM
appears in brackets if it was previously
activated for that specific VM.

2.2.6.10.6.1. Manage a Docker V2 connector  📝

🟢note

If you selected Scan with Broker VM mode, you
can't change to a different scan mode (such as
Cloud Discovery or Scan with Outpost) when
you edit the instance.
When editing an instance configured for Scan
with Broker VM, you must re-enter your
authentication credentials, including Username,
Password, and CA certificate.

2.2.6.10.7. Connect GitLab container registry 

☑️

🟣pr…

Ensure an Outpost is connected to your tenant.

☑️

🟣pr…

Set up and configure Broker VM
Configure High Availability Cluster

📝

🟢note

When both the group ID and project ID are
provided, the system retrieves container images
from all projects within the specified group as
well as from the specified project.
If neither the group ID nor the project ID is
provided, the system retrieves container images
from all registries (across all groups and
projects) accessible to the authenticated...

📝

🟢note

When both the group ID and project ID are
provided, the system retrieves container images
from all projects within the specified group as
well as from the specified project.
If neither the group ID nor the project ID is
provided, the system retrieves container images
from all registries (across all groups and
projects) accessible to the authenticated...

📝

🟢note

If you choose Azure as the Cloud Provider, you
must also select the Tenant Id. The Tenant Id is
required to approve Cortex as an enterprise
application in your Azure tenant.

📝

🟢note

If you choose Azure as the cloud provider, only
Outposts associated with the selected tenant
ID are displayed.

📝

🟢note

When both the group ID and project ID are
provided, the system retrieves container images
from all projects within the specified group as
well as from the specified project.
If neither the group ID nor the project ID is
provided, the system retrieves container images
from all registries (across all groups and
projects) accessible to the authenticated...

📝

🟢note

When both the group ID and project ID are
provided, the system retrieves container images
from all projects within the specified group as
well as from the specified project.
If neither the group ID nor the project ID is
provided, the system retrieves container images
from all registries (across all groups and
projects) accessible to the authenticated...

📝

🟢note

The list of Broker VMs displays only VMs that
support registry scanning.
The list of high-availability Clusters displays
only clusters that contain at least one VM
supporting registry scanning.
The registry scanning status for each VM
appears in brackets if it was previously
activated for that specific VM.

📝

🟢note

When both the group ID and project ID are
provided, the system retrieves container images
from all projects within the specified group as
well as from the specified project.
If neither the group ID nor the project ID is
provided, the system retrieves container images
from all registries (across all groups and
projects) accessible to the authenticated...

📝

🟢note

When both the group ID and project ID are
provided, the system retrieves container images
from all projects within the specified group as
well as from the specified project.
If neither the group ID nor the project ID is
provided, the system retrieves container images
from all registries (across all groups and
projects) accessible to the authenticated...

2.2.6.10.7.1. Manage a Gitlab Container Registry
connector  📝
🟢note

If you selected Scan with Broker VM mode, you
can't change to a different scan mode (such as
Cloud Discovery or Scan with Outpost) when
you edit the instance.
When editing an instance configured for Scan
with Broker VM, you must re-enter your
authentication credentials, including Username,
Password, and CA certificate.

2.2.6.10.8. Connect Harbor registry 

☑️

🟣pr…

Ensure an Outpost is connected to your tenant.

☑️

🟣pr…

Ensure one of the following is configured:
Set up and configure Broker VM.
Configure High Availability Cluster.

📝

🟢note

If you choose Azure as the Cloud Provider, you
must also select the Tenant Id. The Tenant Id is
required to approve Cortex as an enterprise
application in your Azure tenant.

📝

🟢note

If you choose Azure as the cloud provider, only
Outposts associated with the selected tenant
ID are displayed.

📝

🟢note

The list of Broker VMs displays only VMs that
support registry scanning.
The list of high-availability Clusters displays
only clusters that contain at least one VM
supporting registry scanning.
The registry scanning status for each VM
appears in brackets if it was previously
activated for that specific VM.

2.2.6.10.8.1. Manage a Harbor connector  📝

🟢note

If you selected Scan with Broker VM mode, you
can't change to a different scan mode (such as
Cloud Discovery or Scan with Outpost) when
you edit the instance.
When editing an instance configured for Scan
with Broker VM, you must re-enter your
authentication credentials, including Username,
Password, and CA certificate.

2.2.6.10.9. Connect JFrog container registry 

☑️

🟣pr…

Ensure an Outpost is connected to your tenant.

☑️

🟣pr…

Set up and configure Broker VM
Configure High Availability Cluster

📝

🟢note

If you choose Azure as the Cloud Provider, you
must also select the Tenant Id. The Tenant Id is
required to approve Cortex as an enterprise
application in your Azure tenant.

📝

🟢note

If you choose Azure as the cloud provider, only
Outposts associated with the selected tenant
ID are displayed.

📝

🟢note

The list of Broker VMs displays only VMs that
support registry scanning.
The list of high-availability Clusters displays
only clusters that contain at least one VM
supporting registry scanning.
The registry scanning status for each VM
appears in brackets if it was previously
activated for that specific VM.

2.2.6.10.9.1. Manage a JFrog connector  📝

🟢note

If you selected Scan with Broker VM mode, you
can't change to a different scan mode (such as
Cloud Discovery or Scan with Outpost) when
you edit the instance.
When editing an instance configured for Scan
with Broker VM, you must re-enter your
authentication credentials, including Username,
Password, and CA certificate.

2.2.6.10.10. Connect Sonatype Nexus registry 

☑️

🟣pr…

Ensure an Outpost is connected to your tenant.
Outposts

☑️

🟣pr…

Ensure one of the following is configured:
Set up and configure Broker VM.
Configure High Availability Cluster.

📝

🟢note

If you are using a CA certificate, enter the
server IP address instead of the registry url.

📝

🟢note

If you choose Azure as the Cloud Provider, you
must also select the Tenant Id. The Tenant Id is
required to approve Cortex as an enterprise
application in your Azure tenant.

📝

🟢note

If you choose Azure as the cloud provider, only
Outposts associated with the selected tenant
ID are displayed.

📝

🟢note

If you are using a CA certificate, enter the
server IP address instead of the registry URL.

📝

🟢note

The list of Broker VMs displays only VMs that
support registry scanning.
The list of high-availability Clusters displays
only clusters that contain at least one VM
supporting registry scanning.
The registry scanning status for each VM
appears in brackets if it was previously
activated for that specific VM.

📝

🟢note

If you are using a CA certificate, enter the
server IP address instead of the registry URL.

2.2.6.10.10.1. Manage a Sonatype connector  📝

🟢note

If you had selected Scan with Broker VM mode,
you can't change to a different scan mode
(such as Cloud Discovery or Scan with
Outpost) when you edit the instance.
When editing an instance configured for Scan
with Broker VM, you must re-enter your
authentication credentials, including Username,
Password, and CA certificate.

2.2.6.11. Cloud service provider permissions 

2.2.6.11.1. Amazon Web Services provider
permissions 

2.2.6.11.2. Google Cloud Platform provider
permissions 

2.2.6.11.3. Microsoft Azure provider permissions 

2.2.6.11.4. Oracle Cloud Infrastructure provider
permissions 

2.2.7. Onboard the Kubernetes Connector 

📝

🟢note

This option is not supported for Fargate.

📝

🟢note

Enabling Realtime Protection installs the agent
on your Kubernetes clusters as a DaemonSet.

📝

🟢note

For Fargate, you must provide the cluster
resource identifier.
The format of the identifier is arn:aws:eks:
<region>:<account-id>:cluster/<cluster-name>.

📝

🟢note

Basic authentication is only supported in
Posture Management. If deploying Realtime
Protection, select None .

📝

🟢note

If you install the latest version but select the N-
1 strategy, this policy will take effect starting
from the next upgrade cycle (it will not
immediately downgrade your current
installation).

2.2.7.1. What's new in Kubernetes Connector? 

2.2.7.2. Supported Kubernetes distributions  📝

🟢note

In Google Container-Optimized OS release 100
and earlier, where the FANOTIFY EXEC flag is
not supported, the Kernel configuration may be
partial for the user mode agent to properly
function. In such cases, the agent will fallback
to asynchronous mode.
In RHCOS version 4.12 and earlier, the Kernel
configuration may be partial for the user mode
agent to...

2.3. Post-deployment steps 

2.3.1. Set up your environment 

2.3.1.1. Configure server settings 

📝

🟢note

Keyboard shortcuts, timezone, and timestamp
format are not set universally and only apply to
the user who sets them.

📝

🟢note

The Cortex Agentic Assistant and AI case
summarization are currently available for users
in limited regions. For more information, see
Cortex Agentic Assistant.

📝

🟢note

If the Password Protection (for downloaded
files) setting under Settings → Configuration →
General → Server Settings is enabled, enter the
password 'suspicious' to download the file.

🧭

➡️n…

Settings → Configurations → General → Server
Settings

2.3.1.2. Configure security settings 

2.3.1.3. Data and log forwarding 

2.3.1.3.1. Forward logs and data from Cortex
Cloud to external services 

2.3.1.3.1.1. Configure external applications for
forwarding 

📝

🟢note

No prior configuration is required to send data
or logs to an email distribution list.
Only cases and issues can be forwarded to
Slack, Amazon S3, Amazon SQS, Splunk, and
Webhook.
To forward data to Amazon SQS, the following
permissions are required:
sqs:GetQueueAttributes, sqs:ListQueues,
sqs:SendMessage, sqs:SendMessageBatch,
tag:GetResources, and...

📝

🟢note

Only a user with Account Admin or Instance
Admin permissions can configure egress. For
more information, see Egress
configurations.Egress configurations

📝

🟢note

For Amazon SQS or Amazon S3, you need to
provide the queue/bucket name. For Webhook
or Splunk, you need to provide the domain,
including any subdomain. Note that the path
does not include HTTP or HTTPS.

📝

🟢note

You can also configure external applications
when you create a new forwarding
configuration. After defining the configuration
and setting the scope, you can Add Application
and follow the instructions below for any of
these destinations.

📝

🟢note

If egress has not been configured in the Cortex
Gateway, verification will fail and a message
will display that the endpoint does not match
any approved routes.

📝

🟢note

If egress has not been configured in the Cortex
Gateway, verification will fail and a message
will display that no approved routes were found.

📝

🟢note

If egress has not been configured in the Cortex
Gateway, verification will fail and a message
will display that the endpoint does not match
any approved routes.

📝

🟢note

If egress has not been configured in the Cortex
Gateway, verification will fail and a message
will display that the endpoint does not match
any approved routes.

2.3.1.3.1.2. Configure notification forwarding 

☑️

🟣pr…

Before you can select an external service for
notification forwarding, you must integrate the
external service with Cortex Cloud. For more
information, see Configure external
applications for forwarding. No prior
configuration is required to send data to an
email distribution list.

📝

🟢note

Forwarding destinations: Only issues and cases
can be forwarded to Slack, Splunk, Amazon
SQS, Amazon S3, or Webhook.
Notification forwarding by domain: To
configure notification forwarding for issues by
domain, select Issues and filter the Issues table
by Issue Domain.
Alert vs. issue format:By default, new
configurations use the issue format, but you...

📝

🟢note

Not all data and log types can be sent to all
external services. For more information, see
Forward logs and data from Cortex Cloud to
external services.Forward logs and data from
Cortex Cloud to external services

📝

🟢note

The Grouping Timeframe defines the time
frame, in minutes, of how often Cortex Cloud
sends notifications. Every 20 issues or 20 events
aggregated within this time frame are sent
together in one notification, sorted according
to severity. To send a notification when one
issue or event is generated, set the time frame
to 0.

2.3.1.3.1.3. Monitor administrative activity 

2.3.1.3.2. Data and log notification formats 

📝

🟢note

Issues can be forwarded to email, syslog
servers, and Slack in the alert format, if you
prefer. The alert format can be selected when
you configure your forwarding notification.

2.3.1.3.2.1. Management audit log messages 

2.3.1.3.2.2. Issue notification format 

2.3.1.3.2.3. Agent Audit log notification format 

2.3.1.3.2.4. Management Audit log notification
format 

2.3.1.3.2.5. Log format for IOC and BIOC issues 

2.3.1.3.2.6. Analytics log format 

2.3.2. Cortex MCP server  2.3.2.1. Cortex MCP server overview 

📝

🟢note

This feature is in Beta.

📝

🟢note

The Cortex MCP Server empowers you to
integrate AI into your security workflows using
natural language. When using LLM-based
suggestions, always review and approve actions
suggested by the AI before they're executed.
We recommend deploying the Cortex MCP
server in a secure environment where access is
limited to authorized users.

2.3.2.1.1. Install the Cortex MCP server 

☑️

🟣pr…

If you are running the Cortex MCP server in a
Poetry virtual environment, you must have
Python 3.13 or higher.
If you plan to run the Cortex MCP server in a
Docker container, you must have Docker
installed.

📝

🟢note

The MCP Server uses public APIs to
communicate and is limited by the license
quotas available in your tenant. This is
particularly relevant when running XQL queries.
For more information on running XQL query
APIs, see Run XQL query APIs.

📝

🟢note

It is critical to avoid assigning excessive
permissions when creating an API key for the
Cortex MCP Server. Since the key has both read
and write capabilities, overly broad permissions
can lead to unintended actions and potentially
compromise your environment. Ensure the key
follows the principle of least privilege and is
granted only the minimum...

📝

🟢note

When using Docker, we recommend using an
.env file to set the Cortex API credentials as
environment variables. While the credentials
can be provided in the MCP client configuration
settings, the .env file provides safer handling of
API credentials and makes your configuration
easily reproducible.

📝

🟢note

By default, stdio (standard input/output) is
used. You can also configure Streamable HTTP,
to send requests directly to the tenant instead
of through the MCP client. Streamable HTTP
can be useful for testing in the browser without
a MCP client and to bypass limits that may be in
place for your MCP client. For Docker, you can
include the Streamable HTTP...

👀

🟠i…

To configure the Cortex MCP Server, you need
the Cortex API URL, Cortex API key, and Cortex
API key ID. You will not be able to view the API
key again after you complete this step. Ensure
that you copy the API key before closing the
notification.

2.3.2.1.2. Configure the MCP client  🧭

➡️n…

Settings → Developer → Edit Config

2.3.2.1.3. Use the Cortex MCP server  📝

🟢note

The built-in tools retrieve information, but do
not write to the tenant. You can create your
own tools that include write actions. The
examples below include both.

2.3.2.1.4. Create custom Cortex MCP server

tools  📝
🟢note

Any new or updated components provided by
Cortex are automatically downloaded into the
remote_components folder. During each
update, the folder is fully replaced and all
existing contents are recreated. Do not add
custom tools to this directory, as it is managed
entirely by Cortex and is overwritten at every
update.

2.3.3. Manage user roles and access
management 

☑️

🟣pr…

Managing users, roles, scopes, user groups,
authentication settings in Cortex Cloud Access
Management requires View/Edit RBAC
permissions for Access Management (under
Configurations). Account Admin and Instance
Administrator roles are granted this permission
by default. For more information, see
Predefined user roles in Set up users and roles.

📝

🟢note

For more information on assigning user roles
when generating an API key, see Manage API
keys.

📝

🟢note

Some features are license-dependent.
Accordingly, users may not see a specific
feature if the feature is not supported by the
license type or if they do not have access
based on their assigned role or scope.

📝

🟢note

For features where scoping is not applicable,
Role-Based Access Control (RBAC) is used and
can be configured when managing user roles.
For more information, see Manage user roles.

2.3.3.1. Manage user roles 

☑️

🟣pr…

Managing user roles in Cortex Cloud Access
Management requires View/Edit RBAC
permissions for Access Management (under
Configurations). Account Admin and Instance
Administrator roles are granted this permission
by default. For more information, see
Predefined user roles in Set up users and roles.

🧭

➡️n…

Settings → Configurations → Access
Management → Roles

2.3.3.2. Manage user access 

☑️

🟣pr…

Managing users, roles, scopes, user groups,
authentication settings in Cortex Cloud Access
Management requires View/Edit RBAC
permissions for Access Management (under
Configurations). Account Admin and Instance
Administrator roles are granted this permission
by default. For more information, see
Predefined user roles in Set up users and roles.

📝

🟢note

You can only reduce the permissions of an
Account Admin user via Cortex Gateway.

📝

🟢note

User permissions for components and datasets
are based on the access permissions set in the
user role. For more information on editing these
user role permissions, see Manage user roles.

📝

🟢note

Visibility of Security domain Issues that refer to
assets with agents is controlled by the
Endpoints scoping configuration.

👀

🟠i…

Before configuring, ensure that you review
Understand scoping in the Manage user scope
section.

👀

🟠i…

By default, Enable Scope Based Access
Control is disabled in Settings →
Configurations → General → Server Settings,
and granular scoping is not enforced. Before
enabling SBAC, we recommend that an
administrator or a user with Access
Management permissions first ensures that the
users, user groups, and API Keys defined in
Cortex Cloud are granted...

💡

🔵tip

To apply the same settings to multiple users,
select them, and then right-click and select Edit
User Permissions.

💡

🔵tip

To apply the same settings to multiple users,
select them, and then right-click and select Edit
User Permissions.

💡

🔵tip

To apply the same settings to multiple users,
select them, and then right-click and select Edit
User Permissions.

🧭

➡️n…

Settings → Configurations → Access
Management → Users

2.3.3.2.1. User access reference information 

2.3.3.3. Manage user scope 

☑️

🟣pr…

Configuring user scopes in Cortex Cloud
Access Management requires View/Edit RBAC
permissions for Access Management (under
Configurations). Account Admin and Instance
Administrator roles are granted this permission
by default. For more information, see
Predefined user roles in Set up users and roles.
By default, Enable Scope Based Access
Control is...

📝

🟢note

This configuration can impact the visibility of
the related Security domain in the Cases and
Issues scope area, but with not affect asset
visibility.

📝

🟢note

XQL-based dashboard widgets may require a
few hours to initially reflect changes to the list
or definitions of asset groups used for scoping.
To view the most current data immediately,
refresh the dashboard or its XQL widgets.

📝

🟢note

For Cases and Issues domains, a workaround is
to create a Dataset View for each required
combination of domains, and allow the relevant
entity access only to this Dataset View, not to
the underlying cases and issues datasets.

📝

🟢note

Make sure to assign the required default
granular scoping for users. This depends on the
structure and divisions within your organization
and the particular purpose of each
organizational unit to which scoped users
belong.

📝

🟢note

Visibility of Security domain Issues that refer to
assets with agents is controlled by the
Endpoints scoping configuration.

👀

🟠i…

Some areas and features in Cortex Cloud do
not respect SBAC. In these cases, use RBAC
permissions to restrict access.

👀

🟠i…

Action Required: Recent security enhancements
enforce stricter default permissions for cases
and issues that lack specific asset or endpoint
references. This notice explains how to include
access to these items if your users' visibility has
been impacted.

👀

🟠i…

Before configuring, ensure that you review the
Understand scoping section.

2.3.3.4. Manage access to objects  2.3.3.4.1. Manage access to custom dashboards 

☑️

🟣pr…

Configure tenant-level settings: An
administrator must first establish the sharing
framework under Settings → Configurations →
Access Management → Objects.
The configuration of these settings defines the
authorized sharing workflows for custom
dashboards:
Enable "Owners can Share objects they
created": Grants owners the ability to share...

📝

🟢note

If you're designated as an Editor, you can
always duplicate the widget and make your
changes on the copy.

📝

🟢note

Only Account Administrators and Instance
Administrators have the authority to change the
owner of an object.

2.3.4. Configure the Cortex Agentic Assistant 

2.3.4.1. Agentic Assistant components and
concepts  📝
🟢note

Agents are bound by the same rules and robust
permissions as a human user. In addition, you
can mark actions that make real-world changes
in production systems as sensitive, requiring a
quick manual review and confirmation, ensuring
peace of mind before critical system changes
are made.

2.3.4.2. Agents Hub 

📝

🟢note

To manage agents in the Agents Hub, you must
have the proper permissions. For more
information, see Agentic Assistant role-based
access control.

2.3.4.2.1. Manage actions 

📝

🟢note

To manage actions in the Agents Hub, you must
have the correct permissions. For more
information, see Agentic Assistant role-based
access control.

💡

🔵tip

System actions may rely on content packs that
need to be installed and configured.

2.3.4.2.2. Register actions 

📝

🟢note

To register actions, ensure you have the correct
permissions. For more information, see Agentic
Assistant role-based access control.

📝

🟢note

Mandatory arguments cannot be deselected.

👀

🟠i…

A single content item can be registered as
different actions, with each action using
different inputs and outputs from the same
script. Only register the same script, command,
or AI prompt as a new action if it is required for
your use case, as providing an agent with many
actions with overlapping abilities can reduce
the ability of the agent to choose...

2.3.4.2.3. Manage agents 

📝

🟢note

In some cases, an agent may include actions
with content items that are not relevant for all
licenses. If that occurs, the grey circle appears,
but you are not able to install the related
content.

💡

🔵tip

System agents can include actions that require
additional content packs to be installed and
configured. To view all actions assigned to a
system agent, including actions not available
due to missing content, click on the system
agent in the Agents Hub. There may be actions
assigned to a system agent that are not
relevant to your organization. For...

2.3.4.2.4. Build agents 

📝

🟢note

To build agents in the Agents Hub, you must
have view/edit permissions. For more
information, see Agentic Assistant role-based
access control.

📝

🟢note

If you clear the checkbox for a role, all actions
associated with that role are also cleared. The
exception is if another role is also selected,
which is associated with the same actions.
If you clear the checkbox for an action, all roles
associated with that action are cleared. For
example, if you select the Investigator role, and
Send Mail and Tavily...

2.3.4.2.5. Expand agent capabilities with MCP
integrations 

📝

🟢note

For agentic actions to be created from tools on
an MCP server, the tools must have input and
output descriptions on the MCP server. These
descriptions are required for Cortex Cloud to
understand the tools' capabilities and the
correct use cases.

📝

🟢note

If you are using OAuth-based authentication,
the Test button returns an error containing the
command to run in the playground in order to
test the connection.

📝

🟢note

If an MCP tool is removed from the MCP server,
the action will be unavailable due to missing
content. If the tool is restored on the MCP
server, the action is automatically reenabled.

2.3.4.3. Agentic Assistant role-based access
control  📝
🟢note

Agents are limited by the individual permissions
of the user. For example, if users do not have
sufficient permissions to isolate an endpoint,
they cannot use an agent to isolate an endpoint.

2.3.5. XQL query management 

☑️

🟣pr…

Setting query limits requires View/Edit
permissions for Configurations → Query
Management.

📝

🟢note

A scheduled query or report is run on behalf of
the user that originally created it, even if it is
edited and run by another user.

📝

🟢note

Queries run by correlation rules are not
restricted by the query limit.
Very short queries do not count towards
concurrent queries.

📝

🟢note

To ensure optimal system performance, all
queries (user-generated and otherwise) adhere
to a default timeout limit of 60 minutes that is
defined by Palo Alto that takes priority over the
administrator defined value. Therefore,
regardless of the value specified in this field,
queries will be stopped after 60 minutes.
You can override the default timeout...

📝

🟢note

Query visibility is subject to Role-Based Access
Control (RBAC); users can't see queries for
datasets they do not have permission to access.

🧭

➡️n…

You can find Query Management options under
Settings → Configurations → General → Query
Management

2.3.6. Customize cases and issues  2.3.6.1. Customize cases and issues 

2.3.6.1.1. Set up case scoring 

📝

🟢note

For scoped users, a small lock icon indicates
that you don't have permissions to edit a rule.

💡

🔵tip

Right-click an issue field to add it as match
criteria.

2.3.6.1.2. Create a starring configuration  💡

🔵tip

Right-click an issue field to add it as match
criteria.

2.3.6.1.3. Create custom case statuses and
resolution reasons



📝

🟢note

Before you create a custom status, please
review the built-in options. For more
information, see Resolution reasons for cases
and issues.
We recommend using the built-in statuses and
resolution reasons where possible. Custom
statuses and resolution reasons might not be
supported by all content, and status syncing
can take time.
In addition, custom...

📝

🟢note

After creation, custom statuses and resolution
reasons cannot be deleted or modified.

2.3.6.1.4. Create a sync profile 

📝

🟢note

If an issue is using bi-directional syncing, you
need to provide both an Inbound and an
outbound sync profile.

📝

🟢note

Blank fields are skipped.
You must define exact values.
Custom status values are not currently
supported.
Support is currently limited to a specific set of
fields.

2.3.7. Dashboards and reports 

3. Endpoint security 

3.1. Endpoint protection 

3.1.1. Malware protection 

3.1.2. Exploit protection 

3.1.3. File analysis and protection flow  📝

🟢note

To enhance performance and efficiency, hash
verdict requests from the Cortex XDR agent
will be routed to the WildFire service with the
lowest latency. File uploads for analysis will
strictly adhere to the designated Cortex Cloud
and WildFire regions, ensuring data remains
within the appropriate geographical boundaries.

3.1.4. Endpoint protection capabilities 

3.1.5. Endpoint protection modules 

3.1.6. Processes protected by exploit security
policy



3.1.7. File Integrity Monitoring (FIM) 

🔒

🟣c…

File Integrity Monitoring usage requires a
dedicated FIM add-on or the Cloud Runtime
Security add-on.

📝

🟢note

Google Kubernetes Engine (GKE) is currently
not supported

📝

🟢note

It is recommended to create a policy that
targets only the necessary files and folders.

📝

🟢note

Platform cannot be changed once a rule group
has been created

📝

🟢note

A rule group can contain up to 100 rules.

📝

🟢note

When importing a policy, select whether to
enable the associated policy targets. Rules
within the imported policy are managed as
follows:
New rules are added to the top of the list.
Default rules override the default rule in the
target tenant.
Rules without a defined target are disabled
until the target is specified.

📝

🟢note

It is possible to ingest up to 15,000 events per
day (24 hours) for each host/container.

🧭

➡️n…

Creating, modifying and viewing FIM rule
groups and rules is be done in the Rule Groups
page, located at the Inventory → Endpoints →
File Integrity Monitoring menu

In Inventory → Endpoints → Policy management
→ Extensions → Profiles, select +Add Profile
and then select either Create New or Import
from File

3.1.8. WildFire analysis concepts  📝

🟢note

In cases when the Cortex Cloud agent gets a
failed status from the WF service due to a
general error or unsupported file type, and the
Local Analysis is set to disabled or not
applicable, Cortex Cloud will not generate an
alert on the file.

3.1.9. Guidelines for keeping Cortex XDR agents
and content updated 

📝

🟢note

The Cortex XDR agent keeps the name of the
original installation package after every
upgrade.

📝

🟢note

Custom dashboards that include upgrade
status widgets, and the All Endpoints page
display upgrade status.
During the upgrade process, the endpoint
operating system might request a reboot.
However, you do not have to perform the
reboot for the Cortex XDR agent upgrade
process to complete it successfully.
After you upgrade on an endpoint with Cortex...

📝

🟢note

These profiles can be configured on one or
more endpoints, static/dynamic groups, tags, IP
ranges, endpoint names, or other parameters
that allow the creation of logical endpoint
groups. See how to define endpoint group.

📝

🟢note

These profiles can be configured on one or
more endpoints, static/dynamic groups, tags, IP
ranges, endpoint names, or other parameters
that allow the creation of logical endpoint
groups.

👀

🟠i…

When planning Cortex XDR agent upgrades
and content updates, consult with the
appropriate stakeholders and teams and follow
the change management strategy in your
organization.

⚠️

🟡c…

Before enabling Auto-Update for Cortex XDR
agents, make sure to consult with all relevant
stakeholders in your organization.

3.1.10. About content updates  📝

🟢note

Cortex Cloud delivers the content update to
the agent in parts and not as a single file,
allowing the agent to retrieve only the updates
and additions it needs.

3.1.11. Endpoint data collection 

📝

🟢note

For specific files only and only if the file was
written.

👀

🟠i…

Registry key is collected as a real key name, and
not as a symbolic link.
Instead of
HKEY_LOCAL_MACHINE\System\CurrentControl
Set, which is a symbolic link,
KEY_LOCAL_MACHINE\System\ControlSet001
will be collected.
Instead of HKEY_CURRENT_USER, HKEY_USERS\
<SID> will be collected, where SID is a SID of
the current user.

3.2. Install and manage endpoints 

3.2.1. Set up endpoint protection 

3.2.1.1. Set up endpoint profiles and exception
rules 

📝

🟢note

Profiles associated with one or more targets
that are beyond the scope of your defined user
permissions are locked, and cannot be edited.

3.2.1.1.1. Set up malware prevention profiles 

🔒

🟣c…

Requires the ITDR add-on.

📝

🟢note

In this profile, the Report options configure the
endpoints to report the corresponding
suspicious files, actions, processes, or
behaviors to Cortex Cloud, without blocking
them. The Disabled options configure the
endpoints to neither analyze nor report the
corresponding malware or behavior.

📝

🟢note

New profiles based on imported profiles are
added, and do not replace existing ones.

📝

🟢note

As part of the anti-malware security flow, the
Cortex XDR agent leverages the operating
system's capability to identify revoked
certificates for executables, and DLL files that
attempt to run on the endpoint by accessing
the Windows Certificate Revocation List (CRL).
To allow the Cortex XDR agent access the CRL,
you must enable internet access over...

📝

🟢note

The Quarantine Malicious Executables feature
is not available for malware identified on
network drives.

📝

🟢note

The Quarantine Malicious Script Files feature is
not available for malware identified on network
drives.

📝

🟢note

If on-write actions were configured in earlier
versions of Cortex Cloud, the same
configuration has been preserved and applied
globally for all file types.
On-write file protection may have an impact on
the resources required by the Cortex XDR
agent.

📝

🟢note

On-demand file protection may have an impact
on the resources required by the Cortex XDR
agent.

📝

🟢note

We recommend that you disable scheduled
scanning. VDI machine scans are based on the
golden image and additional files will be
examined upon execution.

📝

🟢note

When periodic scanning is enabled in your
profile, the Cortex XDR agent initiates an initial
scan when it is first installed on the endpoint,
regardless of the periodic scanning scheduling
time.

📝

🟢note

This feature only comes into effect after a
restart.

📝

🟢note

When you are actively evaluating with BAS
tools, it is recommended to enable the BAS
mode setting only for the duration of your
evaluation, and for a limited number of agents.

📝

🟢note

BAS tools mode with content older than version
1850 cannot be configured, the agent will be
treated as Enabled.

📝

🟢note

New profiles based on imported profiles are
added, and do not replace existing ones.

📝

🟢note

We recommend that you disable scheduled
scanning. VDI machine scans are based on the
golden image and additional files will be
examined upon execution.

📝

🟢note

When periodic scanning is enabled in your
profile, the Cortex XDR agent initiates an initial
scan when it is first installed on the endpoint,
regardless of the periodic scanning scheduling
time.

📝

🟢note

The Quarantine Malicious Mach-O Files feature
is not available for malware identified on
network drives.

📝

🟢note

This module is supported by Cortex XDR agent
8.1.0 and later releases.

📝

🟢note

The Quarantine Malicious Executables feature
is not available for malware identified on
network drives.

📝

🟢note

When you are actively evaluating with BAS
tools, it is recommended to enable the BAS
mode setting only for the duration of your
evaluation, and for a limited number of agents.

📝

🟢note

BAS tools mode with content older than version
1850 cannot be configured, the agent will be
treated as Enabled.

📝

🟢note

New profiles based on imported profiles are
added, and do not replace existing ones.

📝

🟢note

The Quarantine Malicious ELF Files feature is
not available for malware identified on network
drives.

📝

🟢note

This module is supported by Cortex XDR agent
8.1.0 and later releases.

📝

🟢note

Enabling on-demand scanning will
automatically scan these core system
directories: /etc, /tmp, /home, /usr, /bin, /sbin,
/lib, /var, /opt, /dev, /root, /boot.

📝

🟢note

We recommend that you disable scheduled
scanning. VDI machine scans are based on the
golden image and additional files will be
examined upon execution.

📝

🟢note

When periodic scanning is enabled in your
profile, the Cortex XDR agent initiates an initial
scan when it is first installed on the endpoint,
regardless of the periodic scanning scheduling
time.

📝

🟢note

When you are actively evaluating with BAS
tools, it is recommended to enable the BAS
mode setting only for the duration of your
evaluation, and for a limited number of agents.

📝

🟢note

BAS tools mode with content older than version
1850 cannot be configured, the agent will be
treated as Enabled.

⛔

🔴w…

For optimal user experience, we recommend
that you set the action mode to either Allow or
Run Local Analysis.

⛔

🔴w…

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that you set the action mode to either Allow or
Run Local Analysis.

⛔

🔴w…

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that you set the action mode to either Allow or
Run Local Analysis.

⛔

🔴w…

For optimal user experience, we recommend
that you set the action mode to either Allow or
Run Local Analysis.

3.2.1.1.2. Set up exploit prevention profiles 

📝

🟢note

In this profile, the Report options configure the
endpoints to report the corresponding exploit
attempts to Cortex Cloud, without blocking
them. The Disabled options configure the
endpoints to neither analyze nor report the
corresponding malware or behavior.

📝

🟢note

New profiles based on imported profiles are
added, and do not replace existing ones.

📝

🟢note

In exploit prevention profiles, if you change the
action mode for processes, you must restart
the protected processes for the following
security modules to take effect on the process
and its forked processes:
Brute Force Protection
Java Deserialization
SO Hijacking

📝

🟢note

This step provides a temporary workaround for
the following publicly known information-
security vulnerabilities and exposures: CVE-
2021-24074, CVE-2021-24086 and CVE-2021-
24094.

📝

🟢note

If you have Windows endpoints in your network
that are unpatched and exposed to a known
vulnerability, we strongly recommend that you
upgrade to the latest Windows Update that has
a fix for that vulnerability.

📝

🟢note

New profiles based on imported profiles are
added, and do not replace existing ones.

📝

🟢note

In exploit prevention profiles, if you change the
action mode for processes, you must restart
the protected processes for the following
security modules to take effect on the process
and its forked processes:
Brute Force Protection
Java Deserialization
SO Hijacking

📝

🟢note

New profiles based on imported profiles are
added, and do not replace existing ones.

📝

🟢note

In exploit prevention profiles, if you change the
action mode for processes, you must restart
the protected processes for the following
security modules to take effect on the process
and its forked processes:
Brute Force Protection
Java Deserialization
SO Hijacking

⛔

🔴w…

This workaround applies only to the specific
Windows versions listed as exposed to these
CVEs, and requires a Cortex XDR agent release
7.1 or later and content 167-51646 or later. This
workaround is not recommended for non-
persistent, stateless, or linked-clone
environments. In some cases, enabling this
workaround can affect the network
functionality...

3.2.1.1.3. Set up agent settings profiles 

🔒

🟣c…

Requires a Cortex XDR Pro per Endpoint
license. When you enable this feature, a Cortex
XDR Pro per Endpoint license is consumed.

🔒

🟣c…

Requires Host Insights add-on.

🔒

🟣c…

Requires Forensics Add-on.

🔒

🟣c…

Requires a Cortex XDR Pro per Endpoint
license. When you enable this feature, a Cortex
XDR Pro per Endpoint license is consumed.

🔒

🟣c…

Requires Host Insights add-on.

🔒

🟣c…

Requires Forensics Add-on.

🔒

🟣c…

Requires a Cortex XDR Pro per Endpoint
license. When you enable this feature, a Cortex
XDR Pro per Endpoint license is consumed.

🔒

🟣c…

Requires Host Insights add-on

☑️

🟣pr…

User Space Agent mode requires Linux kernel
5.0.0 or higher.

📝

🟢note

New profiles based on imported profiles are
added, and do not replace existing ones.

📝

🟢note

Disabling Device Control Violation notifications
is only supported on endpoints running Cortex
XDR agent version 8.6 and above.

📝

🟢note

You can enable the option to maintain a
persistent notification regarding the
disconnection of the endpoint from the
network. The settings Persistent Isolation
Notification and Blocked Connectivity
Notification must be enabled. Until the threat
on the endpoint has been removed, the
endpoint remains disconnected from the
network.

📝

🟢note

In Traps 5.0.6 and later releases, when
protection is enabled, access will be read-only.
In earlier Traps releases, enabling protection
disables all access to services, processes, files,
and registry values.

📝

🟢note

If you choose the Enable option, you must also
enable XDR Agent Tampering Protection in the
malware profile and set it to Block. Ensure that
both profiles are assigned to the same
endpoints.

📝

🟢note

When you enable Cortex XDR agent
registration with the Windows Security Center,
Windows automatically shuts down Microsoft
Defender on Windows-based workstation
endpoints. If you still want to allow Microsoft
Defender to run on a workstation endpoint
where Cortex Cloud is installed, you must use
the Disable option. However, Palo Alto
Networks does not...

📝

🟢note

To enable access to these options, scroll down
to Network Location Configuration, and set
Action Mode to Enabled.

📝

🟢note

When using Nmap, the Cortex XDR agent
downloads an Nmap driver for the duration of
the scan and removes the driver upon
completion. If an Nmap scan is in process,
Cortex Cloud identifies the Nmap driver and
places any additional scans in a queue.

📝

🟢note

If you disable content updates for a newly
installed agent, the agent retrieves the content
for the first time from Cortex Cloud, and then
disables content updates on the endpoint.
When you add a Cortex XDR agent to an
endpoint group with a disabled content auto-
upgrades policy, the policy is applied to the
added agent as well.

📝

🟢note

Automatic upgrades are not supported with
non-persistent VDI and temporary sessions.

📝

🟢note

The delay timer starts from the date of the
target agent version's availability in the tenant.

📝

🟢note

Limitations in the content download process:
When you install the Cortex XDR agent, the
agent retrieves the latest content update
version available. A freshly installed agent can
take between five to ten minutes (depending on
your network and content update settings) to
retrieve the content for the first time. During
this time, your endpoint is not...

📝

🟢note

If the Cortex XDR agent is initially unable to
communicate without the local store,
enforcement is not enabled and the agent will
show as partially protected.

📝

🟢note

New profiles based on imported profiles are
added, and do not replace existing ones.

📝

🟢note

Disabling Device Control Violation notifications
is only supported on endpoints running Cortex
XDR agent version 8.6 and above.

📝

🟢note

You can enable the option to maintain a
persistent notification regarding the
disconnection of the endpoint from the
network. The settings Persistent Isolation
Notification and Blocked Connectivity
Notification must be enabled. Until the threat
on the endpoint has been removed, the
endpoint remains disconnected from the
network.

📝

🟢note

If you choose the Enabled option, you must
also set Anti Tampering Protection in the
malware security profile to Block, and ensure
that both profiles are assigned to the same
endpoints.

📝

🟢note

When protection is enabled, access to services,
processes, files, and registry values will be
read-only.

📝

🟢note

If you disable content updates for a newly
installed agent, the agent retrieves the content
for the first time from Cortex Cloud, and then
disables content updates on the endpoint.

📝

🟢note

Automatic upgrades are not supported with
non-persistent VDI and temporary sessions.
When a Cortex XDR agent is added to an
endpoint group, it inherits the group's policy,
including the disabled content auto-upgrades
setting.

📝

🟢note

The delay timer starts from the date of the
target agent version's availability in the tenant.

📝

🟢note

Limitations in the content download process:
When you install the Cortex XDR agent, the
agent retrieves the latest content update
version available. A freshly installed agent can
take between five to ten minutes (depending on
your network and content update settings) to
retrieve the content for the first time. During
this time, your endpoint is not...

📝

🟢note

If the Cortex XDR agent is initially unable to
communicate without the local store,
enforcement is not enabled and the agent will
show as partially protected.

📝

🟢note

New profiles based on imported profiles are
added, and do not replace existing ones.

📝

🟢note

If you disable content updates for a newly
installed agent, the agent retrieves the content
for the first time from Cortex Cloud, and then
disables content updates on the endpoint.
When you add a Cortex XDR agent to an
endpoint group with a disabled content auto-
upgrades policy, the policy is applied to the
added agent as well.

📝

🟢note

Automatic upgrades are not supported with
non-persistent VDI and temporary sessions.

📝

🟢note

Automatic upgrades are not supported for XDR
agents running on K8s.

📝

🟢note

The delay timer starts from the date of the
target agent version's availability in the tenant.

📝

🟢note

Limitations in the content download process:
When you install the Cortex XDR agent, the
agent retrieves the latest content update
version available. A freshly installed agent can
take between five to ten minutes (depending on
your network and content update settings) to
retrieve the content for the first time. During
this time, your endpoint is not...

📝

🟢note

User Space Agent mode requires Linux kernel
5.0.0 or higher. Endpoints running an older Linux
kernel version with this fallback enabled, will
not start using User Space Agent mode, and
will operate asynchronously.

⛔

🔴w…

If you disable or delay automatic-content
updates provided by Palo Alto Networks, it may
affect the security level in your organization.

⛔

🔴w…

If you disable or delay automatic-content
updates provided by Palo Alto Networks, it may
affect the security level in your organization.

⛔

🔴w…

If you disable or delay automatic-content
updates provided by Palo Alto Networks, it may
affect the security level in your organization.

3.2.1.1.4. Set up restrictions prevention profiles 

📝

🟢note

New profiles based on imported profiles are
added, and do not replace existing ones.

📝

🟢note

Configure custom BIOC prevention rules here:

📝

🟢note

New profiles based on imported profiles are
added, and do not replace existing ones.

📝

🟢note

Configure custom BIOC prevention rules here:

📝

🟢note

New profiles based on imported profiles are
added, and do not replace existing ones.

📝

🟢note

Configure custom BIOC prevention rules here:

📝

🟢note

New profiles based on imported profiles are
added, and do not replace existing ones.

📝

🟢note

Wildcards are supported.

3.2.1.1.5. Set up exception profiles and rules 

3.2.1.1.5.1. Exception configuration 

👀

🟠i…

If you don't migrate the legacy exceptions, you
can continue to create exceptions through the
profiles.
Add a new exceptions security profile
Add a global endpoint policy exception
Set up exploit prevention profiles
Set up malware prevention profilesSet up
malware prevention profiles
Set up restrictions prevention profiles

🧭

➡️n…

Settings → Exception Configuration → Legacy
Exceptions and click Start Simulation

3.2.1.1.5.2. Issue exclusions 

📝

🟢note

The agent continues to generate excluded
issues on the endpoint, but they are not saved
or displayed in Cortex Cloud. Configuration of
an issue exclusion does not remove or delete
any of the logs that would have triggered the
issue notification.
You can also set up issue exceptions by
creating global endpoint policy exceptions. For
more information,...

🧭

➡️n…

The Settings → Exception Configuration →
Issue Exclusions page displays the issue
exclusion rules in Cortex Cloud

3.2.1.1.5.3. Add an IOC or BIOC rule exception 

📝

🟢note

Cortex Cloud only supports exceptions with
one attribute. See Add an alert exclusion rule to
create advanced exceptions based on your
filtered criteria.Add an alert exclusion rule

🧭

➡️n…

Settings → Exceptions Configuration →
IOC/BIOC Suppression Rules

3.2.1.1.5.4. Add a disable prevention rule for
endpoints 

👀

🟠i…

All applicable prevention actions are skipped
for the files and process that match the
properties defined in the rule.
Consider the consequences of disabling a
prevention rule before you add the exception,
and monitor it over time.
You can only apply a Disable Prevention Rule to
endpoints running Cortex XDR agents version
7.9 and later.

🧭

➡️n…

Settings → Exception Configuration → Disable
Prevention Rules

3.2.1.1.5.5. Add a disable injection and prevention
rule 

👀

🟠i…

Exceptions are limited up to 48 hours by
default and configurable up to one week.
Consider the consequences of disabling a
prevention rule before you add the exception
and monitor it over time.
You can only apply a Disable Prevention Rule to
agents version 7.9 and later.

🧭

➡️n…

Settings → Exception Configuration → Disable
Injection and Prevention

3.2.1.1.5.6. Add a support exception rule for
endpoints 

👀

🟠i…

If you don't migrate the legacy exceptions, you
can continue to create exceptions through the
profiles.
Add a new exceptions security profile
Add a global endpoint policy exception
Set up exploit prevention profiles
Set up malware prevention profilesSet up
malware prevention profiles
Set up restrictions prevention profiles

🧭

➡️n…

Your migrated rules are displayed on the
Settings → Exception Configurations →
Support Exception Rules page

From Settings → Exception Configuration →
Support Exception Rules, click + Import from
file

3.2.1.1.5.7. Add a legacy exception rule for
endpoints 

📝

🟢note

This exception prevents the agent from
examining the specified file. Use with caution,
as it may unintentionally allow unwanted or
malicious behavior to go undetected.

👀

🟠i…

If you don't migrate the legacy exceptions, you
can continue to create exceptions through the
profiles.
Add a new exceptions security profile
Add a global endpoint policy exception
Set up exploit prevention profiles
Set up malware prevention profilesSet up
malware prevention profiles
Set up restrictions prevention profiles

🧭

➡️n…

Your migrated rules are displayed on the
Settings → Exception Configurations → Legacy
Agent Exceptions page

3.2.1.2. Define endpoint groups 

📝

🟢note

Configuration based on user granular policy is
optimized for VDI and session-persistent
environments; it is not recommended for
decentralized or traditional endpoint
architectures.

📝

🟢note

Disconnecting Cloud Identity Engine in your
Cortex Cloud deployment can affect existing
endpoint groups and policy rules based on
Active Directory properties.

3.2.1.3. Configure global agent settings 

📝

🟢note

Disabling this capability takes immediate effect
on new hashes, fresh agent installations, and
existing security policies. It could take up to a
week to take effect on existing agents in your
environment pending agent caching.

📝

🟢note

If you have any Cortex Cloud filters, starring
policies, exclusion policies, scoring rules, log
forwarding queries, or automation rules
configured for XSOAR/3rd party SIEM, we
advise you to update those to support the
changes before activating the feature. For
example, change the query to include the
previous description that is still available in the...

3.2.1.4. Apply profiles to endpoints 

📝

🟢note

When importing a policy, select whether to
enable the associated policy targets. Rules
within the imported policy are managed as
follows:
New rules are added to the top of the list.
Default rules override the default rule in the
target tenant.
Rules without a defined target are disabled
until the target is specified.

📝

🟢note

The exported file is encoded in Base64 and
cannot be edited.

3.2.1.5. Create an agent installation package 

📝

🟢note

The profile will be applied if the security policy
cannot be retrieved in real-time.

📝

🟢note

The Cortex serverless library must be imported
after other libraries to activate the hooks that
enable auditing.

3.2.1.5.1. Manage an agent installation package  📝

🟢note

Since Cortex Cloud relies on the installation
package ID to approve agent registration during
the installation, we recommend that you don't
delete the installation package of active
endpoints. If you install the Cortex XDR agent
from a package after you delete it, Cortex
Cloud denies the registration request leaving
the agent in an unprotected state....

3.2.1.6. Harden endpoint security 

🧭

➡️n…

From Inventory → Endpoints → Policy
Management → Extensions → Profiles, you can
create profiles for the following hardened
endpoint security capabilities

To apply the profiles, from Inventory →
Endpoints → Policy Management → Extensions
→ Policy Rules, you can view all the policy
rules per operating system

3.2.1.6.1. Device control 

📝

🟢note

Operating systems report on devices in
different ways. Sometimes, the same BLE device
will report different services and interfaces,
depending on the host's operating system. This
may have an effect on the specific BLE services
that are blocked for each operating system.
Depending on your defined user scope
permissions, creating device profiles,...

📝

🟢note

For network printer print jobs, ensure that you
also configure the Agent Settings profile,
Network Location Configuration option. This
setting must be set to Enabled, and configured.
If you do not enable and configure this setting,
all network printer operations will be treated as
internal network print jobs.
For network printer print jobs, ensure that...

📝

🟢note

Currently, the default is set to Use Default
(Allow), however, Palo Alto Networks may
change the default definition at any time.

📝

🟢note

You cannot edit or delete the default profiles
pre-defined in Cortex Cloud.

📝

🟢note

You cannot edit or delete the predefined
profiles in Cortex Cloud.

📝

🟢note

When importing a policy, select whether to
enable the associated policy targets. Rules
within the imported policy are managed as
follows:
New rules are added to the top of the list.
Default rules override the default rule in the
target tenant.
Rules without a defined target are disabled
until the target is specified.

📝

🟢note

Permanent exceptions apply across platforms,
allowing the devices on all operating systems.

📝

🟢note

Disabling Device Control Violation notifications
is only supported on endpoints running Cortex
XDR agent version 8.6 and above.

🧭

➡️n…

Endpoints → Policy Management → Settings →
Device Management

3.2.1.6.2. Host firewall 

3.2.1.6.2.1. Host firewall for Windows 

📝

🟢note

A rule is always part of a rules group. It cannot
stand on its own.
A rule can belong to one rules group only and
cannot be reused in different groups.

📝

🟢note

When selecting ICMP protocol, you must enter
a the ICMP Type and Code. Without these
values the ICMP protocol is ignored by the
Windows and macOS Cortex XDR agents.

📝

🟢note

When importing a policy, select whether to
enable the associated policy targets. Rules
within the imported policy are managed as
follows.
New rules are added to the top of the list.
Default rules override the default rule in the
target tenant.
Rules without a defined target are disabled
until target is specified.

📝

🟢note

The data is aggregated and reported
periodically every 60 minutes since the first
time the host firewall policy was enforced on
the endpoint, not every round hour.
The table lists enforcement events only for
rules set to Report Matching Traffic.

🧭

➡️n…

From Inventory → Endpoints → Policy
Management → Extensions and select + Add
Profile or Import from File

3.2.1.6.2.2. Host firewall for macOS 

📝

🟢note

When importing a policy, select whether to
enable the associated policy targets. Rules
within the imported policy are managed as
follows:
New rules are added to the top of the list.
Default rules override the default rule in the
target tenant.
Rules without a defined target are disabled
until the target is specified.

🧭

➡️n…

From Inventory → Endpoints → Policy
Management → Extensions Profiles → Profiles,
select + New Profile or Import from File

Additionally, to monitor the communication on
your macOS endpoint, you can use the
following operating system utilities: From the
endpoint System Preferences → Security and
Privacy → Firewall → Firewall options, you can
view the list of blocked and allowed
applications in the firewall

3.2.1.6.3. Disk encryption 

📝

🟢note

You must select the same encryption method
configured by the Microsoft Windows Group
Policy in your organization for the target
endpoints. Otherwise, if you select a different
encryption method than the one already
applied through the Windows Group Policy,
Cortex Cloud displays errors.

📝

🟢note

When importing a policy, select whether to
enable the associated policy targets. Rules
within the imported policy are managed as
follows:
New rules are added to the top of the list.
Default rules override the default rule in the
target tenant.
Rules without a defined target are disabled
until the target is specified.

🧭

➡️n…

You can monitor the Encryption Status of an
endpoint in the Inventory → Endpoints → Disk
Encryption Visibility table

3.2.1.6.4. Host Inventory 

📝

🟢note

If you have Cloud Posture Security, Cloud
Runtime Security, or Cortex XSIAM Premium
licenses, go to Inventory → Host Insights →
Host Inventory.

📝

🟢note

Cortex Cloud includes only the first 10,000
results per endpoint.
Cortex Cloud lists only users that belong to
each group directly, and does not include users
who belong to a group within the main group.
If a local users group includes a domain user
(whose credentials are stored on the Domain
Controller server and not on the endpoint),
Cortex Cloud...

🧭

➡️n…

Inventory → Endpoints → Host Inventory

3.2.1.6.5. Vulnerability Assessment 

☑️

🟣pr…

The following are prerequisites for Cortex
Cloud to perform a vulnerability assessment.
Requirement
Description
Licenses and Add-ons
Supported Platforms
Windows
Cortex XDR agent 8.3 or a later release.
Cortex Cloud collects all the information about
the operating system and the installed
applications, and calculates CVE.
CVEs that apply to...

📝

🟢note

The CVE will be removed/reinstated to all
views, filters, and widgets after the next
vulnerability recalculation.

📝

🟢note

Starting with macOS 10.15, Mac built-in system
applications are not reported by the Cortex
XDR agent and are not part of the Cortex
Cloud Application Inventory.

💡

🔵tip

You can click each individual CVE to view in-
depth details about it on a panel that appears
on the right.

💡

🔵tip

You can click each individual endpoint to view
in-depth details about it on a panel that
appears on the right.

3.2.1.7. Set a Cortex XDR agent Critical
Environment version 

📝

🟢note

To ensure the stability of the line, CE versions
maintenance release cadence is longer than in
the standard line, we recommend that
deployment is adjusted accordingly.

🧭

➡️n…

Settings → Configurations → General → Agent
Configurations

Inventory → Endpoints → All Endpoints

3.2.1.8. Set an application proxy for Cortex XDR
agents  ⛔
🔴w…

If adding a proxy to the allow list be aware of
the following:
This allows any application on the isolated
endpoint to communicate through that proxy,
potentially allowing a threat actor or malware
to bypass the isolation and reach external
resources.

Ensure that no other non-security applications
are utilizing this proxy if you choose to allow it.

3.2.2. Manage endpoint protection 

3.2.2.1. Move agents between managing servers 

☑️

🟣pr…

Consider the following before making changes:
Endpoint type is not Kubernetes Node.
Installation type is not VDI.
Ensure you have administrator privileges for
Cortex Cloud in the hub.

📝

🟢note

Cortex Cloud does not support moving agents
between FedRamp and commercial tenants.

🧭

➡️n…

Inventory → Endpoints → Agent Installations

3.2.2.2. Manage endpoint tags 

📝

🟢note

The following explanations use Windows
operating system installation parameters and
Cytool argument examples.

📝

🟢note

Tag names are case sensitive.
In Windows and Mac, a tag name can contain
spaces.

Linux does not support tag names with spaces
as command line arguments to the shell
installer. Instead, tags can be set in the
/etc/panw/cortex.conf configuration file, which
supports all Linux installers.

📝

🟢note

Tag names are case sensitive and can contain
spaces.

📝

🟢note

Tag names are case sensitive and can contain
spaces.

📝

🟢note

When SBAC is enabled, assigning tags may
impact user permissions.

📝

🟢note

If you remove the tag and there are assigned
users or user groups with scope settings, this
can impact user permissions in the system.

3.2.2.3. Set an alias for an endpoint  💡

🔵tip

If you change your mind in the future, you can
select Endpoint Control → Change Endpoint
Alias again, and delete the required aliases.

3.2.2.4. Manage endpoint prevention profiles 

📝

🟢note

The filter is stored within the policy definition
and assessed during runtime to extract the
functions that match the filter criteria.

🧭

➡️n…

From Inventory → Endpoints → Policy
Management → Prevention → Profiles, locate
the profile that you want to remove

Inventory → Endpoints → Policy Management
→ Prevention → Profiles

3.2.2.5. Upgrade Cortex XDR agents 

📝

🟢note

The Cortex XDR agent keeps the name of the
original installation package after every
upgrade.

📝

🟢note

It is possible to cancel an upgrade if the Status
is Pending. Once the Status is In Progress, the
action is already received by the agent locally
and cannot be canceled from the management
console.
Custom dashboards that include upgrade
status widgets, and the All Endpoints page
display upgrade status.
During the upgrade process, the endpoint
operating...

⛔

🔴w…

You must ensure that the System Extensions
were approved on the endpoint. Otherwise, if
the extensions were not approved, after the
upgrade the extensions remain on the endpoint
without any option to remove them which could
cause the agent to display unexpected
behavior. To check whether the extensions were
approved, you can either verify that the...

👀

🟠i…

The following list includes important points to
take into account when upgrading the Cortex
XDR agent:
Review the recommended guidelines for
keeping Cortex XDR agents and content
updated. Read more here.
You cannot upgrade the Cortex XDR agent on
VDI endpoints or a Golden Image.
You must reinstall (uninstall and install again)
the relevant agent...

3.2.2.6. Restart agent 

3.2.2.7. Uninstall the Cortex XDR agent 

📝

🟢note

Before upgrading a Cortex XDR agent running
on macOS 10.15.4 or later, you must ensure that
the System Extensions were approved on the
endpoint. Otherwise, if the extensions were not
approved, after the upgrade the extensions
remain on the endpoint without any option to
remove them which could cause the agent to
display unexpected behavior. To check...

💡

🔵tip

If needed, use the filter to filter the list of
endpoints by attribute or group name.

3.2.2.8. Clear agent database 

📝

🟢note

Clearing the agent database is supported on all
platforms with Cortex XDR agent version 7.9 or
later, and is available only when using the
debugging mode.

📝

🟢note

You can only right-click to cancel the clear
agent database for actions with a pending
status.

3.2.2.9. Delete Cortex XDR agents  📝

🟢note

To reinstate an endpoint, you have to uninstall
and reinstall the agent.

3.2.2.10. Manage agent tokens 

📝

🟢note

Agent tokens are only supported for Windows
and Mac.

📝

🟢note

You can select a one or multiple agents to add
a temporary token.

3.2.2.11. Retrieve support file password  🧭

➡️n…

Retrieve Support File Password from Inventory
→ Endpoints → All Endpoints

3.2.2.12. Monitor agent operational status 

📝

🟢note

Any of the listed items could lead to a partially
protected state. Refer to the Cortex Cloud
management console for specific reasons for
the state.

⚠️

🟡c…

Status can have the following implications on
the endpoint:
*(Status): The exploit protection module is not
running.
**(Status):
XDR data collection is not running
Behavioral threat protection is not running
Anti-malware flow is asynchronous
Local privilege escalation protection is
asynchronous
**(Status):
XDR data collection is not...

🧭

➡️n…

You can monitor the Cortex XDR agent
Operational Status in Inventory → Endpoints →
All Endpoints

3.2.2.13. Monitor agent activity 

3.2.2.14. Monitor agent upgrade status  🧭

➡️n…

From Inventory → Endpoints → All Endpoints,
you can view the upgrade status of any Cortex
XDR agent that you manage

3.3. Endpoint DLP 

3.3.1. Cortex Data Loss Prevention (DLP) module
overview



☑️

🟣pr…

Endpoint DLP add-on license
Cortex agent 9.1 and above for Windows and
macOS

🧭

➡️n…

Settings → Configuration → Access
Management → Roles

3.3.2. Personas workflow for DLP 

3.3.3. Best Practices 

📝

🟢note

A good example would be to add the
description from the Raised Issue Name of the
specific Data-in-Motion rule.

📝

🟢note

When adding a source, it becomes a mandatory
requirement that must be met for the rule to
trigger. You should only apply this setting if you
want the rule to take action exclusively when
files originate from those specific locations.

🧭

➡️n…

Refer to Modules → Data Security → Data
Security Issues → Threats

Modules → Data Security → Endpoint Data-in-
Motion Rules → Endpoint DLP Settings to
configure the tenant settings for DLP

3.3.4. Configure DLP end-to-end  3.3.4.1. Onboarding checklist for DLP 

3.3.4.1.1. Install DLP browser extension on your
endpoint 

📝

🟢note

The extensions are not enabled for Incognito or
InPrivate modes in Chrome and Edge. It is
recommended to disable these modes in the
organization.

📝

🟢note

It is highly recommended to refer to Install the
Cortex XDR Agent using JAMF for a
comprehensive overview of setting up a JAMF
profile.Install the Cortex XDR Agent Using JAMF

👀

🟠i…

Endpoints should be linked to the domain.

🧭

➡️n…

Modules → Data Security → Endpoint Data-in-
Motion Rules → Endpoint DLP Settings

3.3.4.1.2. Create endpoint applications  🧭

➡️n…

Modules → Endpoint Data-in-Motion Rules →
Endpoint Applications

3.3.4.1.3. Create endpoint application groups  📝

🟢note

The catalog contains a list of all supported and
tested applications. If an application you need
is not on the list, contact support for assistance.

3.3.4.1.4. Create data-in-motion rules 

📝

🟢note

Third-party application behaviour:
When a file upload is blocked, the local
application may display its own generic error
message in response to the DLP restriction. In
similar cases, some third-party applications
might still proceed by sending a dummy file or
an error placeholder instead of the actual data.

📝

🟢note

To maximize data security, if you use Microsoft
Purview for extensive manual and automated
file classification tagging and are looking to
integrate those labels directly with the DLP
policies to trigger protective actions based on
a file's sensitivity, refer to How to use
information protection labels in Cortex Cloud
Data Security.

📝

🟢note

Distribution of the Endpoint DLP package is
restricted to agents assigned to the data-in-
motion policy. This ensures that only endpoints
requiring DLP functionality are affected, rather
than all eligible endpoints in the tenant.

3.3.4.1.5. Configure endpoint DLP settings 

📝

🟢note

When a fail-close action occurs, the system
creates a Data movement blocked by Endpoint
DLP fail-close action issue.

📝

🟢note

The agent does not force-install the extension
if it is already managed by the MDM on the
endpoint.
The XDR agent ensures full coverage by force-
installing the extension on both managed and
unmanaged browsers. But if a browser becomes
officially managed by the organization later,
the extension must be redeployed through the
central management console...

📝

🟢note

In the case of MDM, the extension is user-
managed, so Cortex doesn't remove an
installed MDM extension, it only disables
communication with the DLP extension.

📝

🟢note

You can specify the end-user message per rule.

3.3.5. DLP status in all endpoints 

📝

🟢note

It is recommended to work with the latest
security content.

📝

🟢note

Close and reopen your browser to activate the
DLP extension.

3.3.6. Cortex DLP threat detection and issues 

📝

🟢note

Access to this page is restricted to users with
the roles: Data Security Admin, Instance
Administrator, and Account Admin.

📝

🟢note

If the default action configured in Endpoint DLP
Settings is set to Block file movement (fail-
close), an issue is raised where the assigned
severity is set to low, and includes the Name
Data movement blocked by Endpoint DLP
default action

🧭

➡️n…

Data Security → Data Security Issues → Threats

From Data Security → Data Security Issues →
Threats, you can view the DLP issues

4. Threat management 

4.1. Detection rules  4.1.1. What are detection rules? 

4.1.1.1. What's an IOC? 

📝

🟢note

Cortex Cloud supports a maximum of
4,000,000 IOCs.

4.1.1.1.1. IOC rule details 

📝

🟢note

Field cannot exceed 36 characters.

🧭

➡️n…

In the Threat Management → Detection Rules
→ IOC page, you can view all configured or
uploaded indicators of compromise (IOCs

4.1.1.1.2. Create an IOC rule 

📝

🟢note

To ensure your IOC rules generate issues
efficiently and do not overcrowd your Issues
table, Cortex Cloud automatically does the
following:
Disables any IOC rules that reach 5000 or more
hits over 24 hours.
Creates a rule exception based on the
PROCESS SHA256 field for IOC rules that hit
more than 100 endpoints over 72 hours.

🧭

➡️n…

In Threat Management → Detection Rules →
IOC, select + Add IOC

4.1.1.2. What's a BIOC? 

4.1.1.2.1. BIOC rule details  🧭

➡️n…

Manage your behavioral indicator of
compromise (BIOC) rules in Threat
Management → Detection Rules → BIOC

4.1.1.2.2. Create a BIOC rule 

🔒

🟣c…

Custom prevention rules are supported on
Cortex Cloud agent 7.2 and later versions and
enable you to configure and apply user-
defined BIOC rules to Restriction profiles
deployed on your Windows, Mac, and Linux
endpoints.

📝

🟢note

To ensure your BIOC rules generate issues
efficiently and do not overcrowd your Issues
table, Cortex Cloud automatically does the
following:
Disables BIOC rules that reach 5000 or more
hits over a 24-hour period.
Creates a rule exception based on the
PROCESS SHA256 field for BIOC rules that hit
more than 100 endpoints over a 72 hour period.

📝

🟢note

A cloud_audit_log dataset requires a Cortex
XDR Pro per GB license.
Currently, you cannot create a BIOC rule on
customized datasets and only the filter stage,
alter stage, and functions without any
aggregations are supported for XQL queries
that define a BIOC.
For BIOC rules, the field values in XQL are
evaluated as case insensitive (config...

📝

🟢note

To demonstrate the expected behavior of the
rule before you save it, Cortex Cloud tests the
BIOC on historical logs. After you save a BIOC
rule, it will operate both on historical logs (up
to 10,000 hits) and on new data received from
your log sensors.

📝

🟢note

To demonstrate the expected behavior of the
rule before you save it, Cortex Cloud tests the
BIOC on historical logs. After you save a BIOC
rule, it will operate on both historical logs (up
to 10,000 hits) and new data received from your
log sensors.

📝

🟢note

You can only import files that were exported
from Cortex Cloud. You can not edit an
exported file.

📝

🟢note

You can only add to existing profiles you
created, Cortex Cloud Default profiles will not
appear as an option.
When you want to add to restrictions profile,
you can only use fields or options that exist in
pre-built process-type BIOCs

📝

🟢note

Auto-disable will turn off both the BIOC rule
detection and the BIOC prevention rule.

4.1.1.2.3. Manage Global BIOC Rules 

4.1.1.3. What's a correlation rule? 

🧭

➡️n…

After you configure your correlation rules, you
can manage them in Threat Management →
Detection Rules → Correlations, and view and
analyze the generated issues in Cases and the
Issues Table

4.1.1.3.1. Correlation rule details 

📝

🟢note

Only an administrator can create and view
queries built with an unknown dataset that
currently does not exist in Cortex Cloud .

📝

🟢note

For more information on troubleshooting server
errors in scheduled correlation rules,
Troubleshoot server errors in scheduled
correlation rules.

📝

🟢note

Certain fields are exposed and hidden by
default. An asterisk (*) is beside every field that
is exposed by default.

📝

🟢note

Only an administrator can create and view
queries built with an unknown dataset that
currently does not exist in Cortex Cloud.

🧭

➡️n…

If you are assigned a role that enables
Investigation → Rules privileges, you can
manage all user-defined Correlation Rules from
Threat Management → Detection Rules →
Correlations

4.1.1.3.2. Create a correlation rule 

📝

🟢note

When you open the New Correlation Rule
editor from XQL Search, this XQL Search field is
already populated with the XQL query that you
defined.
An administrator can create and view queries
built with an unknown dataset that currently
does not exist in Cortex Cloud . All other users
can only create and view queries built with an
existing dataset.

📝

🟢note

The XQL features for call, top, and wildcards in
datasets (dataset in (<dataset prefix>_*)) are
currently not supported in Correlation Rules. If
you add them to the XQL definition, you will not
be able to Create or Save the Correlation Rule.
The XQL features for transaction in datasets
(dataset in (<dataset prefix>_*)) are currently
not supported in...

📝

🟢note

If a generated issue has severity Medium or
above, a case is automatically opened.
Low severity issues generated by correlation
rules are not grouped into cases.

📝

🟢note

There is no validation or auto complete for
these parameters and the values can be null or
empty. In these scenarios, Cortex Cloud does
not display the null or empty values, but adds
the text NULL or EMPTY in the descriptions.

📝

🟢note

When more than one XDM field is listed, Cortex
Cloud looks for the field value according to the
order of the fields listed.

📝

🟢note

When Use the Cortex Cloud default incident
enrichment is not selected and you have not
mapped any issue fields, the issue is dispatched
into a new incident.

👀

🟠i…

The maximum size of a lookup dataset is 50 MB.
If the data exceeds this limit, the add to lookup
action fails.

💡

🔵tip

When creating your correlation rule, you can
use predefined values for different fields in the
editor, such as Alert Name, Alert Description,
and Drill-Down Query. For more information,
see Field replacement syntax in correlation
rules.

🧭

➡️n…

You can create a new correlation rule from
either the Threat Management → Detection
Rules → Correlation Rules page or when
building a query in XQL Search

4.1.1.3.3. Field replacement syntax in correlation
rules 

4.1.1.3.4. Manage correlation rules 

4.1.1.3.5. Monitor correlation rules  📝

🟢note

In the dataset, the Query start time and Query
end time indicate the time frame of the data
that was queried. The actual start and end
times of the correlation rule execution are
recorded in the _time field for the Initiated and
Completed entries.

4.1.1.3.6. Troubleshoot server errors in scheduled

correlation rules 

4.1.1.4. Manage IOC and BIOC rules 

📝

🟢note

For Analytics BIOC rules, you can only disable
and enable rules.

📝

🟢note

You cannot delete global BIOCs delivered with
content updates.

📝

🟢note

For BIOC rules that are applied to prevention
profiles:
If you disable a rule only on the agent,
detection on the server works as usual.
If you disable a rule only on the server,
prevention on the agent works as usual.
If you disable a rule only on the agent,
detection on the server works as usual.
If you disable a rule only on the server,
prevention...

📝

🟢note

When a BIOC rule is disabled automatically by
Cortex Cloud, for example due to the server
anti flooding mechanism, prevention on the
agent works as before.

🧭

➡️n…

Threat Management → Detection Rules → BIOC

4.1.1.5. Attack surface rules 

4.2. Analytics  4.2.1. Analytics overview 

4.2.1.1. Analytics engine 

4.2.1.2. Analytics sensors 

4.2.1.3. Coverage of MITRE Attack tactics 

4.2.1.4. Review MITRE ATT&CK framework
coverage

 📝

🟢note

When a contributing data source type is active,
it does not imply that all the rules and
detectors associated with the data source type
are active. Rule applicability is dependent on
the data source's context and configuration. To
enable an active status, data source types
require the following setup:
Endpoint: Installed Cortex XDR agent.
Network: A...

4.2.1.5. Analytics detection time intervals  📝

🟢note

When establishing a baseline, detectors
compute limits beyond which network activity
will require an issue. In some cases, detectors
do not compute baseline limits; instead they are
predetermined by Cortex Cloud engineers. The
engineers determine the values used for
predetermined limits using statistical analysis of
malicious activity recorded...

4.2.1.6. Analytics issues and Analytics BIOCs 

4.2.1.7. View and manage Analytics rules 

📝

🟢note

The default severity is updated by content
updates. If a content update determines a new
default severity for the rule that's the same as
the value you had previously determined, you
won't have the option to reset to default. For
example, if the default was Informational, and
you changed the severity to Medium, and after
a content update Cortex Cloud...

📝

🟢note

When you select multiple rows, you can only
enable or disable the selected rules.

⛔

🔴w…

Changing the default severity may result in
issues not being triggered or too many issues
being triggered. Please consider this carefully
before you change the severity recommended
by Cortex Cloud. Any responsibility for not
getting issues triggered as a result of changing
the severity will be yours.

4.2.1.8. Identity Analytics  🧭

➡️n…

After configuring your Cloud Identity Engine
instance and Cortex Cloud Analytics, select
Settings () → Configurations → Cortex Cloud -
Analytics, and in the Featured in Analytics
section, Enable Identity Analytics

4.2.1.9. Identity Threat Module (ITDR) 

4.2.1.10. AI Detection & Response in Cortex
Cloud (Beta) 

☑️

🟣pr…

To enable AIDR, you must have one of the
following user roles: Instance Admin or Account
Admin.

🧭

➡️n…

Settings → Configurations → Cortex - Analytics

4.2.1.10.1. Data sources and supported services 

4.2.1.10.2. AI Detection & Response Dashboard 

4.2.1.10.3. Collect prompt logs 

5. Web and API Security (WAAS) 

5.1. Overview  🔒

🟣c…

Requires the Cortex Cloud Runtime Security
add-on.

5.2. Personas workflow 

5.3. Secure your API landscape 

5.3.1. Gain visibility and assess risk of API
endpoints 

📝

🟢note

Data classification findings are only available
for enabled profiles.

📝

🟢note

Indicates that a JSON Web Token was identified,
but it is unknown how to determine its type
from the given string. It could be a non-
standard JSON Web Token creation algorithm.

📝

🟢note

To link APIs to business applications, the two
prerequisites must be met:
The VM must have XDR agent with WAAS
enabled.
Applications must be defined in Application
Security Posture Management
(ASPM).Application Security Posture
Management (ASPM)

5.3.2. Monitor and investigate API threats 

5.3.3. Configure API security from end to end 

5.3.3.1. Third-party integrations 

5.3.3.1.1. Ingest AWS API Gateway 

📝

🟢note

You must generate a new key if you did not save.

📝

🟢note

If no logs are showing, send some API requests
on Postman or CURL.

👀

🟠i…

Leave '–'filter-pattern empty as displayed
above.

5.3.3.1.2. Ingest Azure APIM 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

Microsoft Azure APIM service must be running
before starting to configure the integration.

📝

🟢note

From the URL, save the UUID and the resource
group -

/resource/subscriptions/<UUID>/resourceGroup
s/<ResourceGroup>.
The UUID is the Azure account/subscription ID
and the resource group, which is the group
where the APIM Service is defined.

📝

🟢note

Use the search to navigate to the specific
section.

📝

🟢note

131072 bytes = 128 KB. This value determines
the size (in bytes) of request and response
bodies to send to Cortex. Any bytes beyond this
limit are truncated.

📝

🟢note

If any other inbound policies should be added,
they must be added after these elements.

👀

🟠i…

If you want to add additional data to the
<outbound>, add it at the start of the
<outbound> code.

5.3.3.1.3. Ingest Apigee Proxy 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

The steps include the prerequisites that run the
automated script that deploys files and
configurations to Apigee. For manual
configuration, refer to the section Manual
deployment.

📝

🟢note

If you have an existing hookand would like to
integrate with the shared flow, run the
deploy.sh script, and select n' and exit at the
prompt to create a new hook. Refer to the
section Connect to existing hook.

📝

🟢note

Start with the hook in pre-proxy.

📝

🟢note

Archive deployments are currently in preview
and are subject to change.

👀

🟠i…

This should only be done when there are
already existing hooks, and API security shared
flows can't be hooked as a standalone. Run the
deployment script, but skip step 9 by passing n.
This step publishes API security shared flows to
the desired Apigee environment without setting
them to flow hooks.

5.3.3.1.4. Ingest Kong 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

Contact support to obtain the custom plugin
file.

📝

🟢note

The queue parameters can be updated to
change when the plugin mirrors data to Cortex.

📝

🟢note

Use commas to separate tags.

👀

🟠i…

The size of the buffer must be equal or larger
than the max body size setting in the plugin
configuration, on every data plane node.

5.3.3.1.5. Ingest-F5 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

Extract the files from the F5 plugin to a folder
before selecting to upload to F5.

📝

🟢note

The provider for "context_provider" should
always be uppercase.
Supported providers: AWS, GCP, Azure, On-
prem.

📝

🟢note

Select panw_apisec_set_ssl_data only if your
client SSL profile is enabled.

5.3.3.2. Agent-based protection 

📝

🟢note

Web and API Security (WAAS) profiles and
policies are currently a Beta feature.

5.3.3.2.1. Set up Web and API Security profiles 

📝

🟢note

Web and API Security profiles and policies are
currently a Beta feature.

📝

🟢note

In this profile, the Report options configure the
workload to report the corresponding malicious
applications or APIs to Cortex Cloud, without
blocking them. The Disabled options configure
the workloads to neither analyze nor report the
corresponding malware or behavior.

5.3.3.2.2. Apply Web and API Security profiles
to workloads 

📝

🟢note

Web and API Security profiles and policies are
currently a Beta feature.

📝

🟢note

When importing a policy, select whether to
enable the associated policy targets. Rules
within the imported policy are managed as
follows:
New rules are added to the top of the list.
Default rules override the default rule in the
target tenant.
Rules without a defined target are disabled
until the target is specified.

📝

🟢note

The exported file is encoded in Base64 and
cannot be edited.

5.3.3.2.3. Manage Web and API Security
prevention profiles 

5.3.3.2.4. Add a disable prevention rule for
cloud workloads 

👀

🟠i…

All applicable prevention actions are skipped
for the files and process that match the
properties defined in the rule.
Consider the consequences of disabling a
prevention rule before you add the exception,
and monitor it over time.

🧭

➡️n…

Settings → Exceptions Configuration → Disable
Prevention Rules

5.3.3.2.5. Add a support exception rule for cloud
workloads  🧭
➡️n…

From Settings → Exceptions Configuration →
Support Exception Rules, click + Import from
file

5.3.3.2.6. Add a legacy exception rule for cloud
workloads  🧭
➡️n…

Your rule is created, and can be viewed at the
following location: Settings → Exceptions
Configuration → Legacy Agent Exceptions

5.3.3.2.7. Additional workload management tasks  🧭

➡️n…

Several management activities that can be
performed on workloads are accessible from
Inventory → Endpoints → All Endpoints and
from Inventory → Endpoints → Groups

5.3.4. API specification inventory 

📝

🟢note

Even if you have already imported the
specification, you can edit the API specification
in Cortex Cloud and add or update the server
list.

5.3.4.1. Import API specification  📝

🟢note

Even if you already imported the file, you can
edit the API asset and add or update the server
list.

6. Review inventory and explore
your cloud environment  6.1. Asset management 

6.1.1. All Assets 

📝

🟢note

When working with the Asset Inventory page,
consider the following:
To maintain an accurate and clutter-free asset
inventory, an automated cleanup process
periodically removes outdated assets in the
background.

📝

🟢note

The options available are dependent on your
license.

6.1.1.1. Container Images  🧭

➡️n…

All Assets → Compute → Container Images

6.1.1.2. Kubernetes Cluster 

📝

🟢note

The Vulnerabilities tab is only available if the
cluster you wish to analyze has a K8s connector.

🧭

➡️n…

Inventory → All Assets → Compute →
Kubernetes Cluster for a Kubernetes Clusters
assets overview

6.1.1.3. Virtual Machine Image Scanning 

6.1.1.3.1. VM Images Page 

📝

🟢note

The data in the Scanners column is accurate
only for Core Image assets. Ignore the Scanners
value for assets categorized as Registry, Build,
or Runtime images, as it may not reflect an
accurate scan status.

📝

🟢note

The Last Scan value is only accurate for Core
Image assets; ignore the Last Scan values for
Registry, Build, and Runtime images, as they
may be incorrect.

🧭

➡️n…

Inventory → All Assets → Compute → VM
images for Agentless VM Image scanning
overview

6.1.1.3.2. VM Image Details Page  🧭

➡️n…

Inventory → All Assets → Compute → VM
images

6.1.1.4. Cortex Cloud discovery catalog 

6.1.2. Network configuration  6.1.2.1. Configure your network parameters 

📝

🟢note

You can add a range that is fully contained in
an existing range, however, you cannot add a
new range that partially intersects with another
range.

🧭

➡️n…

In Cortex Cloud , select Assets → Network
Configuration → Internal Domain Suffixes

6.1.3. Asset Groups 

📝

🟢note

When you create or edit an Asset Group, the
changes are applied immediately to new assets
and to existing assets that have been updated.
However, it may take a few hours for the
changes to appear on existing assets that have
not been updated.

🧭

➡️n…

Inventory → Assets → Groups → Add Group

6.1.4. Asset Roles 

📝

🟢note

Asset Roles are available only if the Identity
Threat Module add-on is enabled.

6.1.4.1. Manage Asset Roles for Endpoints 

📝

🟢note

Endpoint Role Management is available only if
the Identity Threat Module add-on is enabled.

📝

🟢note

Deleting a manually added endpoint removes
the endpoint from the Included Endpoints list. If
this endpoint is detected automatically as
having this asset role in the future, it will appear
in the Included Endpoints list.
Excluding a manually added endpoint ensures
that even if in the future the endpoint is
detected as having this asset role, this...

6.1.4.2. Manage Asset Roles for Users 

📝

🟢note

User Role Management is available only if the
Identity Threat Module add-on is enabled.

📝

🟢note

Deleting a manually added user removes the
user from the Included Users list. If this user is
detected automatically as having this asset role
in the future, it will appear in the Included Users
list again.
Excluding a manually added user ensures that
even if in the future the user is detected as
having this asset role, this detection is
overridden and...

6.1.4.2.1. Honey user  ☑️

🟣pr…

The honey user role is available only if the
Identity Threat Module add-on is enabled.

6.1.5. Manage Asset Scores 

📝

🟢note

The Hosts tab is available if the Identity Threat
Module add-on is enabled.

📝

🟢note

As new alerts are associated with incidents, the
User and Host Scores are recalculated. You can
view the latest User and Host Scores on the
Asset Scores page, or track the Score trend on
the User Risk View and Host Risk View.

📝

🟢note

Some User Associated Insights may not appear
as part of the User Associated Incidents due to
the insight generation mechanism. For example,
when an insight related to one of the assets in
an incident is generated a few days after the
associated incident, the insight may not be
associated with the incident.

6.1.6. Vulnerability Assessment 

☑️

🟣pr…

The following are prerequisites for Cortex
Cloud to perform a vulnerability assessment.
Requirement
Description
Licenses and Add-ons
Supported Platforms
Windows
Cortex XDR agent 8.3 or a later release.
Cortex Cloud collects all the information about
the operating system and the installed
applications, and calculates CVE.
CVEs that apply to...

📝

🟢note

The CVE will be removed/reinstated to all
views, filters, and widgets after the next
vulnerability recalculation.

📝

🟢note

Starting with macOS 10.15, Mac built-in system
applications are not reported by the Cortex
XDR agent and are not part of the Cortex
Cloud Application Inventory.

💡

🔵tip

You can click each individual CVE to view in-
depth details about it on a panel that appears
on the right.

💡

🔵tip

You can click each individual endpoint to view
in-depth details about it on a panel that
appears on the right.

🧭

➡️n…

You can access the Vulnerability Assessment
panel from Inventory+Endpoints → Host
Inventory → Vulnerability Assessment

7. Review and prioritize posture
issues 

7.1. Cases and issues 

7.1.1. Overview of cases 

7.1.1.1. What are cases? 

7.1.1.2. Resolving cases with AI 

7.1.1.3. Case lifecycle 

7.1.1.4. Case thresholds 

7.1.1.5. Case scope and impact 

7.1.1.6. Case and issue domains 

7.1.2. Case concepts 

7.1.2.1. Issues, findings, and events 

7.1.2.1.1. Issues 

7.1.2.1.2. Findings and events 

7.1.2.2. Case grouping 

7.1.2.3. Case scoring 

7.1.2.4. Case starring  🧭

➡️n…

You can manage all starring configurations
under Case & Issues → Case Configuration →
Starred Issues

7.1.2.5. What is Causality?  📝

🟢note

There are no CGOs in the Cloud Causality
View, when investigating cloud Cortex Cloud
alerts and Cloud Audit Logs, or SaaS Causality
View, when investigating SaaS-related alerts for
501 audit events, such as Office 365 audit logs
and normalized logs.

7.1.3. Analyze and resolve cases 

7.1.3.1. Review all cases  📝

🟢note

The legacy view is also available for users who
prefer this format. From the Actions menu
select Switch to legacy view.

7.1.3.2. Start case analysis 

7.1.3.2.1. Agentic Assistant- Case Investigation

agent 

7.1.3.3. Establish case context 

7.1.3.3.1. AI-generated case summaries 

📝

🟢note

The AI-generated title and description is a
calculated value that is regenerated each time
you open a case.

This value is not a saved static description,
therefore it is not reflected in the saved case
names in the list of cases in the Split view , or in
the Case Name and Case Description columns
in the Table view.

🧭

➡️n…

Configurations → General → Server Settings →
AI Configuration and enable the following
settings

7.1.3.3.2. Assess case severity and score 

7.1.3.3.3. Update case attributes  📝

🟢note

The defined values are shown in the Case Name
and Case Description columns in the Table view,
and saved to the cases dataset. The case title is
also shown the list of cases in the Split view.
These values do not replace the AI-generated
case title and summary. If the Summarize with
AI toggle is enabled, AI-generated case
summaries are automatically...

7.1.3.4. Analyze case details 

📝

🟢note

If you prefer a tabular or legacy layout, switch
the case card to the Detailed view.
This view preserves the legacy tab based
format and custom layouts, ensuring full
backward compatibility. You can switch
between the new case experience and the
legacy view based on personal workflow
preferences. For more information, see Detailed
View.

7.1.3.4.1. Grouping graph  📝

🟢note

Similar issues are displayed as individual entities
rather than in a parent/child hierarchy.

7.1.3.4.2. Evidence 

7.1.3.4.3. Issue feed 

7.1.3.4.4. Associated assets and artifacts  📝

🟢note

If you do not have permissions to access an
asset of a case (which is shown as grayed out
and locked), check your scoping permissions in
Manage Users or Manage User Groups.

7.1.3.4.5. MITRE ATT&CK tactics and techniques  📝

🟢note

This component is available for cases
associated with the Security domain or custom
domains.

7.1.3.4.6. Detailed View  📝

🟢note

When an issue is resolved, it remains linked to a
case. Once all of the issues in a case are
resolved, the case is automatically closed.

7.1.3.4.7. Issue card  📝

🟢note

This section is context-specific and shows data
according to the issue context.

7.1.3.5. Resolve the case 

7.1.3.5.1. Resolution Center  📝

🟢note

Tasks that are already In Progress but still
require your input will appear in both the
Pending and In Progress tabs.

7.1.3.5.2. Collaborative notes and comments 

7.1.3.5.3. Resolve a case  📝

🟢note

If a case is resolved with the status Resolved -
Auto Resolved, Cortex Cloud can reopen the
case for up-to six hours if a new issue is
triggered that matches the case. The six-hour
period is defined by the timestamp of the last
issue that was grouped into the case. After the
six-hour period, any new issues are linked to a
new case for a new...

7.1.3.5.4. Resolution reasons for cases and issues 

📝

🟢note

Cases and issues resolved as True Positive and
False Positive help Cortex Cloud to identify
real threats in your environment by comparing
future cases and associated issues to the
resolved cases. Therefore, the handling and
scoring of future cases is affected by these
resolutions.

📝

🟢note

Cases and issues resolved as True Positive and
False Positive help Cortex Cloud to identify
real threats in your environment by comparing
future cases and associated issues to the
resolved cases. Therefore, the handling and
scoring of future cases is affected by these
resolutions.

7.1.3.6. Additional case actions 

7.1.3.6.1. Create a case 

📝

🟢note

To create a case manually, you must have
View/Edit permission for Cases and Issues
selected under Settings → Configurations →
Access Management → Roles → Components
→ Cases & Issues.

📝

🟢note

This option is only relevant for certain domains.

📝

🟢note

This option is only relevant for certain domains.

📝

🟢note

You can't attach files to manually created
cases.

💡

🔵tip

The issues that you link to a case can be linked
to multiple cases, and the issue domains do not
need to match the case domain.

7.1.3.6.2. Merge a case 

7.2. Investigation and response 

7.2.1. Investigate issues 

7.2.1.1. Overview of the Issues page  📝

🟢note

Every 12 hours, the system enforces a cleanup
policy to remove the oldest issues once the
maximum limit is exceeded. The default issue
retention period in Cortex Cloud is 186 days.

7.2.1.2. Link or unlink issues from a case 

7.2.1.3. Run an automation on an issue 

📝

🟢note

In addition to automation, some playbooks
contain manual tasks that prompt the analyst
for input. This enables you to enhance an
automation workflow with analyst input.

📝

🟢note

You can also manually select a playbook to run
from the Issue Work Plan tab.

7.2.1.4. Use the War Room in an investigation 

📝

🟢note

The case War Room is usually used for
communication capabilities, but unlike the issue
War Room, it does not include playbook
specific entries. The case War Room enables
you to investigate an entire case, not just an
issue.

📝

🟢note

Cortex Cloud does not index notes and chats.

💡

🔵tip

In the Playground, you can clear the context
data, if needed, which deletes everything in the
Playground context data, but does not affect
the actual issue or case. To clear the context,
run !DeleteContext all=yes' from the CLI or
click Clear Context Data while viewing the
context data.

💡

🔵tip

You can use the up/down arrow buttons in the
CLI to do a reverse history search for previous
commands with the same prefix.

🧭

➡️n…

Investigation & Response → Automation →
Playground

7.2.1.5. Use the Work Plan in an investigation  👀

🟠i…

A playbook will not continue its execution path
if a prior task has failed; you must resolve the
failed task before subsequent tasks can run.

7.2.1.6. Issue syncing 

☑️

🟣pr…

You must set up an integration before you can
sync issues. For more information, see Set up an
integration for mirroring issues.

☑️

🟣pr…

You must set up an integration before you can
sync issues. For more information, see Set up an
integration for mirroring issues.

☑️

🟣pr…

You must set up an integration before you can
sync issues. For more information, see Set up an
integration for mirroring issues.

📝

🟢note

Multiple tickets can be linked to an issue with
outbound syncing. Issues with inbound syncing
can be linked to a single ticket only.

📝

🟢note

Using issue fields as variables is not currently
supported.

📝

🟢note

You can only define a single inbound profile. If
you change the inbound sync profile the
current profile is overwritten.
You can define multiple outbound profiles; one
issue can update multiple tickets.

📝

🟢note

Using issue fields as variables is not currently
supported.

📝

🟢note

You can only define a single inbound profile. If
you change the inbound sync profile the
current profile is overwritten.
You can define multiple outbound profiles; one
issue can update multiple tickets.

📝

🟢note

The inbound syncing flow runs every two
minutes, and the outbound syncing flow runs
every five minutes.
In a bi-directional set-up, if the same field is
updated in both tickets, the most recently
updated value is used.
In the external ticket, the logged history shows
updates to the ticket. The user name that is
logged with the history reflects the user...

📝

🟢note

After an issue is resolved, ticket syncing
remains active for up-to seven days. Therefore,
you still update, change, or reopen the issue or
external ticket and the tickets will continue to
sync.

📝

🟢note

If you change the selected inbound sync
profile, the original sync profile is immediately
overwritten.

📝

🟢note

You can only define a single inbound profile. If
you change the inbound sync profile the
current profile is overwritten.
You can define multiple outbound profiles; one
issue can update multiple tickets.

⛔

🔴w…

If you leave this field blank, all configured
instances will be used.

⛔

🔴w…

If you leave this field blank, all configured
instances will be used.

💡

🔵tip

You can find a sync profile ID under Settings →
Configurations → Object Setup → Issues →
Sync Profiles. By default the ID field is not
displayed in the table. Click the three dot menu
and add it to the table layout.

7.2.1.7. Causality view 

💡

🔵tip

To view statistics for files on VirusTotal, you
can pivot from the Initiator MD5 or SHA256
value of the file on the Files tab.

7.2.1.7.1. Cloud causality view 

7.2.1.7.2. SaaS causality view 

7.2.1.7.3. Timeline  📝

🟢note

The Timeline view is not available when
investigating cloud Cortex Cloud issues and
cloud audit logs or SaaS-related issues for 501
audit events, such as Office 365 audit logs and
normalized logs. Only the applicable cloud
causality view and SaaS causality view is
available for this data.

7.2.1.7.4. Network causality view 

📝

🟢note

The network causality view displays only the
information it collects from the detectors. It is
possible that the CI may not show some of the
firewall or agent processes.

💡

🔵tip

To view statistics for files on VirusTotal, you
can pivot from the Initiator MD5 or SHA256
value of the file on the Files tab.

7.2.1.7.5. Causality icons key 

7.2.1.8. Issue investigation actions 

7.2.1.8.1. Copy issues 

7.2.1.8.2. Update issue fields 

📝

🟢note

You must create a custom status before you
can select it.

💡

🔵tip

You can create custom issue statuses and
resolution reasons, and use the setIssueStatus
command to set these custom statuses for
issues.
For example, when a user starts investigating an
issue, the issue status is automatically changed
from New to Under Investigation. In some cases,
it is useful to create an interim status, such as
Triage. After you...

7.2.1.8.3. Create profile exceptions  🧭

➡️n…

Inventory+Endpoints → Policy Management →
Profiles

7.2.1.8.4. Investigate contributing events 

7.2.1.8.5. Create a featured field 

📝

🟢note

Featured Active Directory values are displayed
in the User and Host fields accordingly.

🧭

➡️n…

Cases & Issues → Case Configuration →
Featured Fields and select a type of featured
field

7.2.1.8.6. View generating BIOC or IOC rule 

7.2.1.8.7. Add a file path to a malware profile
allow list  🧭
➡️n…

Inventory+Endpoints → Policy Management →
Prevention → Profiles and locate the malware
profile you selected

7.2.1.8.8. Retrieve additional issue details  💡

🔵tip

If you require assistance from Palo Alto
Networks support to investigate the issue, make
sure to provide the downloaded ZIP file.

7.2.1.8.9. Export issue details to a file 

7.2.1.8.10. Exclude an issue 

7.2.1.8.11. Query case and issue data 

7.2.2. Review findings  7.2.2.1. Findings card  📝

🟢note

The information in this card is context specific,
therefore some sections are not available for all
findings.

7.2.3. Investigate artifacts and assets 

7.2.3.1. Investigate an IP address  📝

🟢note

Requires a license key. Select Settings →
Configurations → Integrations → Threat
Intelligence.

7.2.3.2. Investigate an asset  📝

🟢note

The Asset view is available for hosts with a
Cortex XDR agent installed.

7.2.3.3. Investigate a file and process hash  📝

🟢note

Requires a license key. Go to Settings →
Configurations → Integrations → Threat
Intelligence.

7.2.3.4. Investigate a user 

🔒

🟣c…

If the Identity Threat module is enabled, you
can open the User Risk View. This view displays
insights and profiling information to help you
investigate issues and cases. Viewing anomalies
in the context of baseline behavior facilitates
risk assessment and shortens the time you
require for making verdicts.
If the Identity Threat module is not enabled...

🔒

🟣c…

The User Risk view is available with the ITDR
add-on. Depending on your permissions, some
information may be limited by your scope.

📝

🟢note

Cortex Cloud normalizes and displays case and
issue times in your time zone. If you're in a half-
hour time zone, the activity in the Issues &
Insights Heatmap is displayed in the whole-hour
time slot preceding it. For example, if you're in a
UTC +4.5 time zone, the time displayed for the
activity will be UTC +4.5, however, the
visualization in the...

💡

🔵tip

You can also see a list of all users under
Inventory → Assets → Asset Scores.

7.2.4. Investigate endpoints 

7.2.4.1. Overview of the Action Center 

📝

🟢note

Blocking files on endpoints is enforced by the
endpoint malware profile. To block a hash
value, ensure the hash value is configured in the
Malware security profile.
Select Override Report mode to allow the
agent to block hashes, even if the Malware
Profile is set to Report.

🧭

➡️n…

Investigation & Response → Response → Action
Center

7.2.4.1.1. Initiate and monitor endpoint actions 

7.2.4.1.2. Action Center reference information 

7.2.4.2. Manage endpoints 

📝

🟢note

For the Include endpoints from auto upgrade
action, you cannot enable auto upgrade for
Mobile, VDI, and TS installations.

📝

🟢note

If an endpoint is excluded, the auto upgrade
profile configuration is not available.
If you exclude the endpoint from auto upgrade
while the auto upgrade action is In progress,
the ongoing upgrade will still take place.

📝

🟢note

Content Status is calculated every 30 minutes.
Therefore, there might be a delay of up to 30
minutes in displaying the data.

📝

🟢note

Only supported for Windows and macOS.

📝

🟢note

Changes to the agent status can take up to ten
minutes to display on Cortex Cloud .

🧭

➡️n…

Settings → Configurations → General → Agent
Configurations → Endpoint Administration
Cleanup

7.2.4.3. Retrieve files from an endpoint 

📝

🟢note

You cannot define a path using environment
variables on Mac and Linux endpoints.

📝

🟢note

If the Password Protection (for downloaded
files) setting under Settings → Configuration →
General → Server Settings is enabled, enter the
password 'suspicious' to download the file.

📝

🟢note

Disabling File Retrieval does not take effect on
file retrieval actions that are in progress.

7.2.4.4. Retrieve support logs from an endpoint  🧭

➡️n…

Investigation & Response → Response → Action
Center → + New Action

7.2.4.5. Scan an endpoint for malware  🧭

➡️n…

Investigation & Response → Response → Action
Center → New Action

7.2.5. Investigate files 

7.2.5.1. Manage file execution 

7.2.5.2. Manage quarantined files  📝

🟢note

This will restore all files with the same hash on
all of your endpoints.

7.2.5.3. Review WildFire analysis details  📝

🟢note

Cortex Cloud displays the preview of WildFire
reports that were generated within the last
couple of years. To view a report that was
generated more than two years ago, you can
download the report.

7.2.5.4. Import file hash exceptions 

7.2.6. Cortex Assistant 

📝

🟢note

If you are in an eligible region and have enabled
the Cortex Agentic Assistant, the Cortex
Agentic Assistant replaces the Cortex Assistant.
The Cortex Assistant is available if you do not
have access to the Cortex Agentic Assistant
based on the tenant region or you have not
enabled it. For more information, see Cortex
Agentic Assistant.

🧭

➡️n…

To increase usability, you can create a
personalized keyboard shortcut: Settings →
Configurations → Server Settings → Keyboard
Shortcuts and choose the shortcut you want to
use

7.2.6.1. Cortex Assistant layout 

7.2.6.2. Cortex Assistant capabilities 

📝

🟢note

In some cases, if the prompt does not include at
least one recognizable entity such as an IP,
hash, user, asset, domain, case, or XQL query,
no response is returned.

📝

🟢note

When you choose an option from the Respond
column, Cortex Assistant will always prompt
you to approve the action before executing.

7.2.7. Automation 

7.2.7.1. Automation in Cortex Cloud  📝

🟢note

You can build end-to-end automation
workflows from within the playbook editor,
including creating automation rules,
configuring integration instances, and creating
and editing tasks. For more information, see
Playbooks.

7.2.7.2. Quick Actions  📝

🟢note

Quick Actions appear as War Room entries, but
do not appear in the Work Plan.

7.2.7.3. Automation Exclusion Center 

📝

🟢note

By default, all users have read and edit
permissions to lists. When creating a list of
critical assets, we recommend limiting the read
and edit permissions to specific roles.

7.2.7.3.1. Manage automation exclusion policies 

📝

🟢note

For the IAM User Hard Remediation and User
Soft Remediation policies, we recommend
including username, email, and ID for each user
you want to exclude. Example: username1,
user@example.com, userID112.

📝

🟢note

You can also right click on a policy from the
main Automation Exclusion Center page to
disable or enable the policy.
If you click on a list name in the Exclude
column, that list opens in the Lists page.

7.2.7.4. Playbooks 

☑️

🟣pr…

To provide playbook access, first set the Cases
& Issues RBAC permission to View or View/Edit
and then set the Playbooks permission to View
or View/Edit.
To completely restrict playbook access, first
set the Cases & Issues RBAC permission to
None and then set the Playbooks permission to
None.

7.2.7.4.1. Playbooks overview 

7.2.7.4.2. Playbook development checklist 

7.2.7.4.3. Plan your playbook 

7.2.7.4.4. Manage playbooks  📝

🟢note

The library by default shows only playbooks
that are not adopted. Click the Show Adopted
checkbox to show the adopted playbooks,
indicated by an Adopted mark.
The library shows the most updated playbook
version. Adopting an older version than shown
should be done through Marketplace.
Adopting a playbook does not make it run.
Some content packs include...

7.2.7.4.5. Build your playbook 

7.2.7.4.5.1. Task 1. Choose from existing
playbooks or create your own



📝

🟢note

When a task inside of a sub-playbook is not
configured, the alert is propagated to the main
playbook. If multiple sub-playbooks are nested,
and any of the sub-playbooks have non-
configured components, the alert appears in
the main playbook as well as in the sub-
playbooks. Alerts also appear for the individual
non-configured tasks within the...

💡

🔵tip

If there are additional relevant playbooks in
Marketplace that are not in your Org repository,
you can click Explore them now to see them in
the Playbook Catalog and choose to adopt.

💡

🔵tip

To open multiple playbooks at the same time,
edit the first playbook and then click New next
to the playbook name to create a new tab. You
can either create a new playbook, or add an
existing one.
You can view recently modified or deleted
playbooks by clicking version history for all
playbooks .

🧭

➡️n…

the Investigation & Response → Automation →
Playbooks page to find an existing playbook,
customize it, or create a playbook

7.2.7.4.5.2. Task 2. Configure playbook settings 

📝

🟢note

If the playbook has inputs and outputs, the
Playbook Starts task will show back and forth
arrows. Clicking them opens the Playbook
Settings pane Inputs/Outputs tab.

📝

🟢note

This rule will trigger the playbook to run if no
other Automation Rule triggers the playbook
first. You can view and edit the order the rules
run in the Automation Rules page.

📝

🟢note

If you do not add any fields, the group will be
deleted when you click Save.

7.2.7.4.5.3. Task 3. Add objects from the Task
Library



7.2.7.4.5.4. Task 4. Add custom playbook features 

7.2.7.4.5.5. Task 5. Test and debug the playbook 

7.2.7.4.5.6. Task 6. Manage playbook content 

7.2.7.4.6. Customize your playbook 

7.2.7.4.6.1. Configure a sub-playbook loop 

📝

🟢note

Consider the following when adding a loop:
The maximum number of loops (default is 100).
A high number of loops or a high wait time
combined with a large number of issues may
affect performance.
Periodically check looping conditions to ensure
they are still valid for the data set.
If you want a sub playbook task to loop over an
array passed into its...

💡

🔵tip

Balance between the number of iterations and
the interval so you do not overload the server.

7.2.7.4.6.2. Filter and transform data 

⚠️

🟡c…

You can change the context data root to filter,
but it is not recommended to select a different
root, as it affects the filter results. The drop-
down list displays the filter root for backward
compatibility.

🧭

➡️n…

Investigation & Response → Automation →
Scripts → New Script

7.2.7.4.6.3. Extend context 

7.2.7.4.6.4. Extract indicators 

📝

🟢note

This configuration may delay playbook
execution (issue creation).
While indicator creation is asynchronous,
indicator extraction and enrichment are run
synchronously. Data is placed into the issue
context and is available via the context for
subsequent tasks.

📝

🟢note

When using out of band, the extracted
indicators do not appear in the context. If you
want the extracted indicators to appear select
inline.

7.2.7.4.6.5. Update issue fields with playbook
tasks

 📝

🟢note

The setIssue script includes all available fields;
use the scroll bar to see all the fields.
The name field has a limit of 600 characters. If
there are more than 600 characters, you can
shorten the name field to under 600 characters
and then include the full information in a long
text field such as the description field.

7.2.7.4.7. Test your playbook 

📝

🟢note

The debugger does not support using
parentIncidentFields.

📝

🟢note

Using an existing issue in the debugger does not
affect the original issue or change the original
context data.

7.2.7.4.7.1. Troubleshoot playbook performance 

7.2.7.4.8. Manage playbook content 

📝

🟢note

Opening a playbook does not immediately lock
it. A playbook is considered to be in edit mode
only when a user makes a modification that
causes the Save button to become available.

👀

🟠i…

Manually unlocking a playbook may cause
version conflicts. This turns off concurrent
editing protection, and if the original user is still
editing, their changes might be lost or
overwritten.

🧭

➡️n…

Investigation & Response → Automations →
Playbooks

7.2.7.4.9. Best practices 

📝

🟢note

If you reattach a detached playbook, any
customizations you have made to the playbook
will be overwritten when the playbook updates
to the current version.

📝

🟢note

Retries are not supported for data collection
tasks that have errors sending emails (indicated
by a server timeout). This is because retries only
work on automation execution failures, not on
email delivery issues.

7.2.7.5. AI Prompts 

🧭

➡️n…

On the AI Prompts page (Investigation →
Response → Automation → AI Prompts), you
can view, edit, and manually create prompts

7.2.7.5.1. AI prompts role-based access control 

7.2.7.5.2. Use existing prompts  🧭

➡️n…

Investigation & Response → Automation → AI
Prompts and in the Prompts Library search for
the prompt you want to use

7.2.7.5.3. Create a prompt  🧭

➡️n…

Investigation & Response → Automation → AI
Prompts and click + New Prompt

7.2.7.6. Create an automation rule 

📝

🟢note

You can also define the conditions that trigger
a specific playbook in the playbook editor. For
more information, see Task 2. Configure
playbook settings

📝

🟢note

Quick Actions, by default, run using all
available integration instances that contain the
command. When selecting a Quick Action for
an automation rule, you can instead choose one
specific integration instance to use.

👀

🟠i…

Automation rules apply to Medium and higher
severity issues. They also apply to Low severity
Analytic issues and Low severity ABIOC issues
that are tagged with Identity or Cloud.
Rules are evaluated in order, and only the first
rule that matches the trigger conditions is
executed.

7.2.7.7. Scripts 

7.2.7.7.1. Use existing scripts  🧭

➡️n…

Investigation & Response → Automation →
Scripts and in the Scripts Library search for the
script you want to use

7.2.7.7.2. Create a script 

📝

🟢note

You can enable/disable a script in the Settings
without having to duplicate the script.
You can view recently modified or deleted
scripts by clicking the version history for all
scripts .

👀

🟠i…

If you choose Python, from the Agentic
Assistant you can use the Automation Engineer
agent.

💡

🔵tip

If there is an error, you can copy the error
message from the test result into the Agentic
Assistant prompt and ask the Automation
Engineer agent to correct the error.

7.2.7.7.3. Use the Automation Engineer agent to
accelerate script development and deployment 

📝

🟢note

The Automation Engineer agent is available
with the Cortex Agentic Assistant, for users
with script editing permissions. For more
information, see Cortex Agentic Assistant and
Agentic Assistant role-based access control.

📝

🟢note

If you make manual edits in the script and don't
save the changes and you then modify the
script with the Automation Engineer agent, you
are prompted to confirm overwriting the
manual edits.

💡

🔵tip

Use detailed prompts, for example, Get failed
logins from last 24 hours and return as a table.
Clearly define argument names, types, and
default values.
Mention if you expect the output in a specific
format, such as a table, JSON, or plain text.

💡

🔵tip

If there is an error, you can copy the error
message from the test result into the Agentic
Assistant prompt and ask the Automation
Engineer agent to correct the error.

7.2.7.7.4. Change the Docker image in a script  7.2.7.7.4.1. Connect an engine to an image registry  📝

🟢note

This procedure uses the --username and --
password command line options to pass the
username and password directly. For
environments where command history or logs
are visible to others, consider more secure
methods like Docker configuration files for
handling authentication in production or CI/CD
environments. For more details, see docker
login or...

7.2.7.8. Context data 

7.2.7.8.1. Issue context data 

7.2.7.8.2. Case context data 

7.2.7.8.3. Search context data 

7.2.7.8.4. Add context data to an issue 

7.2.7.8.5. Add context data to a case  📝

🟢note

If you run the command in the issue War Room,
the data is added to the following places:
The case context data.
The issue context data under the case tab.
If you run the command in the Case War Room,
the data is added to the case context data only.

7.2.7.8.6. Delete context data from a case 

7.2.7.8.7. Use context data in a playbook  ⚠️

🟡c…

Users with Trigger Playbook permissions on a
given issue may still be able to modify the
parent case via commands and scripts, even
without full access to the case.

7.2.7.9. Lists 

📝

🟢note

The maximum size of a list is 209715 characters.

7.2.7.9.1. Create a list 

📝

🟢note

If you intend to use the list as part of an
Automation Exclusion Policy, we recommend
choosing specific roles for read and edit
access. For more information on using lists in
exclusion policies, see Manage automation
exclusion policies.Manage automation exclusion
policies

📝

🟢note

If you want to edit a list from a content pack,
you need to duplicate or detach a list.
Detached lists do not receive updated content
in subsequent Cortex Cloud content releases.
To retain an updated list, reattach it.

7.2.7.9.2. List commands 

7.2.7.9.3. Use cases: JSON lists  🧭

➡️n…

Settings → Configurations → Object Setup →
Lists → Add a List

Investigation & Response → Automation →
Playbooks → New Playbook

7.2.7.9.4. Transform a list into an array  🧭

➡️n…

Investigation & Response → Automation →
Playbooks and create or edit a playbook

7.2.7.10. Integrations 

7.2.7.10.1. Integrations in Cortex Cloud 

7.2.7.10.2. Add an integration instance 

7.2.7.10.3. Use integration commands in the CLI  💡

🔵tip

In the Playground, you can clear the context
data, if needed, which deletes everything in the
Playground context data, but does not affect
the actual issue or case. To clear the context,
run !DeleteContext all=yes' from the CLI or
click Clear Context Data while viewing the
context data.

7.2.7.10.4. Troubleshoot Integrations 

7.2.7.10.5. Manage credentials  🧭

➡️n…

Settings → Configurations → Integrations →
Credentials → New Credential

7.2.7.11. Engines 

7.2.7.11.1. What is an engine? 

📝

🟢note

You cannot share a multiple-engine installation
with a single-engine installation.

📝

🟢note

When you add an engine to a load-balancing
group, you cannot use that engine separately.
The engine does not appear in the engines
menu when configuring an integration instance,
but you can choose the load-balancing group.

7.2.7.11.2. Engine requirements 

📝

🟢note

The Cron package is required to install engines
on a Linux machine.

📝

🟢note

CentOS 8.x reached End of Life (EOL) on
December 31, 2021, and is no longer supported
as an operating system.
CentOS 7.x reached End of Life (EOL) on June
30, 2024, and is no longer supported as an
operating system.

📝

🟢note

If you have configured a range of Approved IP
Ranges under Allowed Sessions on the Security
Settings page, the engine must communicate
through one of the approved IPs.

7.2.7.11.3. Install an engine 

📝

🟢note

If you are using DEB, RPM, or Zip installation,
install Docker or Podman.
Natively running Python or PowerShell
integrations/scripts on Windows or Linux is not
supported on Cortex Cloud engines.

📝

🟢note

When upgrading an engine that was installed
using the Shell installation, you can use the
Upgrade Engine feature in the Engines page. For
Amazon Linux 2 type engines, you need to
upgrade these engine types using a zip-type
engine and not use the Upgrade Engine feature.
If you use the shell installer, Docker/Podman is
automatically installed. We...

📝

🟢note

Use DEB and RPM installation when the shell
installation is not available. You need to
manually install Docker or Podman and any
dependencies.

📝

🟢note

If you receive a permissions denied error, it is
likely that you do not have permission to access
the /tmp directory.
If the installer fails to start due to a permissions
issue, even if running as root, add one of the
following two arguments when running the
installer:
--target <path> - Extracts the installer files
into the specified custom...

👀

🟠i…

For DEB/RPM engines, Python (including 3.x)
and the containerization platform
(Docker/Podman) must be installed and
configured. For Docker or Podman to work
correctly on an engine, IPv4 forwarding must be
enabled.

💡

🔵tip

For Linux systems, we recommend using the
shell installer. If using Amazon Linux 2, use the
zip installer (see step 4).

7.2.7.11.3.1. Docker 

📝

🟢note

This section is relevant when installing an engine.

📝

🟢note

Docker images can be downloaded together
with their relevant content packs for offline
installation.

7.2.7.11.3.2. Podman  📝

🟢note

When upgrading an engine, the engine keeps
the previously used container management
type (regardless of distribution version).

7.2.7.11.4. Manage engines 

7.2.7.11.5. Upgrade an engine 

7.2.7.11.6. Remove an engine 

7.2.7.11.7. Configure engines 

📝

🟢note

A day is defined as 24 hours and may not
exactly correspond to calendar days due to
daylight savings, leap seconds, etc.

📝

🟢note

In addition, to support engine upgrades from
the UI, edit the /usr/local/demisto/upgrade.conf
file on the engine to include the SERVER_URLS
setting with the new tenant's address. Include
only the host, without https:// or any additional
path at the end of the host name. For example:
SERVER_URLS="api-
example.us.paloaltonetworks.com"

📝

🟢note

If the engine loses communication for longer
than this time, it will disconnect and you need
to restart the service.

7.2.7.11.7.1. Configure the engine to use a web
proxy



📝

🟢note

You need to configure Docker to use a proxy.
When using a BlueCoat proxy, ensure you
encode the values correctly.

📝

🟢note

In an environment with a single engine, go to
/usr/local/demisto/upgrade.conf. In an
environment with multiple engines, go to
/usr/local/demisto/<engine-
name>/upgrade.conf, replacing <engine-name>
with the name of the engine.
Note that the key is in the upgrade.conf file and
must be https_proxy, even if your proxy address
starts with http://.

7.2.7.11.7.2. Configure the engine to call the
server without using a proxy



7.2.7.11.7.3. Configure an engine to use custom
certificates 

7.2.7.11.8. Use an engine in an integration  📝

🟢note

Long-running integrations should not run on
load-balancing groups.

7.2.7.11.9. Run a script using an engine  🧭

➡️n…

Investigation & Response → Automation →
Scripts

7.2.7.11.10. Troubleshoot engines 

📝

🟢note

Ensure that pop-ups are not blocked by your
browser.

📝

🟢note

If the installer fails to start due to a permissions
issue, even if running as root, add one of the
following two arguments when running the
installer:
--target <path> - Extracts the installer files
into the specified custom path.
--keep - Extracts the installer files into the
current working directory (without cleaning at
the end).
If using installer...

📝

🟢note

If the Allow running multiple engines on the
same machine option is selected, run the
command:
sudo systemctl status d1_<Engine _name>

📝

🟢note

Any changes made to variables in the
upgrade_engine.sh file are reset after each
upgrade. We recommend instead using the
upgrade.conf file to set variables.

📝

🟢note

You can ignore the following error: Cannot
create folder '/var/lib/demisto'

📝

🟢note

If the Allow running multiple engines on the
same machine option is selected, run the
command:
sudo systemctl status d1_<Engine _name>

📝

🟢note

If the installer fails to start due to a permissions
issue, even if running as root, add one of the
following two arguments when running the
installer:
--target <path> - Extracts the installer files
into the specified custom path.
--keep - Extracts the installer files into the
current working directory (without cleaning at
the end).
If using installer...

📝

🟢note

If the Allow running multiple engines on the
same machine option is selected, run the
command:
sudo systemctl status d1_<Engine _name>

📝

🟢note

If the Allow running multiple engines on the
same machine option is selected, run the
command:
sudo systemctl status d1_<Engine _name>

7.2.7.11.11. Troubleshoot integrations running on
engines 

📝

🟢note

If the Allow running multiple engines on the
same machine option is selected, run the
command:
sudo systemctl restart d1_<Engine _name>

📝

🟢note

Docker CE installations typically run Docker,
while Docker EE installations typically run
dockerroot.

📝

🟢note

If the Allow running multiple engines on the
same machine option is selected, run the
command:
sudo systemctl restart d1_<Engine _name>

📝

🟢note

If the Allow running multiple engines on the
same machine option is selected, run the
command:
sudo systemctl restart d1_<Engine _name>

7.2.8. Response actions 

7.2.8.1. Initiate a Live Terminal session 

📝

🟢note

You can run PowerShell 5.0 or a later release on
Live Terminal of Windows.

📝

🟢note

Network drives are not supported.

📝

🟢note

On Windows endpoints, you cannot run GUI-
based cmd commands like winver or appwiz.cpl.

📝

🟢note

Disabling Live Terminal does not take effect on
sessions that are in progress.

💡

🔵tip

You can fine-tune the Live Terminal session
visibility on the endpoint by adjusting the User
Interface options in your Agent Settings Profile.

7.2.8.2. Isolate an endpoint 

📝

🟢note

IP-based file storage protocol traffic will also
be blocked. This might affect endpoint
functionality if the endpoint uses such mounts.

📝

🟢note

If file system operations become unresponsive
during isolation, such as being unable to list
folder content, unmount the mounted network
shares.

💡

🔵tip

If needed, Filter the list of endpoints.

7.2.8.3. Pause endpoint protection 

📝

🟢note

Pausing your endpoint protection modules
leaves your machines exposed to risks.

🧭

➡️n…

Investigation & Response → Response → Action
Center and locate Action Type Pause Endpoint
Protection or Resume Endpoint Protection

7.2.8.4. Remediate changes from malicious
activity 

☑️

🟣pr…

To initiate remediation suggestions, you must
have the following system requirements:
An App Administrator, Privileged Responder, or
Privileged Security Admin role permissions
which include the remediation permissions.
EDR data collection enabled.
Agent version 7.2 or above on Windows
endpoints.

📝

🟢note

Endpoints that are part of the Case view and
do not meet the required criteria are excluded
from the remediation analysis.

7.2.8.5. Search and destroy malicious files 

☑️

🟣pr…

The following are prerequisites to enable
Cortex Cloud to search and destroy files on
your endpoints:
Supported platforms:
Windows: Cortex XDR agent version 7.2 or a
later. If you plan to enable Search and Destroy
on VDI sessions, you must perform the initial
scan on the Golden Image.
Mac: Cortex XDR agent version 7.3 or a later
release running on...

📝

🟢note

The Cortex Cloud agent does not include the
following information in the local files inventory:
Information about files that existed on the
endpoint and were deleted before the Cortex
Cloud agent was installed.
Information about files where the file size
exceeds the maximum file size for hash
calculations that are pre-configured in Cortex
Cloud .
If...

7.2.8.6. Manage external dynamic lists 

🔒

🟣c…

This functionality is not supported for XDR
Prevent.

📝

🟢note

Ensure EDL sizes don't exceed your firewall
model limit.

💡

🔵tip

You can also use the IP list and URL lists as part
of a URL Filtering policy, or the domain list as
part of a custom Anti-Spyware profile.

7.2.8.7. Collect a memory image  🔒

🟣c…

This functionality has the following license
requirements:
Forensics add-on license.

7.2.9. Forensics 

🔒

🟣c…

Requires the Forensics add-on

7.2.9.1. Forensic investigations 

7.2.9.1.1. Manage an investigation 

7.2.9.1.1.1. Create a new investigation  📝

🟢note

To set up user permissions, you must have
Scope-Based Access Control (SBAC) enabled.

7.2.9.1.1.2. Edit an investigation 

7.2.9.1.1.3. Close an investigation  📝

🟢note

When you close an investigation, Palo Alto
Networks has a grace period of 24 hours before
deleting any collections associated with the
investigation. During this timeframe, you have
the option to cancel the close investigation
action.

7.2.9.1.1.4. User permissions  📝

🟢note

Even if a user does not have access to view an
investigation via the Forensics Investigations
page, they can still query the results of the
collections using an XQL query.

7.2.9.1.2. Data collection 

7.2.9.1.2.1. Hunting 

7.2.9.1.2.2. Triage 

7.2.9.1.2.3. Configure collection 

📝

🟢note

When search fields are specified, the search is
limited based on those filters. If more than one
entry is in a search filter field, the search
returns entries that match any of them. For
example: A File Search with two specified paths
("C:\Test*" and "C:\Windows*") will return
results from both the Test and Windows folders.
If you specify multiple...

📝

🟢note

Data collected during a Triage Collection is
categorized into Artifacts, Volatiles, and File
Collection

📝

🟢note

Supported only for the UFW tool (Firewall
managment tool for some Linux distributions
such as Ubuntu)

7.2.9.1.3. Analysis and documentation 

7.2.9.1.3.1. Review alerts 

7.2.9.1.3.2. Investigation timeline 

7.2.9.1.3.3. Key assets & artifacts 

7.2.9.1.4. Export  📝

🟢note

You can export a collection more than once.

7.2.10. Build XQL queries 

📝

🟢note

If you have the Cortex Agentic Assistant, you
can use natural language prompts to create
and run XQL queries within the chat interface.
For more information, see Create and run XQL
queries with Agentic Assistant chat.Cortex
Agentic Assistant

7.2.10.1. About the Query Builder 

📝

🟢note

Schema changes to datasets may not be
reflected in the autocomplete suggestions and
definitions as you type in real time the XQL
query, and can appear with a slight delay.

💡

🔵tip

When creating XQL queries, you can:
Use the up and down arrow keys to navigate
through the auto-suggestion commands and
definitions.
Select an auto-suggestion command by
pressing either the Enter or Tab key.
Press Shift+Enter to add a new line, and easily
ignore the auto-suggestion output.
Close the auto-suggestion output by pressing
the Esc key.

7.2.10.2. How to build XQL queries 

📝

🟢note

Users with different dataset permissions can
receive different results for the same XQL query.
An administrator or a user with a predefined
user role can create and view queries built with
an unknown dataset that currently does not
exist in Cortex Cloud. All other users can only
create and view queries built with an existing
dataset.
When you have more...

📝

🟢note

You can build a query that investigates data in
both a cold dataset and a hot dataset in the
same query. In addition, as the hot storage
dataset format is the default option and
represents the fully searchable storage, this
format is used throughout this guide for
investigation and threat hunting. For more
information on hot and cold storage, see...

👀

🟠i…

Forensic datasets are not inlcuded by default in
XQL query results, unless the dataset query is
explicitly defined to use a forensic dataset.

7.2.10.2.1. Get started with XQL queries 

7.2.10.2.2. Useful XQL user interface features 

📝

🟢note

Schema changes to datasets may not be
reflected in the autocomplete suggestions and
definitions as you type in real time the XQL
query and can appear with a slight delay.

💡

🔵tip

When creating XQL queries, you can:
Use the up and down arrow keys to navigate
through the auto-suggestion command
suggestions and definitions.
Select an auto-suggestion command by
pressing either the Enter or Tab key.
Press Shift+Enter to add a new line, and easily
ignore the auto-suggestion output.
Close the auto-suggestion output by pressing
the Esc...

7.2.10.2.3. XQL Query best practices 

7.2.10.2.4. Expected results when querying fields 

7.2.10.2.5. Create XQL query 

📝

🟢note

Whenever the time period is changed in the
query window, the config timeframe is
automatically set to the time period defined for
the entire query, including queries that are part
of the join stage. Yet, this won't be visible as
part of the query. Only if you manually type in
the config timeframe will this be seen in the
query.

These time picker...

💡

🔵tip

When creating XQL queries, you can:
Use the up and down arrow keys to navigate
through the auto-suggestion command
suggestions and definitions.
Select an auto-suggestion command by
pressing either the Enter or Tab key.
Press Shift+Enter to add a new line, and easily
ignore the auto-suggestion output.
Close the auto-suggestion output by pressing
the Esc...

💡

🔵tip

While the query is running, you can navigate
away from the page. A notification is sent when
the query has finished. You can also Cancel the
query or run a new query, where you have the
option to Run only new query (cancel previous)
or Run both queries.

7.2.10.2.6. Review XQL query results 

📝

🟢note

It's also possible to graph the results displayed.
For more information, see Graph query results.

📝

🟢note

Real time query results are available only in the
Query Builder and in free text query.
Real time results are displayed only for queries
run on hot datasets.
The Sort option is available only after all the
data is retrieved.
When you formulate complex queries, the
results will be displayed when the query has
finished running completely, and not in real...

📝

🟢note

Results are received incrementally for the first
100K records, or up to 100MB worth of records,
whichever comes first. After that, the next
update is when the query has finished running
completely.

📝

🟢note

In order for Cortex Cloud to provide a
histogram for a field, the field must not contain
an array or a JSON object.

7.2.10.2.7. Translate to XQL  👀

🟠i…

This feature is still in a Beta state and you will
find that not all Splunk queries can be
converted to XQL. This feature will be improved
upon in the upcoming releases to support
greater Splunk query translations to XQL.

7.2.10.2.8. Graph query results 

🔒

🟣c…

Building Cortex Query Language (XQL) queries
in the Query Builder requires a Data Collection
add-on.

📝

🟢note

To display the result of as a time duration,
choose the graph type Single Value and enable
Show as Time. You can then select the Time
Unit (millisecond, second, minute, or hour) and
the Display format.

7.2.10.3. XQL query entities 

7.2.10.3.1. Create authentication query 

7.2.10.3.2. Create event log query 

7.2.10.3.3. Create file query 

7.2.10.3.4. Create image load query 

7.2.10.3.5. Create network connections query 

7.2.10.3.6. Create network query  📝

🟢note

When you run the query, depending on the
outcome of the results, the value specified in
this field might be displayed in the dst_ip field
in the query results. This occurs if an RDP event
is recorded whereby a user connected from the
source IP to the destination IP.

7.2.10.3.7. Create process query 

7.2.10.3.8. Create registry query  👀

🟠i…

Ensure the KEY NAME is entered as a real
registry key name, and not as a symbolic link.
Otherwise, the query will not retrieve results.
Instead of
HKEY_LOCAL_MACHINE\System\CurrentControl
Set, which is a symbolic link, use
KEY_LOCAL_MACHINE\System\ControlSet001.
Instead of HKEY_CURRENT_USER, use
HKEY_USERS\<SID>, where SID is either a SID of
the current...

7.2.10.3.9. Query across all entities 

7.2.10.4. Overview of the Query Center 

📝

🟢note

Very short queries might not be listed.
You cannot cancel correlation queries.
The default retention period for historic queries
is aligned with issue retention.

7.2.10.4.1. Edit and run queries in Query Center 

📝

🟢note

If query limits are applied to your tenant, the
number of concurrent running queries is limited
per user. If query usage is reaching the defined
limit, a system message warns you that a high
query load is impacting performance. If you
exceed the limit, new queries are blocked until
query usage drops. You can view all active
queries under Query Center...

📝

🟢note

You can cancel your own queries. To cancel
queries run by other users, you must have
View/Edit permissions for Configurations →
Query Management. By default, Instance
administrators have View/Edit permission.

📝

🟢note

Cancelled queries show a Canceled status. You
can see details of all canceled queries in the
Query History tab.
You cannot cancel correlation rule queries.
If you cancel a scheduled query, only the
current query is cancelled. Future recurrences
of the scheduled query are not affected.

7.2.10.4.1.1. Query Center reference information  📝

🟢note

Certain fields are exposed and hidden by
default. An asterisk (*) is beside every field that
is exposed by default.

7.2.10.5. Manage scheduled queries 

7.2.10.5.1. Scheduled Queries reference

information  📝
🟢note

Certain fields are exposed and hidden by
default. An asterisk (*) is beside every field that
is exposed by default.

7.2.10.6. Manage your personal query library 

7.2.11. Quick Launcher  📝

🟢note

For hosts, Cortex Cloud displays results for
exact matches but supports the use of
wildcard (*) which changes the search to return
matches that contain the specified text. For
example, a search of compy-7* will return any
hosts beginning with compy-7 such as compy-
7000, compy-7abc, and so forth.

7.2.12. Research a known threat 

7.3. Customize cases and issues  7.3.1. Customize cases and issues 

7.3.1.1. Set up case scoring 

📝

🟢note

For scoped users, a small lock icon indicates
that you don't have permissions to edit a rule.

💡

🔵tip

Right-click an issue field to add it as match
criteria.

7.3.1.2. Create a starring configuration  💡

🔵tip

Right-click an issue field to add it as match
criteria.

7.3.1.3. Create custom case statuses and
resolution reasons



📝

🟢note

Before you create a custom status, please
review the built-in options. For more
information, see Resolution reasons for cases
and issues.
We recommend using the built-in statuses and
resolution reasons where possible. Custom
statuses and resolution reasons might not be
supported by all content, and status syncing
can take time.
In addition, custom...

📝

🟢note

After creation, custom statuses and resolution
reasons cannot be deleted or modified.

🧭

➡️n…

Configurations → Object Setup → Cases →
Properties

7.3.1.4. Create a sync profile 

📝

🟢note

If an issue is using bi-directional syncing, you
need to provide both an Inbound and an
outbound sync profile.

📝

🟢note

Blank fields are skipped.
You must define exact values.
Custom status values are not currently
supported.
Support is currently limited to a specific set of
fields.

8. Agentic Assistant chat 

8.1. Get started with Agentic Assistant chat  📝

🟢note

The Cortex Agentic Assistant is currently
available in limited regions. For more
information see Cortex Agentic Assistant If your
tenant is not within one of those regions, you
have access to the Cortex Assistant.
To enable the Cortex Agentic Assistant, go to
Settings → Configurations → General → Server
Settings → AI Configuration.

8.2. Choose an Agentic Assistant agent  📝

🟢note

Requires the Exposure Management add-on.

8.3. Chat with an Agentic Assistant agent 

📝

🟢note

Case data is only loaded when you send your
first message. Opening the chat interface
without sending a prompt does not provide the
agent with the case context.

📝

🟢note

An agent's proposed plans and results may
contain inaccuracies or errors. Always review
the results carefully to ensure you fully
understand the proposed action before
proceeding.

💡

🔵tip

You can quickly jump to different product
pages within Cortex Cloud by typing / in the
prompt area. This shortcut is a built-in
navigation feature that is available even if the
Cortex Agentic Assistant is disabled.

8.4. Create and run XQL queries with Agentic
Assistant chat 

📝

🟢note

The TextToXQL action is a hidden system action
and does not appear in the Agents Hub.

📝

🟢note

If you are in a non-supported region and ask an
agent to create an XQL query, it may attempt
to generate the query using general logic. We
do not recommend this, as it may result in
inaccurate syntax and unreliable results.

📝

🟢note

Running XQL queries manually through agent
does not consume compute units. This includes
scenarios where you prompt the agent to
create and execute a multi-step plan.

8.5. Manage chat history 

9. Review and report your security

posture and progress

 9.1. Monitor dashboards and reports 

9.1.1. About dashboards 

📝

🟢note

Dashboards that are based on custom
infrastructure cannot be exported.
When importing a dashboard that already
exists in the system, the imported dashboard
overwrites the existing dashboard. To avoid
overwriting the existing dashboard, duplicate
and rename the existing dashboard before
importing the new dashboard.

👀

🟠i…

The ability to create, edit, or share custom
dashboards is governed by access
management. If certain options are unavailable,
contact your administrator. For more
information, see Manage access to custom
dashboards.

9.1.1.1. Command Center dashboards 

📝

🟢note

Access to these dashboards requires RBAC
permissions under Dashboards & Reports. The
Dashboards component must be Enabled in
your role and requires View permissions for
Command Center Dashboards. If certain
options are unavailable, contact your
administrator. For more information, Manage
access to custom dashboards.
The dashboards are available in dark...

9.1.1.1.1. Cortex Agentic Assistant dashboard 

☑️

🟣pr…

The Cortex Agentic Assistant is only available
after it is enabled, and for tenants in certain
regions. For more information, see Cortex
Agentic Assistant.

📝

🟢note

Access to this dashboard requires RBAC
permissions under Dashboards & Reports. The
Dashboards component must be Enabled in
your role and requires View permissions for
Command Center Dashboards. If certain
options are unavailable, contact your
administrator. For more information, see
Manage access to custom dashboards.

9.1.1.1.2. Cloud Security Operations  9.1.1.1.2.1. Cloud Security Operations  📝

🟢note

Command Center data may not match the
counts on the Issues page, and you may
observe inconsistencies. This is because
dashboard data is a snapshot of issues
identified, whereas the Issues page provides the
most up-to-date view of risks across your
cloud assets. In addition, the Issues pages do
not support all the currently available filters on
the...

9.1.1.1.3. Cortex Cloud Command Center 

9.1.1.2. Predefined dashboards 

📝

🟢note

Select the star in the right corner of a widget to
filter the data for cases that match case-
starring policies.
A purple star indicates that the widget is
displaying only starred cases. The starring filter
is persistent and continues to show the filtered
results until you clear the star.

📝

🟢note

Users can access all information on the
dashboard when their user access is scoped to
view All assets or assigned to the Instance
Administrator role. Otherwise, users with
granular scoping set to No assets or Select
asset groups will have limited access to the
dashboard. For more information on Scope-
Based Access Control (SBAC), see Manage
user...

9.1.1.3. Reports  9.1.1.3.1. Report templates 

9.1.2. Build custom dashboards and reports 

9.1.2.1. Build a custom dashboard  📝

🟢note

For agent-related widgets, you can apply an
endpoint scope to refine the displayed data to
only show results from specific endpoint
groups. Select the menu on the top right corner
of the widget, select Groups, and select one or
more endpoint groups.
For case-related widgets, you can refine the
displayed data to only show results from cases
that match a...

9.1.2.2. Manage your Widget Library 

📝

🟢note

Unlike dashboards, individual widgets cannot
be shared with other users directly from the
Widget Library. Yet, if you add a Restricted
widget to a dashboard and then share that
dashboard, other users who have access to the
dashboard will also be able to see that specific
widget.

📝

🟢note

Any dashboards or reports that include the
widget are affected by the changes.

9.1.3. Fine-tune dashboards and reports 

9.1.3.1. Create a custom widget using a script 

📝

🟢note

If you have added arguments to the script,
these appear when creating a widget.

📝

🟢note

Available options are Pie, Column, Line, and
Single Value.
To display the result of the script as a time
duration, choose the graph type Single Value
and enable Show as Time. You can then select
the Time Unit (millisecond, second, minute, or
hour) and the Display format.

9.1.3.1.1. Script-based widget examples 

📝

🟢note

Add the widget tag in the script settings to
make the script available for use in script-
based widgets. For more information, see
Create a script.

📝

🟢note

If your script returns a time duration, configure
the widget with the graph type Single Value
and enable Show as Time..

9.1.3.2. Create a text widget  📝

🟢note

From the Dashboard Manager, you can right-
click an existing dashboard and choose Save as
report template. You can also add text-based
widgets directly to new or existing report
templates, as described below.

9.1.3.3. Create custom XQL widgets 

📝

🟢note

Cortex Query Language (XQL) queries
generated from the Widget Library do not
appear in the Query Center. The results are
used only for creating the custom widget.

📝

🟢note

Whenever the time period is changed in the
query window, the config timeframe is
automatically set to the time period defined,
but this won't be visible as part of the query.
Only if you manually type in the config
timeframe will this be seen in the query.
These time picker options are available in XQL
queries when using the Query Builder, XQL
Widgets,...

📝

🟢note

To display the result of as a time duration,
choose the graph type Single Value and enable
Show as Time. You can then select the Time
Unit (millisecond, second, minute, or hour) and
the Display format.

💡

🔵tip

You can create a generic dashboard for
multiple views of the same dataset by using an
asterisk (*) when defining the dataset in the
XQL widget as dataset = <dataset_name>*. The
placement of the asterisk in the dataset name
ensures that any view containing this prefix text
is displayed in the results.
The dataset in a query is defined as:
If there are...

9.1.3.3.1. Configure filters and inputs for custom
XQL widgets 

☑️

🟣pr…

Fixed filters are based on parameters that are
defined in custom XQL widgets. Before you can
configure fixed filters, take the following steps:
Create custom XQL widgets with parameters.
For more information, see Create custom XQL
widgets.
Add the widgets to a custom dashboard. For
more information, see Build a custom
dashboard.

📝

🟢note

This option only appears if the dashboard
contains custom XQL widgets with defined
parameters.

📝

🟢note

You can define up to four parameter filters on a
single dashboard or report.

📝

🟢note

If you specify more than one field, only the first
field value is used.

💡

🔵tip

After the initial setup, when you access your
dashboard the filters and inputs might need
further refinement. You can make changes to
the configured parameters in the XQL widgets,
and update the Filters & Inputs on your
dashboard until you are satisfied with the
results.

9.1.3.3.2. Configure dashboard drilldowns 

☑️

🟣pr…

To configure drilldowns your dashboard must
contain custom XQL widgets. In addition, if you
want to configure in-dashboard drilldowns your
custom XQL widget must contain one or more
parameters. For more information about
configuring parameters in custom XQL widgets,
see Create custom XQL widgets.

📝

🟢note

If the selected parameter is configured in other
XQL widgets on the dashboard, these widgets
are also affected by the drilldown.

9.1.3.3.2.1. Variables in drilldowns 

9.1.4. Run or schedule reports 

📝

🟢note

Report templates that are based on custom
infrastructure cannot be exported.
If you import a report template that already
exists in the system, the imported template will
overwrite the existing template. If you do not
want to overwrite the existing template,
duplicate and rename the existing template
before importing the new template.
You can also...

📝

🟢note

The report name and description will be
displayed in the report header and are not
editable during customization.

📝

🟢note

For agent-related widgets, you can apply an
endpoint scope to refine the displayed data to
only show results from specific endpoint
groups.

Select the menu on the top right corner of the
widget, select Groups, and select one or more
endpoint groups.
For agent-related widgets, you can apply an
endpoint scope to refine the displayed data to
only show...

📝

🟢note

To send reports to Slack, Slack must be
configured as an external application in Cortex
Cloud. For more information, see Integrate
Slack for outbound notifications

🧭

➡️n…

Under Settings → Configurations → General →
Notifications, click + Add Forwarding
Configuration

10. Discovery Engine  10.1. What is the Discovery Engine? 

11. Cortex Cloud AI Security 

11.1. What is Cortex Cloud AI Security? 

11.2. Supported services in Cortex Cloud AI
Security 

11.3. Cortex Cloud AI Security concepts 

11.4. Cortex Cloud AI Security use cases 

11.5. How to perform advanced AI Security
investigations using XQL  📝
🟢note

For more information, see Build XQL queries.

12. Cortex Cloud Application

Security 

12.1. Onboard Data Sources 

12.1.1. Onboard version control systems 

📝

🟢note

Cortex Cloud Application Security (which
includes IaC and Secrets scanning), is an add-
on to a license (such as Posture Security) that
must be purchased separately.

📝

🟢note

Disclaimer: When onboarding with third-party
data sources, we outline the required steps for
setup, but we do not monitor these external
resources, and they may change over time.
Always refer to the relevant third-party
documentation for the most current integration
steps.

💡

🔵tip

Navigate to Settings → Data Sources &
Integrations → + Add New → and enter your
VCS data source in the search bar.

12.1.1.1. AWS CodeCommit 

☑️

🟣pr…

Before you begin, ensure the following:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
AWS user permissions: To onboard
CodeCommit, the user logged into the AWS
Console must have permissions to deploy the
CloudFormation stack and authorize the...

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

💡

🔵tip

The instance ID is identical to the stack ID on
the AWS platform.

🧭

➡️n…

Modules → Application Security → Periodic
scans

12.1.1.2. Azure DevOps 

☑️

🟣pr…

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
Azure DevOps permissions: Ensure the user
performing the integration holds one of the
following roles in Azure DevOps:
Project Administrator: This permission is
required to subscribe...

📝

🟢note

This forces Azure to bypass browser cookies
and issue a token for the correct directory.

📝

🟢note

PATs are static. To onboard a different tenant,
you must log in to that specific environment to
generate a new token.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

👀

🟠i…

When redirected to the Microsoft login screen,
do not immediately enter your email.

🧭

➡️n…

Settings → Data Sources & Integrations → +
Add New

User Settings → Personal access tokens → +
New Token

12.1.1.2.1. Azure DevOps onboarding system
architecture 📝 
🟢note

Microsoft has announced the deprecation of
legacy OAuth methods by 2026, making Entra
ID the required standard for long-term support.

12.1.1.3. Bitbucket Cloud 

☑️

🟣pr…

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
In Bitbucket, grant the user performing the
Cortex application authorization the following
permissions. The level of access required
depends on the modules you intend to use:
For code...

📝

🟢note

Ensure that you receive the Instance
Successfully Created message on this step,
indicating successful instance creation.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.1.4. Bitbucket Data Center 

☑️

🟣pr…

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
In Bitbucket, grant the user performing the
Cortex application authorization the following
permissions:
Administrator permissions for projects
Administrator permissions for...

📝

🟢note

By default, the permissions of the access token
are set according to your current access level.
It is essential to define two levels of
permissions, Project and Repository
permissions. The Repository permissions inherit
from Project permissions, requiring Repository
permissions to match or exceed Project
permissions
Providing read and write permissions...

📝

🟢note

For additional security, it is recommended to
set an expiry automatically. The expiry date of
a token cannot be changed after it is created.
You can see the expiry dates for all your tokens
on Profile picture → Manage account →
Personal access tokens.

📝

🟢note

For more information about the Transporter,
including setup instructions, refer to
Transporter over Broker VM.

📝

🟢note

Ensure that you receive the Instance
Successfully Created message on this step,
indicating successful instance creation.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

👀

🟠i…

Always refer to the Bitbucket documentation
for information relating to creating a PAT.

12.1.1.5. GitHub Cloud 

☑️

🟣pr…

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
In GitHub, grant the user performing the Cortex
application authorization the following
permissions:
Organization Owner: Only an Organization
Owner can directly authorize and install...

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

🧭

➡️n…

Verify integration: On Data Sources, select
Code Providers → GitHub SaaS → View more
and confirm that the status of your integrated
GitHub instance is 'Connected

12.1.1.6. GitHub Enterprise (On-Prem) 

☑️

🟣pr…

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
In GitHub, you must have Organization Owner
permissions to install the Cortex application.
Users with only repository-level admin
permissions cannot complete the installation
unless the...

📝

🟢note

The domain is the hostname associated with
your GitHub Enterprise (On-Prem) instance.

📝

🟢note

For more information about the Transporter,
including setup instructions, refer to
Transporter over Broker VM.

📝

🟢note

Ensure that you receive the Instance
Successfully Created message on this step,
indicating successful instance creation.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.1.7. GitLab SaaS 

☑️

🟣pr…

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
In Gitlab, the following permissions are required
to integrate the application:
Maintainer (Project-level). Grants sufficient
permissions to configure external integrations,
manage...

📝

🟢note

A repository can only be integrated with a
single instance. The first instance that connects
with the repository will be the one that the
repository is assigned to. This means that if
multiple integrations attempt to connect to the
same repository, only the first integration to
establish the connection will be associated with
that repository.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.1.8. GitLab Self Managed (On-Prem) 

☑️

🟣pr…

Authorize the user integrating Cortex Cloud
Application Security with your GitLab Self
Managed (On-Prem) instances with the
following permissions:
Maintainer permissions. Grants sufficient
permissions to configure external integrations,
manage repository access, and adjust CI/CD
settings
api: Grants full read and write access to the
API, including...

📝

🟢note

The domain is the hostname associated with
your GitLab Self Managed (On-Prem) instance.

📝

🟢note

For more information about the Transporter,
including setup instructions, refer to
Transporter over Broker VM.

📝

🟢note

Ensure that you receive the Instance
Successfully Created message on this step,
indicating successful instance creation.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.2. Onboard CI/CD systems 

12.1.2.1. CircleCI for CI/CD pipeline scans 

☑️

🟣pr…

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
CircleCI user requirements:
Permissions: To enable Cortex Cloud visibility
for all CircleCI projects, a version control
system (VCS) user with integration permissions
must be...

📝

🟢note

This integration utilizes a Personal Access
Token (PAT) for authentication
CircleCI onboarding offers both code and
CI/CD scanning. A single integrated instance
supports either code or CI scanning, but not
both. If you require both code and CI scanning
for your CircleCi environment, you must create
two separate integrations, selecting the
appropriate...

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.2.2. Jenkins for CI/CD pipeline scans 

☑️

🟣pr…

Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
In Jenkins:
To install and configure the Cortex plugin in
Jenkins, you must be a Jenkins Administrator
with Overall/Administer permissions
Ensure the build server allows outbound HTTPS
(Port 443...

📝

🟢note

Jenkins onboarding offers both code and CI/CD
scanning. A single integrated instance supports
either code or CI scanning, but not both. If you
require both code and CI scanning for your
Jenkins servers, you must create two separate
integrations, selecting the appropriate scanning
type for each. To onboard Jenkins for code
scans, refer to Jenkins for...

📝

🟢note

The integration is added on the console but
integration is pending, and will only be
completed after completing step 5 below. You
can view the pending integration on the Jenkins
Instances page: Select Data Sources → Jenkins
→ View Details. The type of integration is
Pipeline Risks

📝

🟢note

Always refer to the official Jenkins
documentation when installing plugins on
Jenkins servers.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.3. Integrate CI tools 

🧭

➡️n…

Settings → Data Sources & Integrations →
hover over a CI tool → View Details

Delete an instance: Right-click on an instance
of the CI tool → Delete instance → Delete

12.1.3.1. AWS CodeBuild 

☑️

🟣pr…

Before you begin:
User permissions: Ensure the user performing
the integration has permissions to edit pipeline
configurations (such as YAML files) and
manage secrets/credentials within the CI
platform to store the Cortex Cloud API key
securely

📝

🟢note

Do not change the names of the environment
variables provided by Cortex Cloud. They are
required for proper integration and
functionality.

📝

🟢note

The code is only a reference. Replace the
placeholder values with your build-specific
values.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.3.2. CircleCI for code scans 

☑️

🟣pr…

Before you begin:
User permissions: Ensure the user performing
the integration has permissions to edit pipeline
configurations (such as YAML files) and
manage secrets/credentials within the CI
platform to store the Cortex Cloud API key
securely

📝

🟢note

CircleCI onboarding offers both code and
CI/CD scanning. A single integrated instance
supports either code or CI scanning, but not
both. If you require both code and CI scanning
for your CircleCi environment, you must create
two separate integrations, selecting the
appropriate scanning type for each. To onboard
CircleCI for CI/CD scans, refer to...

📝

🟢note

Do not change the names of the environment
variables provided by Cortex Cloud. They are
required for proper integration and
functionality.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

👀

🟠i…

The cortex-secrets naming convention for the
context is mandatory to ensure functionality
and must not be changed.

12.1.3.3. Connect Cortex CLI 

☑️

🟣pr…

System requirements:
macOS (Intel Core i7, such as Sequoia): To
ensure all functionalities work correctly, you
must install the vectorscan dependency via
Homebrew, using this command: brew install
vectorscan

RHEL 8.10 and Red Hat UBI9. The following
prerequisites must be met:
Install patchelf
Install zstd
Ubuntu 20 requires the prefetch...

📝

🟢note

The Cortex CLI requires an API key with the
Standard security level.

📝

🟢note

On macOS arm 64 architecture you must
unpack the downloaded file to retrieve the
executable.

📝

🟢note

To add an additional CLI instance, navigate to
Settings → Data Sources → select the menu
for your connected CLI instance → + New
Instance, and repeat the onboarding steps.

📝

🟢note

For more information about CLI usage for CWP,
refer to Cortex CLI for Cloud Workload
Protection
For more information about CLI usage for API
Security, refer to Cortex CLI for API Security
For more information about CLI usage for
Cortex Cloud Application Security, refer to
Cortex CLI usage for Cortex Cloud Application
Security

12.1.3.4. GitHub Actions 

☑️

🟣pr…

Before you begin:
User permissions: Ensure the user performing
the integration has permissions to edit pipeline
configurations (such as YAML files) and
manage secrets/credentials within the CI
platform to store the Cortex Cloud API key
securely

📝

🟢note

Do not change the names of the environment
variables provided by Cortex Cloud. They are
required for proper integration and
functionality.

📝

🟢note

The code is only a reference. Replace the
placeholder values with your build-specific
values.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.3.5. Jenkins for code scans 

☑️

🟣pr…

Grant Administrator permissions to the user
integrating Cortex Cloud Application Security
with Jenkins
Create an egress path to establish the
designated route for outbound data
transmission from Cortex Cloud to third party
services. For more information about
configuring egress paths, refer to Egress
configurationsEgress configurations

☑️

🟣pr…

For Cortex Cloud Application Security CI tools,
you must store secrets in Jenkins Credentials
for use in your Jenkins pipelines using either of
these methods:
Plain text storage: Store secrets directly as
plain text in Jenkins Credentials. Access them in
your pipeline using the credentials function,
which retrieves the secret directly as plain...

📝

🟢note

Jenkins onboarding offers both code and CI/CD
scanning. A single integrated instance supports
either code or CI scanning, but not both. If you
require both code and CI scanning for your
Jenkins servers, you must create two separate
integrations, selecting the appropriate scanning
type for each. To onboard Jenkins for CI/CD
scans, refer to Jenkins for...

📝

🟢note

This step is only required for new pipelines
For private repositories, ensure the necessary
credentials are configured in Jenkins Credentials

📝

🟢note

This ensures your build runs within a Docker
environment. If a node without Docker is used,
the build will fail.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.3.6. Terraform Cloud (Run Tasks) 

☑️

🟣pr…

Before you begin:
User permissions: Ensure the user performing
the integration has permissions to edit pipeline
configurations (such as YAML files) and
manage secrets/credentials within the CI
platform to store the Cortex Cloud API key
securely
Procure a Terraform cloud license that is either
a trial license or a TF Cloud license at the
TEAM &...

📝

🟢note

Skip this step if you plan on using an existing
token.

📝

🟢note

Cortex Cloud Application Security performs a
scan of Terraform templates on selected
workspaces based on the Run Stage.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.3.7. Terraform Enterprise (Run Tasks) 

☑️

🟣pr…

Before you begin:
Ensure access to a Terraform Enterprise
console to enable you to provide a user or
team token that authorizes Cortex Cloud
Application Security to access workspaces and
helps regulate run configurations
Terraform Enterprise version compatibility:
Ensure Run Tasks for workspaces on is
compatible with version 1.1.9 and above
Terraform...

📝

🟢note

For container image vulnerabilities, Cortex
Cloud Application Security performs 'Image
Referencer' scans within Terraform Enterprise
(Run Tasks), as full SCA scans are not currently
supported.

📝

🟢note

Skip this step if you plan on using an existing
token.

📝

🟢note

Cortex Cloud Application Security performs a
scan of Terraform templates on selected
workspaces based on the Run Stage.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

☑️ 12.1.4. CLI pipeline code snippets 

🟣pr…

User permissions: Ensure the user performing
the integration has permissions to edit pipeline
configurations (such as YAML files).

12.1.5. Onboard private package registries 

12.1.5.1. JFrog Artifactory 

12.1.5.2. Onboard JFrog Artifactory 

☑️

🟣pr…

Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
JFrog permissions:
The permissions associated with the user
configured during the onboarding process
determine the scope of scan results. Only
repositories and artifacts the user can access
are...

📝

🟢note

To authorize the scanner to connect through
your firewall, select Enable access by IPs, and
copy the displayed source IPs to your
organization's allowlist.

📝

🟢note

For Maven:

Select Mirror Registry if this repository mirrors
an external repository
Use the Mirror Of value to define the
duplication scope:
* mirrors all requests
Type a request [value]: Mirrors only specific
requests (such as central).
For Maven:

Select Mirror Registry if this repository mirrors
an external repository
Use the Mirror Of...

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.6. Ingest third-party data sources 

📝

🟢note

Only onboarded and scanned repositories can
be mapped.

12.1.6.1. Semgrep 

☑️

🟣pr…

Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
Ensure that you have a connected version
control system (VCS) and repositories
Semgrep: A valid Semgrep API token
Note
To create a Semgrep API token, in Semgrep,
navigate to Settings → Tokens →...

📝

🟢note

Mapping establishes relationships between
Semgrep projects and Cortex Cloud code
repositories, simplifying access management
and enabling risk analysis at the repository
level, including displaying findings on the tenant
Only mapped projects are ingested

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

👀

🟠i…

Deletion is irreversible. To restore Semgrep
findings after deletion, reconfigure the
integration and wait for the next polling cycle
to re-ingest findings from Semgrep.

12.1.6.1.1. Semgrep Software Composition
Analysis (SCA) data ingestion 🧭 
➡️n…

Application Security → Issues → Vulnerabilities

12.1.6.1.2. Semgrep Static Application Security
Testing (SAST) data ingestion 🧭 
➡️n…

Application Security → Issues → Code
Weaknesses

12.1.6.2. Snyk 

☑️

🟣pr…

Permissions: The following user permissions are
required:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
Ensure that you have a connected version
control system (VCS) and repositories
Snyk permissions and requirements:
Permissions: The...

📝

🟢note

Select Test Connection to verify that Cortex
Cloud can connect to your Snyk organization.

📝

🟢note

SCA requires Snyk Open Source or Snyk
Container projects configured in the
organization
SAST requires Snyk Code enabled and projects
configured in the organization

📝

🟢note

Mapping establishes relationships between
Snyk applications and Cortex Cloud code
repositories, simplifying access management
and enabling risk analysis at the repository
level, including displaying findings on the tenant
Only mapped applications are ingested

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

👀

🟠i…

Snyk Free tier is not supported. A paid Snyk
plan (Team, Business, or Enterprise) is required
to ingest findings from Snyk.

12.1.6.2.1. Snyk Software Composition Analysis
(SCA) ingestion 🧭 
➡️n…

Modules → Application Security →
Vulnerabilities (under Issues

12.1.6.2.2. Snyk Static Application Security
Testing (SAST) Findings 

12.1.6.3. SonarQube 

☑️

🟣pr…

Permissions: Ensure you have System Admin,
AppSec Admin or GRBAC permissions. For more
information on AppSec Admin permissions,
refer to Code Security user roles and
permissions
Ensure that you have a connected version
control system (VCS) and repositories
Generate and copy a SonarQube API token.
Ensure to assign Web API scope to the API
token. Refer...

📝

🟢note

Mapping establishes relationships between
SonarQube Applications and Cortex Cloud
Application Security code repositories,
simplifying access management and enabling
risk analysis at the repository level, including
displaying findings on the tenant
Only mapped projects will be ingested

📝

🟢note

Verify that the Connector Created Successfully
message is displayed on the page.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.6.4. Veracode 

☑️

🟣pr…

Permissions: The following user permissions are
required:
Cortex Cloud: Instance Admin, AppSec Admin
or GRBAC permissions. For more information on
AppSec Admin permissions, refer to Code
Security user roles and permissions
Veracode: At minimum, Reviewer permissions
are required
Permissions: The following user permissions are
required:
Cortex...

📝

🟢note

Only mapped applications will be ingested.

📝

🟢note

This is the recommended option to ensure
complete coverage and successful operation
of all features.

📝

🟢note

Mapping establishes relationships between
Veracode projects and Cortex Cloud
Application Security code repositories,
simplifying access management and enabling
risk analysis at the repository level, including
displaying findings on the tenant
Only mapped projects will be ingested

📝

🟢note

Verify that the Connector Created Successfully
message is displayed on the page.

📝

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.6.5. Generic 3rd Party AppSec Collector 

👀

🟠i…

File uploads are limited to a maximum size of 10
MB.

12.1.6.5.1. Onboard the 3rd Party AppSec
Collector 

☑️

🟣pr…

Permissions: The following Cortex Cloud user
roles or RBAC permissions are required:
User roles: CortexCortex Cloud Cloud Instance
Admin, AppSec Admin
RBAC: View/Edit permissions for Data Sources
configurations are required when not using a
dedicated user role
For more information about user permissions
and groups, refer to Assign user roles...

📝

🟢note

We recommend using the tool.driver.name from
the SARIF file.

📝

🟢note

This is the newly created generic collector API
URL endpoint.

📝

🟢note

Only the validity of the format is tested. No
findings will be generated from the test file.

⛔

🔴w…

The API key ID and API secret cannot be
retrieved once the wizard is closed.

📝 12.1.7. Manage data source integrations 

🟢note

Requisite RBAC permissions are required for
edit and delete operations.

12.1.8. Transporter over Broker VM 

☑️

🟣pr…

Ensure you have the necessary permissions and
have already set up your Broker VM
Confirm that your Broker is v 28 or above
Whitelist IP addresses to enable access to
Cortex Cloud resources. The IP addresses for
the Transporter are in the Broker VM Resources
section of the Enable access to required PANW
resources document
Open port 4052 (inbound), which...

📝

🟢note

The broker enables multiple connections,
allowing a single Transporter applet to connect
to various VCS instances
Multiple Transporter applets cannot be created
on a single Broker VM. Each Broker VM is
limited to a single Transporter applet instance

⛔

🔴w…

The Transporter applet is not supported for
FedRAMP customers.

12.1.8.1. Set up a Transporter applet on Broker
VM 

☑️

🟣pr…

Permissions: To configure and manage
Transporter applet settings, you must have
permissions to manage Broker Service
configurations (such as an Instance
Administrator)

📝

🟢note

You cannot add a Transporter applet through
Clusters.

🧭

➡️n…

your AppSec Transporter under the App
column → Configure → Add Connection (on
the redirected AppSec Transporter setup page

12.1.8.2. Set up a Transporter on your VCS 

☑️

🟣pr…

Ensure you have user permissions to onboard
data sources.
Note
The dedicated AppSec Admin role includes the
required permissions.
Ensure you have user permissions to onboard
data sources.
Before you begin you must setup a Transporter
applet on your Broker VM. Refer to Set up a
Transporter applet on Broker VM for more
information

📝

🟢note

GitLab Enterprise webhook internal IP
restriction: GitLab Enterprise's security policies
prevent webhook subscriptions to internal IP
addresses (such as broker addresses). These
subscriptions can only be enabled through
specific instance configuration. For more
information, refer to
https://docs.gitlab.com/security/webhooks/.

🧭

➡️n…

Settings → Data Sources & Integrations → +
Add New → , and search for your VCS on-
premises data source

12.2. Cortex Cloud Application Security
dashboard 🧭 
➡️n…

Modules → Application Security → AppSec
Dashboard

12.3. Application Security Posture Management
(ASPM) 

12.3.1. ASPM use cases 

12.3.2. ASPM key features 

🧭 12.3.3. ASPM user roles and permissions 

➡️n…

You can view AppSec Admin permissions in the
tenant by navigating to Settings →
Configurations → Roles → AppSec Admin

12.3.4. Code to Cloud 

🧭

➡️n…

Full Code to Cloud (C2C) coverage is achieved
when Cortex Cloud can resolve the following
chain: Repository → Pipeline → Image →
(optional Registry) → Runtime resources
(including VMs, VM images, and IaC-defined
infrastructure

12.3.4.1. Supported integrations 

🧭 12.3.4.2. Code to Cloud visibility 

➡️n…

Topology graph: Located in the Business
Applications side card, this tab visualizes the
entire path to production (Code → Build →
Deploy → Run), allowing you

📝 12.3.4.3. Code to Cloud troubleshooting 

🟢note

Empty state messages and troubleshooting
guidance for missing lineage are only visible to
users with an active license.

📝 12.3.5.1. ASPM Command Center workflow  12.3.5. ASPM Command Center 

🟢note

Breakdown by type: Issues are further broken
down by type (such as IAC or Secrets).

12.3.6. Applications 

12.3.6.1. Defining Business Applications 

12.3.6.1.1. Defining business applications by
Criteria 

12.3.6.1.2. Define applications by VCS criteria 

☑️

🟣pr…

Data source: Your Version Control System (e.g.,
GitHub, GitLab) must already be onboarded as
a Data Source
Permissions: You must have View/Edit
permissions for Access Management
SBAC: You can only create applications from
VCS entities (Organizations, Projects, or
Repositories) that are already included in your
SBAC Asset Groups

📝

🟢note

This workflow allows you to unify assets across
disparate providers (such as grouping a GitHub
repository and a Bitbucket repository into one
application) if they share naming conventions

12.3.6.1.3. Manage application criteria 

12.3.6.1.4. Define applications by cloud tag-
based criteria 

☑️

🟣pr…

Permissions: You must have View/Edit
permissions for Access Management, or a role
that includes these permissions.

📝

🟢note

You can only create applications based on the
entities from onboarded Cloud accounts listed
in the Cloud card.

📝

🟢note

Kubernetes (K8S) labels: K8S labels are
supported as tags for asset grouping only when
they originate from a supported cloud provider;
AWS, GCP, Azure, or OCI.

12.3.6.1.5. How to manually build an application 

☑️

🟣pr…

Before you begin, ensure you have connected
the necessary data sources. Refer to How to
onboard data sources for more
information.How to onboard data sources

📝

🟢note

These filters are represented by icons displayed
on the Code pane after selecting a VCS.

📝

🟢note

To edit application assets, click the Clear All
icon before clicking Finish. This clears all
application data, allowing you to restart the
application building process from the beginning.

🧭 12.3.6.2. Application management and visibility 

➡️n…

Inventory → All Assets → Business Applications
(under Application

12.3.6.3. Business application assets 

12.3.6.3.1. Business application expanded asset
details 📝 
🟢note

The topology graph is available only when all
application components (code, pipeline, build
and deploy), are configured.

🧭 12.3.6.3.2. Export business application data 

➡️n…

Inventory → All Assets → Business Applications

12.3.6.4. Scope user access to applications
(Application SBAC) 

🧭

➡️n…

Configure user scopes in Cortex Cloud by
navigating to Settings → Configurations →
Access Management

12.3.6.4.1. Enable SBAC in the Cortex Cloud

tenant 

☑️

🟣pr…

RBAC permissions: To configure user scopes
you must have Administrator or View/Edit RBAC
permissions for Access Management (under
Configurations).

📝

🟢note

Exclusions (roles not governed by SBAC):
Certain roles cannot have SBAC applied. For
these roles, access and permissions are
managed through Role-Based Access Control
(RBAC). You must manually ensure that these
roles have all necessary base permissions (for
example Edit/View permissions to assets),
because SBAC is bypassed and does not
impose its usual...

🧭

➡️n…

Settings → Configurations → General → Server
Settings → Enable Scope Based Access Control

12.3.6.4.2. Create an application-based Asset
Group 📝 
🟢note

You cannot create SBAC based on static
groups.

12.3.6.4.3. Scope user access to an application 

📝

🟢note

For more information about User Groups, refer
to User group management.

🧭

➡️n…

Settings → Configurations → User Groups
(under Access Management

Right-click on a group in the table → Edit
Group → select the Scope tab

Assets → Select asset groups → select an
Asset Group associated with applications

Settings → Configurations → Users (under
Access Management

the chevron icon (>) in the Assets field →
Select Asset groups → select the user group
scoped to the application (see above

12.3.6.4.4. Create application-scoped policies 

📝

🟢note

Application-scoped policies apply to both
code and CI/CD configuration policies.

🧭

➡️n…

Modules → Application Security → AppSec
Polices (under Policy Management) → Add
Policy

Add Filter → Business Application Names →
enter the required application name

12.3.7. Repositories as assets 

12.3.7.1. Explore repository assets 

12.3.7.2. In-depth repository asset information 

12.3.7.3. Manage Repository assets 

12.3.7.4. Export Software Bill of Materials
(SBOM) 

📝 12.3.7.5. Manage issues detected in repositories 

🟢note

You can also find the repository issues in the
general issue inventory table, and in the
dedicated inventory of issues for each scanner
type (see step 4 above for details).

🧭 12.3.8.1. Coverage in the user interface  12.3.8. Coverage 

➡️n…

Modules, → Application Security → AppSec
Coverage

12.3.9.1. Urgency metrics  12.3.9. Urgency 

12.3.10. Backlog baseline 

12.3.10.1. Backlog use cases 

12.3.10.2. Issue/Finding classification by scanner 

📝

🟢note

In some cases/vendors, this is not accurate as
findings are deleted every time new findings are
uploaded. In such cases, the feature may not be
accurate or supported
For SAST, the vendor does not support policy in
pull requests

📝

🟢note

Scanner updates and new detections: When a
security scanner is updated to support new
languages, detection rules, or capabilities, any
issues discovered by these new features for
existing code are classified as part of the
backlog.

12.3.10.3. Using Backlog 

12.3.11. Service Lead Agreements (SLA) 

12.3.11.1. Configure and monitor Cortex Cloud
Application Security SLAs 🧭 
➡️n…

Settings → Configurations → Application
Configuration (under Application Security

12.3.12. Compliance for Cortex Cloud
Application Security 

12.3.12.1. Infrastructure-as-Code (IaC)

compliance 

12.3.12.2. Manage IaC compliance 

☑️

🟣pr…

Add these properties to the IaC Rules table
through the Table Settings Menu, as they are
not exposed by default.

🧭

➡️n…

Modules → Application Security → AppSec
Polices → Add Policy

12.3.12.3. CI/CD Compliance 

🧭

➡️n…

To access Compliance, select Posture
Management → Compliance → Reports or
Assessment

12.3.12.3.1. Create CI/CD compliance reports 

📝

🟢note

The CI/CD module supports GitHub and GitLab
provider types.

📝

🟢note

For more information about about Asset
Groups, refer to Asset Groups.

📝

🟢note

CIS GitLab Benchmark, CIS GitHub Benchmark,
and the OWASP Top 10 CI/CD Risks standards
are supported.

📝

🟢note

For more information about assessment
profiles, refer to Use an assessment profile to
run compliance checks on your assets.

🧭

➡️n…

Posture Management → Compliance → Reports

12.3.13. Application Security Policies 

12.3.13.1. User roles and permissions 

📝 12.3.13.2. Policies inventory 

🟢note

To view all out-of-the-box (OOTB) policies,
filter by Policy Owner = System.

12.3.13.3.1. Manage AI-recommended guardrails  12.3.13.3. AI-recommended guardrails 

📝

🟢note

Recommended guardrails are generated based
exclusively on findings from the Cortex Cloud
Application Security Software Composition
Analysis (SCA) scanner. They do not support
SCA data ingested from third-party scanners,
nor do they currently support other finding
categories, such as Secrets or Infrastructure as
Code (IaC).

📝

🟢note

The total amount of available recommended
guardrails is displayed on the AI
Recommendations button.

📝

🟢note

The AI-Recommended tag is only assigned to
recommendations applied as-is, that is, without
modifications. Edited recommendations
generate policies without this label.

12.3.13.4. Create Cortex Cloud Application
Security policies 

📝

🟢note

When an issue is created, only the first policy ID
that triggered the issue is associated with it.

12.3.13.4.1. Create code security policies 

📝

🟢note

For a detailed list of all available attributes and
their values by scanner type, refer to Cortex
Cloud Application Security code policy
Condition attributes .

📝

🟢note

SBAC scope-based limitations do not apply to
Asset Types

📝

🟢note

When selecting Asset Groups, the code policy
is evaluated only on the application and
repository assets included in the group.

📝

🟢note

For each application that matches the criteria,
the policy is evaluated only on the repositories
associated with that application.

📝

🟢note

If you select only Code Weaknesses as the
Finding Type in the condition step (indicating a
third-party scanner), the PR and CI triggers are
disabled, leaving only Periodic Scan enabled.

12.3.13.4.1.1. Cortex Cloud Application Security
code policy Condition attributes 

🧭 12.3.13.4.2. Create IaC Drift Detection policies 

➡️n…

Modules → Application Security → AppSec
Policies → Add Policies

12.3.13.4.3. Manage Cortex Cloud Application
Security policies 

📝

🟢note

You cannot edit out-of-the-box (OOTB)
policies.

📝

🟢note

The duplicated policy will include the word
"clone" in its name and must be renamed.

12.3.14. Application Security Rules 

📝

🟢note

Out-of-the-box rules cannot be modified
directly. However, you can create a custom rule
by cloning the existing one. This allows you to
make changes to the original rule according to
your requirements. Refer to Manage Cortex
Cloud Application Security custom rules for
more information.

12.3.14.1. Roles and permissions 

🧭 12.3.14.2. Rules inventory 

➡️n…

To filter rules relating to Secrets, select filter
icon → Scanner (from the Select field) →
Secrets (from the Value field

12.3.14.3. Create custom Cortex Cloud

Application Security rules 

📝

🟢note

See Configure YAML file properties below for
more details.

📝

🟢note

Scanning/testing behavior is not supported.

12.3.14.4. Manage Cortex Cloud Application
Security custom rules 

12.3.14.5. Configure YAML file properties 

12.3.15. Manage code weaknesses 

📝

🟢note

You can customize policies to define which
findings are categorized as issues.

🧭

➡️n…

Dedicated issues view: To access SAST code
violation issues, under Modules, select
Application Security → Issues → Code
Weaknesses

Issues → AppSec Issues (under All Issues) →
SAST Scanner (as the Detection Method value

12.3.15.1. Code weaknesses issue inventory 

12.3.15.2. Detailed code weakness issue

information 📝 
🟢note

Different issue types include different
properties; therefore, not all properties are
available for every issue.

📝 12.3.15.3. Code weakness findings 

🟢note

Findings on the Cortex platform are not
intended for direct action; but rather represent
data collected by the platform. They must be
promoted to issues to enable mitigation efforts
to secure your codebase.

12.4. CI/CD Security 

12.4.1. CI/CD Security user roles and permissions 

12.4.2. CI/CD Assets 

12.4.2.1. CI/CD Instances as assets 

🧭 12.4.2.1.1. Explore CI/CD Instance assets 

➡️n…

To access CI/CD pipeline instances, under
Inventory, select All Assets → Code →
Category → CI/CD Instances

12.4.2.1.2. In-depth CI/CD pipeline instance asset
information 

12.4.2.1.3. Manage CI/CD pipeline instances 

12.4.2.2. CI/CD Pipelines as assets 

🧭 12.4.2.2.1. Explore CI/CD Pipeline assets 

➡️n…

To access CI/CD pipelines assets, under
Inventory, select All Assets → Code → CI/CD
Pipelines

12.4.2.2.2. In-depth CI/CD pipeline asset
information 📝 
🟢note

The Highlights section and other asset
properties only display attributes when their
corresponding indicators are present. For
example, if an asset is not deployed, its
deployment-related attributes will not show up;
similarly, if there are no detected issues, those
highlights or properties will not appear.

12.4.2.2.3. Manage CI/CD pipeline assets 

12.4.2.3. Version Control System (VCS)

Organizations as assets 

🧭 12.4.2.3.1. Explore VCS Organization assets 

➡️n…

To access VCS Organization assets, under
Inventory, select All Assets → Code → VCS
Organizations

12.4.2.3.2. In-depth VCS Organization asset
information 📝 
🟢note

The Highlights section and other asset
properties only display attributes when their
corresponding indicators are present. For
example, if an asset is not deployed, its
deployment-related attributes will not show up;
similarly, if there are no detected issues, those
highlights or properties will not appear.

12.4.2.3.3. Manage VCS organization assets 

12.4.2.4. VCS Collaborators-as-assets 

12.4.2.4.1. In-depth Collaborator asset
information 

12.4.2.4.2. Manage Collaborator assets 

12.4.3. Supply Chain Inventories 

📝

🟢note

Although attributes are identical across
inventories, their values for the same tool can
differ, most commonly in Risk Factors and Type.
This is because the inventory reflects your live
environment, which may include different
versions or configurations than the catalog—for
example, a package may not have been
upgraded or may be deployed differently.

🧭

➡️n…

In the inventory table, right-click on a tool →
Change Status → select a status

12.4.3.1. Supply Chain Tools 

🧭 12.4.3.1.1. Supply Chain Tools 

➡️n…

To access Supply Chain Tools, select Modules
→ Application Security → Supply Chain Tools
(under 3rd Party Tools

12.4.3.1.2. Expanded Supply Chain tool
information 📝 
🟢note

AppSec Admin user permissions are required to
perform these actions.

🧭 12.4.3.2. Supply Chain Catalog 

➡️n…

To access the Supply Chain Catalog, select
Modules → Application Security → Supply
Chain Catalog (under 3rd party tools

12.4.4. CI/CD Risks 

📝

🟢note

Cortex Cloud Application Security CI/CD
pipeline scans create a comprehensive
inventory of all CI/CD pipelines in your
environment. For more information refer to
CI/CD Pipelines as assets.

🧭 12.4.4.1. CI/CD pipeline issues 

➡️n…

To access CI/CD pipeline risks issues, under
Modules, select Application Security → Issues
→ CI/CD Risks

📝 12.4.4.2. Expanded CI/CD risks issue information 

🟢note

Different issue types include different
properties; therefore, not all properties are
available for every issue.

12.4.4.3. VCS and CI/CD pipeline risk findings 

12.4.5. CI/CD Rules 

📝

🟢note

Out-of-the-box rules cannot be modified
Custom CI/CD rules are not supported

12.4.5.1. CI/CD rules roles and permissions 

12.4.5.2. CI/CD rules inventory 

12.4.6. CI/CD Policies 

💡

🔵tip

For Cortex Cloud Code policies, refer to
Application Security Policies.

12.4.6.1. CI/CD policies user roles and
permissions 

📝 12.4.6.2. CI/CD policies inventory 

🟢note

To view all out-of-the-box (OOTB) policies,
filter by Policy Owner = System.

12.4.6.3. Create custom CI/CD policies 

📝

🟢note

PR Scan and CI Scan triggers are automatically
disabled and unchecked. They can only be
enabled if other scan types (that is non-CI/CD
risk scans such as Secrets) are also selected,
and will only run on non-CI/CD risks types of
scans

📝

🟢note

You can combine multiple conditions to create
complex rules for when the policy should apply.
Create conditions that apply to a CI/CD policy
which detects high severity CI/CD risks on
GitHub: Select Add Filter → Finding Type →
CI/CD Risks → AND → Provider → [VCS/CI/CD
system] → AND → Severity: → High.

📝

🟢note

SBAC scope limitations do not apply to
Matching Criteria
For CI/CD policies, you can filter Matching
Criteria by VCS Organization Name, CI/CD
Pipeline [Name/ID], CI/CD Instance [Name/ID] or
Collaborator [Name/Email/MFA Enabled/Last
Observed]

12.4.6.4. Manage CI/CD policies 

📝

🟢note

You cannot edit out-of-the-box (OOTB)
policies.

📝

🟢note

The duplicated policy will include the word
"clone" in its name and must be renamed.

12.5. Code Security 

12.5.1. Code Security user roles and permissions 

12.5.2. Code Security assets 

12.5.3. Software packages as assets 

12.5.3.1. Explore software package assets 

12.5.3.2. In-depth software package asset
information 

12.5.4. Infrastructure-as-Code (IaC) resources
as assets 

🧭 12.5.4.1. Explore IaC assets 

➡️n…

To access IaC assets, under Inventory, select All
Assets → Code → IaC Resources

12.5.4.2. In-depth IaC resource asset information 

12.5.5. Code Security scanners 

12.5.6. Secrets scanners 

🧭 12.5.6.1. Secrets issues 

➡️n…

Under Modules, select Application Security →
Issues → Secrets

Inventory → All Assets → Code: On the Secrets
tab under Repositories

Inventory → All Assets → Code

Inventory → All Assets → Application → select
an option from the Application menu → select
an item from the inventory → Secrets

12.5.6.2. Secrets issues inventory 

📝 12.5.6.3. Detailed Secrets issue information 

🟢note

Different issue types include different
properties; therefore, not all properties are
available for every issue.

12.5.6.4. Secrets findings 

📝

🟢note

Findings are informational and, as such, are not
directly mitigable. Remediation is performed on
issues derived from findings.

🧭

➡️n…

The Cortex Cloud Application Security Secrets
Findings table is a pre-filtered view of the
comprehensive Findings inventory (located at
Cases & Issues → Issues → Findings Table tab

To access Secrets findings, select Application
Security → Issues → Secrets → click the
Findings tab

12.5.6.5. Manage Secrets issues 

12.5.7. Infrastructure as Code (IaC)

misconfiguration scanner 

12.5.7.1. Supported frameworks and languages 

12.5.7.2. IaC misconfiguration issues 

💡

🔵tip

You can also view IaC misconfiguration issues in
dedicated tabs under other sections when
available:
In code Asset inventories, navigate to Inventory
→ All Assets → Code:
On the Configuration tab under Repository
assets. Refer to In-depth repository asset
information for more information
Under the All Code asset inventory: Select an
asset from...

🧭

➡️n…

To access IaC misconfiguration issues, under
Modules, select Application Security → Issues
→ IaC misconfiguration

12.5.7.3. IaC misconfiguration issues inventory 

12.5.7.4. Detailed IaC misconfiguration issue
information 📝 
🟢note

Different issue types include different
properties; therefore, not all properties are
available for every issue.

12.5.7.5. IaC misconfiguration findings 

📝

🟢note

Findings are informational and, as such, are not
directly mitigable. Remediation is performed on
issues derived from findings.

🧭

➡️n…

To access IaC misconfiguration findings, under
Modules, select Application Security → Issues
→ IaC Misconfigurations → click the Findings
tab

12.5.7.6. Manage IaC misconfiguration issues 

12.5.8. IaC Drift Detection scans 

🧭 12.5.8.1. IaC Drift Detection issues 

➡️n…

Modules → Application Security → IaC Drifts

12.5.8.2. IaC Drift Detection issue inventory 

12.5.8.3. Detailed IaC drift detection issue

information 

🧭 12.5.8.4. IaC Drift Detection findings 

➡️n…

Modules → Application Security → IaC Drifts
(under Issues) → click the Findings tab

12.5.9. Software Composition Analysis (SCA)
scanners



12.5.9.1. Supported Software Composition
Analysis (SCA) frameworks and languages 

12.5.9.2. Software Composition Analysis (SCA)
vulnerability issues 

💡

🔵tip

In Asset inventories, navigate to Inventory → All
Assets → Code:
On the Vulnerabilities tab under Repositories.
Refer to In-depth repository asset information
for more information
Under the All Code inventory: Select an asset
from the table → Vulnerabilities
In Asset inventories, navigate to Inventory → All
Assets → Code:
On the...

🧭

➡️n…

To access CVE vulnerability issues, under
Modules, select Application Security → Issues
→ Vulnerabilities

12.5.9.2.1. Vulnerability issues inventory 

📝 12.5.9.2.2. Detailed vulnerability issue information 

🟢note

Different issue types include different
properties; therefore, not all properties are
available for every issue.

12.5.9.2.3. CVE vulnerabilities findings 

📝

🟢note

Findings are informational and, as such, are not
directly mitigable. Remediation is performed on
issues derived from findings.

🧭

➡️n…

The Cortex Cloud Application Security
Vulnerabilities Findings table is a pre-filtered
view of the comprehensive Findings inventory
(located at Cases & Issues → Issues → Findings
Table tab

To access CVE findings, under Modules, select
Application Security → Issues → Vulnerabilities
→ click the Findings tab

12.5.9.2.4. Manage SCA CVE vulnerability issues 

12.5.9.3. License miscompliance issues 

💡

🔵tip

You can also view license miscompliance issues
in dedicated tabs under other sections when
available:
In Application asset inventories: navigate to
Inventory → All Assets → Application → select
an option from the Application menu → select
an item from the inventory → Package Integrity
In Asset inventories, navigate to Inventory → All...

🧭

➡️n…

To access license miscompliance issues, under
Modules, select Application Security → Issues
→ Licenses

12.5.9.3.1. License miscompliance issues inventory 

12.5.9.3.2. Expanded License miscompliance
issues information 

🧭 12.5.9.3.3. License miscompliance findings 

➡️n…

To access license miscompliance findings,
under Modules, select Application Security →
Issues → Licenses → click the Findings tab

12.5.9.3.4. Open-source software license
categories 

12.5.9.3.5. Manage license miscompliance issues 

12.5.9.4. Package Integrity 

📝 12.5.9.4.1. Package Integrity inventory 

🟢note

You can modify the severity level from the
severity field menu.

12.5.9.4.2. Expanded Package Integrity issues
inventory information 📝 
🟢note

Different issue types include different
properties; therefore, not all properties are
available for every issue.

📝 12.5.9.4.3. Package Integrity findings 

🟢note

Findings cannot be mitigated. They must be
promoted to issues to enable remediation
efforts to secure your software packages.

12.5.9.4.4. Manage Package Integrity issues 

12.5.10. Application Security scans management 

📝 12.5.10.1. Overview 

🟢note

While scans provide a comprehensive inventory
of all issues detected during a scan, dedicated
inventories are also maintained for specific
scan types for more granular management. For
more information, refer to Infrastructure as
Code (IaC) misconfiguration scanner, Secrets
scanners and Software Composition Analysis
(SCA) scanners.

📝 12.5.10.2. Branch periodic scans 

🟢note

The inventory table displays scan issues for
visibility only; remediation is not available here.
To resolve issues, navigate to the dedicated
issue type inventory, where you can manage
and remediate them.

📝 12.5.10.3. Pull Request scans 

🟢note

The inventory table displays scan issues for
visibility only; remediation is not available here.
To resolve issues, navigate to the dedicated
issue type inventory, where you can manage
and remediate them.

📝 12.5.10.4. CI scans 

🟢note

The inventory table displays scan issues for
visibility only; remediation is not available in
scan management. To resolve issues, navigate
to the dedicated issue type inventory, where
you can manage and remediate them.

12.5.10.5. Manage repository scan configurations 

12.5.10.6. Monitor data source instances health 

12.5.11. Application Security CLI 

☑️

🟣pr…

These requirements apply specifically to the
Application Security CLI.
For the Cortex CLI binary:
Ensure you have Node.js v22 installed on your
host machine before running any scans with the
Cortex CLI. This is crucial to prevent runtime
errors, as the CLI depends on Node.js for
executing JavaScript analysis
Note

To check your version of...

📝

🟢note

For detailed information about the Cortex
Cloud CLI, refer to Cortex CLI.

12.5.11.1. Connect Cortex CLI 

☑️

🟣pr…

System requirements:
macOS (Intel Core i7, such as Sequoia): To
ensure all functionalities work correctly, you
must install the vectorscan dependency via
Homebrew, using this command: brew install
vectorscan

RHEL 8.10 and Red Hat UBI9. The following
prerequisites must be met:
Install patchelf
Install zstd
Ubuntu 20 requires the prefetch...

📝

🟢note

The Cortex CLI requires an API key with the
Standard security level.

📝

🟢note

On macOS arm 64 architecture you must
unpack the downloaded file to retrieve the
executable.

📝

🟢note

To add an additional CLI instance, navigate to
Settings → Data Sources → select the menu
for your connected CLI instance → + New
Instance, and repeat the onboarding steps.

📝

🟢note

For more information about CLI usage for CWP,
refer to Cortex CLI for Cloud Workload
Protection

For more information about CLI usage for API
Security, refer to Cortex CLI for API Security
For more information about CLI usage for
Cortex Cloud Application Security, refer to
Cortex CLI usage for Cortex Cloud Application
Security

12.5.11.2. Self-service API keys for CLI scans 

12.5.11.3. Cortex CLI usage for Cortex Cloud
Application Security 

☑️ 12.5.11.4. CLI pipeline code snippets 

🟣pr…

User permissions: Ensure the user performing
the integration has permissions to edit pipeline
configurations (such as YAML files).

12.5.11.5. Cortex CLI Cortex Cloud Application
Security command line reference



📝

🟢note

The repo-id flag must not end with .config, .log
or .ini. -config is acceptable.
-- repo-id foo.config will be blocked
-- repo-id foo-config will pass

👀

🟠i…

The Cortex CLI Cortex Cloud Application
Security only supports single occurrences of
each flag. If the same flag is passed multiple
times, only the last provided value will be used.
For example, in the following command, only TF
CloudFormation will be the scanned framework.
./cortexcli --api-base-url <YOUR_API_URL> --
api-key <YOUR_API_KEY> --auth-id...

12.5.11.6. Cortex CLI common command line

reference guide 📝 
🟢note

For soft fails, a failed check matches the
defined severity threshold. If multiple soft fail
severities are specified, the highest severity acts
as the threshold for determining a soft fail.
However, a successful scan will always return
an exit code of 0, even if block-level findings
(which might trigger soft fails based on
severity) are present.

12.5.11.7. Git Hooks 

12.5.11.7.1. Cortex CLI pre-commit hooks 

☑️

🟣pr…

These common prerequisites are required for
all types of installation (both local and global)
of the Cortex CLI pre-commit hook.
Ensure you have a license for Cortex Cloud
Application Security
Install the Cortex Cloud CLI binary locally.
Refer to Connect Cortex CLI for information
about onboarding the CLIConnect Cortex CLI
Obtain Cortex Cloud API...

☑️

🟣pr…

For local installation: Install the pre-commit
framework version 3.2.0 or greater. Refer to
https://pre-commit.com/ for installation
instructions.

📝

🟢note

It is recommended you configure credentials
for the Cortex CLI using a configuration file.

📝

🟢note

The pre-commit framework is not required for
global mode.

12.5.11.7.1.1. Pre-commit hook usage 

12.5.11.7.2. Cortex CLI pre-receive hooks 

☑️

🟣pr…

Before you begin, ensure you have:
Administrator access to the VCS server and
console
A valid license for Cortex Cloud Application
Security
The Cortex Cloud CLI binary or Docker image
installed on the server (requires GLIBC (GNU C
library) version 2.35 or greater). Refer to
Connect Cortex CLI for information about
onboarding the CLIConnect Cortex...

📝

🟢note

Make sure to create the directory under the
home directory of the Linux user that runs the
Git hooks. This user is typically not the root user.

12.5.11.7.2.1. Pre-receive hook usage 

12.5.12. IDE 

📝

🟢note

Not all remediation options are available for all
findings or all type of scan category.

12.5.12.1. System requirements 

12.5.12.2. Visual Studio (VS) Code and VS Code

compatible IDEs 

☑️

🟣pr…

Before you begin (These prerequisites apply to
both VS Code and VS Code compatibles):
Permissions: CLI Read only permissions. Refer to
Cortex CLI for more information about
permissions
Environment setup
macOS and Windows: Install Python 3.9.x to
3.12.x
Install Node.js version 22 and above for SCA
scans (such as vulnerabilities scans)
Environment...

☑️

🟣pr…

You must insert your API key and API ID values
into the Settings before providing the tenant
URL.

📝

🟢note

Enforcement rules and CA certificates are not
applicable to the open-source project.

👀

🟠i…

When generating an API key, ensure you select
the Standard security level. IDE and CLI scans
will fail if the security level of the API key is set
to Advanced.

💡

🔵tip

You can access Extensions using the keyboard
shortcuts Ctrl+Shift+X (Windows) or
Cmd+Shift+X (macOS).

🧭

➡️n…

Settings → Configurations → API Keys (under
Integration) → click Copy API URL

Install through VS Code IDE Access the Activity
bar → Extensions → Cortex Cloud → Install

Access the Activity bar → Extensions → Cortex
Cloud → Install

12.5.12.2.1. How to use the Cortex Cloud

extension in VS Code 

📝

🟢note

Not all remediation options are available for all
issues.

📝

🟢note

The suppression is scoped to the file.

12.5.12.3. JetBrains 

☑️

🟣pr…

Permissions: CLI Read only permissions. Refer to
Cortex CLI for more information about
permissions
Environment setup
macOS and Windows: Install Python 3.9.x to
3.12.x
Install Node.js version 22 and above for SCA
scans (such as vulnerabilities scans)
Environment setup
macOS and Windows: Install Python 3.9.x to
3.12.x
Install Node.js version 22 and...

☑️

🟣pr…

You must insert your API key and API ID values
into the Settings before providing the tenant
URL.

📝

🟢note

The Cortex AppSec code security plugin
supports all JetBrains products.

👀

🟠i…

When generating an API key, ensure you select
the Standard security level. IDE and CLI scans
will fail if the security level of the API key is set
to Advanced.

🧭

➡️n…

Settings → Plugins → select the Marketplace
tab → search for Prisma Cloud → Install → OK

Settings → Tools → Cortex Cloud

12.5.12.3.1. How to use the JetBrains Cortex

Cloud extension 

📝

🟢note

After selecting Fix Available, the number of
issues displayed in the issue categories (such as
IaC) reflect the number of fixable issues for
that type.

📝

🟢note

The corresponding file simultaneously opens in
the Code editor, highlighting the issue within its
exact code context. See below for more
information.

📝

🟢note

Not all types of remediation are available for all
issue categories. For example, fixes are not
available for License issues.

📝

🟢note

Secrets and Licenses category issues are
typically mitigated by following the guidance in
the Documentation.

📝 12.5.13. Developer Suppressions 

🟢note

The check IDs and rule names are placeholders
for illustration purposes and do not correspond
to actual checks.

12.6. API documentation 

13. Cloud ASM 

13.1. What is Cloud ASM? 

13.2. Cloud ASM concepts 

13.2.1. Scanning 

13.2.1.1. Scanning cadences 

📝 13.2.1.2. Scanning ports and protocols 

🟢note

The following lists are not exhaustive. For
current and complete lists, contact your
customer success team.

13.2.1.3. Scanning activity 

13.2.2. Network mapping 

13.3. Enable Cloud ASM 

📝

🟢note

Discovered services will increase workload
consumption. ASM data will begin to appear in
your tenant and be included in your licensing
workload meter within 72 hours.

🧭

➡️n…

Settings → Configurations → Attack Surface →
Data Management

13.4. Attack Surface Management detections 

🧭 13.4.1. Attack surface rules 

➡️n…

Modules → Attack Surface → Policies →
Attack Surface Rules

📝 13.4.2. Externally inferred CVEs 

🟢note

An externally inferred CVE might impact your
service or asset, but additional investigation is
required to confirm that the CVE is actually
present.

🧭 13.5. Attack surface assets 

➡️n…

The internet-facing assets that were discovered
in an attack surface management (ASM) scan
and attributed to your organization are
available in the inventory on the Inventory →
Assets → All Assets → External Surface pages

🧭 13.6. Review your unmanaged cloud services 

➡️n…

Inventory → Assets → All Assets → External
Surface → Services

13.7. Review unmanaged cloud issues 

13.8. Attack Surface Management FedRAMP
support 

14. Network exposure detection 

14.1. What is Cloud Network Analyzer? 

📝 14.2. Internet exposure detection 

🟢note

Currently, CNA supports only an isolated
architecture with an AWS Gateway Load
Balancer (GWLB) within a single VPC. Other
more centralized topologies including one
security VPC that forwards traffic to other
workload VPCs are currently not supported.

14.3. Outbound exposure detection 

14.4. East-west exposure detection 

14.5. Investigate an internet exposure 

📝

🟢note

The Network tab is currently only available for
virtual machines.

📝

🟢note

The Network tab is only displayed when you
have access to the main asset and associated
ones, such as security groups, VPCs and
subnets. For more information on Scope-Based
Access Control (SBAC) for configuring granular
scoping, see Manage user scope.Manage user
scope

📝

🟢note

The Network tab is currently only available for
virtual machines.

📝

🟢note

The Network tab is only displayed when you
have access to the main asset and associated
ones, such as security groups, VPCs and
subnets. For more information on Scope-Based
Access Control (SBAC) for configuring granular
scoping, see Manage user scope.Manage user
scope

15. Cortex Cloud Data Classification 

🧭 15.1. What is Cortex Cloud Data Classification? 

➡️n…

To access Data Classification management,
click Settings → Configurations → Data
Classification

15.2. How to create and validate a custom data

pattern 

📝

🟢note

Once a custom data pattern is saved, it runs on
all data in the same way as any out-of-the-box
(OOTB) pattern, becoming globally applicable
for all modules using Cortex Cloud Data
Classification.

📝

🟢note

For more information, see How to disable and
enable data patterns in Data Classification.How
to disable and enable data patterns in Data
Classification

15.2.1. Custom data patterns: Guardrails and
syntax guide 

15.3. How to disable and enable data patterns in
Data Classification 

📝

🟢note

Disabling data patterns can cause changes in
your data profile results and stop detection of
these data patterns.

📝

🟢note

For more information about data patterns in
data classification, see What is Cortex Cloud
Data Classification?.

🧭

➡️n…

In the lower left part of the screen, click
Settings → Configurations → Data
Classification → Data Patterns

15.4. How to create and validate a custom data

profile 

📝

🟢note

If you choose two data locations, only the
filters they have in common will be included in
the possible filter options.

📝

🟢note

For more information, see How to disable and
enable data profiles in Cortex Cloud Data
Classification.

👀

🟠i…

Deleting a data profile deletes all past data
associated with it in all modules using Cortex
Cloud Data Classification after a warning
notification is displayed.

👀

🟠i…

Enabling and disabling a data profile removes
or re-adds the data profile results to the data
objects; that is, files and tables.

15.5. How to disable and enable data profiles in
Cortex Cloud Data Classification 

📝

🟢note

Existing results on past scans are updated in
the Asset and Object inventories within two
hours after being disabled or enabled.

📝

🟢note

For more information about data profiles in
Cortex Cloud Data Classification, see What is
Cortex Cloud Data Classification?.What is
Cortex Cloud Data Classification?

🧭

➡️n…

In the lower left part of the screen, click
Settings → Configurations → Data
Classification → Data Profiles

15.6. How to report a false positive in Cortex
Cloud Data Classification 📝 
🟢note

Alternatively, in Cortex Command Center, click
Help → Submit a Support Case.

16. Cortex Cloud Data Security 

16.1. What is Cortex Cloud Data Security? 

16.2. Supported assets in Cortex Cloud Data
Security 

🔒

🟣c…

This feature is included with a Cortex XSIAM
Premium license. It is also included with any
other Cortex XSIAM license that has the Cloud
Posture Security or Cloud Runtime Security
add-on.

📝

🟢note

CMK (customer-managed key) in SQL Server is
only supported in outpost mode.

📝

🟢note

The list above refers to only data classification;
however, Cortex Cloud Data Security discovers
and monitors all cloud assets and services for
usage and misconfigurations.

📝

🟢note

For information about configuring data
classification asset types, see How to configure
the scanning settings for supported services.

16.3. Cortex Cloud Data Security concepts 

16.4. Cortex Cloud Data Security use cases 

16.5. Data Inventory 

📝

🟢note

To clear the filter in the filter panel, click Delete.

🧭

➡️n…

When you select Inventory → All Assets →
Data → All Data Assets, a list of all data assets
appears

16.6. How to review errors in Cortex Cloud Data

Security 

16.7. How to configure the scanning settings for
supported services 📝 
🟢note

If you do not select a scanning cadence, the
default setting is applied. For more information,
contact your Customer Support team.

16.8. How to perform advanced Data Security
investigations using XQL 📝 
🟢note

For more information, see Build XQL queries.

📝 16.9. How to onboard Databricks 

🟢note

If you turn on this feature, both the cloud and
region will be used for scanning, possibly
incurring cost and requiring adherence to
certain compliance policies.

☑️ 16.10. How to onboard Microsoft 365 

🟣pr…

You have generated a Globally Unique Identifier
(GUID), also known as a Universally Unique
Identifier (UUID). You will need this ID for the
tenant you want to use for the Microsoft 365
instance.
In order to use Microsoft 365, you must be
registered with Microsoft Azure.

16.11. How to onboard on-premise file shares to
Cortex Cloud Data Security 

📝

🟢note

Your data is scanned on the Broker VM itself,
and only the metadata and classification
results are transmitted from the on-premise
environment to Cortex Cloud.

📝

🟢note

For information about activating the DSPM
Fileshare applet, see Activate DSPM Fileshare.

16.12. How to onboard Snowflake 

☑️

🟣pr…

In order to use Snowflake, you must be
registered with one of these cloud providers:
Amazon AWS, Microsoft Azure, or Google
Cloud Platform (GCP).
Ensure you have the necessary account
permissions to onboard. It is recommended to
use Account Admin as the role for the
onboarding.

📝

🟢note

The account identifier can be found using the
user information at the bottom left. Hover over
the account you wish to onboard and select
the copy option at the top right. The account
identifier is usually of the format:
(organization).[account]

16.13. How to use information protection labels
in Cortex Cloud Data Security 📝 
🟢note

Only If you have connected a Microsoft 365
instance to Cortex Cloud, the information
protection labels appear. See How to onboard
Microsoft 365.

17. Cortex Cloud Identity Security 

17.1. What is Cortex Cloud Identity Security? 

17.2. Review and improve your Identity Security
posture 💡 
🔵tip

Reducing the number of nonhuman entities that
are either dormant or have administrative
privileges is the most effective way to improve
your security score.

17.3. How does Effective Permission Calculation

work? 

17.4. Cortex Cloud Identity Security functionality 

📝

🟢note

Access to the Weak/Compromised Password
widget and column requires the ITDR add-on.

📝

🟢note

For more information about Effective
Permission Calculation, see How does Effective
Permission Calculation work?.

📝

🟢note

When a policy is analyzed and categorized as
excessive, a relevant finding and highlight is
attached to that policy and the various
identities being granted excessive permissions.

📝

🟢note

Each action must be defined as either being
administrative or not.
The administrative tag is an additional tag for
administrative actions, along with one of the
other access level types. For example, the AWS
action iam:CreateGroup is categorized as
config and has the administrative tag as well.

👀

🟠i…

If you turn off the audit logs, even briefly, this
temporarily impacts the accuracy of the Last
Access data, potentially showing permissions
as unused when actually they were active. Full
accuracy is restored 90 days after you re-
enable the audit logs.

17.5. Achieve the principle of least privilege
access



📝

🟢note

In the case of Amazon AWS, Cortex Cloud
Identity Security also uses AWS Identity and
Access Management (IAM) Access Advisor
insights to expand the coverage of supported
actions.

👀

🟠i…

If you turn off the audit logs, even briefly, this
temporarily impacts the accuracy of the Last
Access data, potentially showing permissions
as unused when actually they were active. Full
accuracy is restored 90 days after you re-
enable the audit logs.

17.6. Explore permissions using the simple and
advanced access tables 

17.7. Create a custom detection rule in Cortex

Cloud Identity Security 🧭 
➡️n…

Posture Management → Rules & Policies →
Rules → Cloud Security

17.8. Perform advanced Identity Security
investigations using XQL 📝 
🟢note

For more information, see Build XQL queries.

☑️ 17.9. Ingest logs and data from Okta 

🟣pr…

Administrator privileges: Your Okta user must
have a role capable of creating API tokens,
such as Read-only Administrator, Super
Administrator, or Organization Administrator.
For more information, see the Okta
Administrators Documentation.

17.10. Enable inactive human identity logs on
Azure in Cortex Cloud Identity Security 📝 
🟢note

For more information, see Ingest logs from
Microsoft Azure Event Hub.

17.11. Manage RBAC and SBAC in Cortex Cloud
Identity Security 👀 
🟠i…

You can use the Cortex Cloud Identity Security
RBAC roles to define access to the various
sections and functionalities of Cortex Cloud
Identity Security, but these roles do not directly
control the specific data a user sees within
those sections. Data visibility is further refined
and limited by scope-based access control
(SBAC) capabilities.

18. Vulnerability management 

18.1. Vulnerability management in Cortex Cloud 

🔒

🟣c…

Requires the Cortex Cloud Posture Security,
Cortex Cloud Runtime Security, Exposure
Management, Cortex XSIAM Premium or ASM
add-on.

📝

🟢note

Cortex Cloud vulnerability management
provides the ability to identify and assess
runtime vulnerabilities in every asset across
traditional IT and cloud environments. For
vulnerabilities detected in your software
development lifecycle through application
security scans, refer to the Cortex Cloud
Application Security documentation.

📝 18.1.1. Cortex Cloud vulnerability concepts 

🟢note

CVE is an acronym for Common Vulnerabilities
and Exposures, which is a list of publicly
disclosed security threats. We often use the
term "CVE" to refer to a vulnerability that has
been a assigned a CVE ID. Cortex Cloud
identifies CVEs and non-CVE vulnerabilities.

18.1.2. Vulnerability Management dashboard 

🧭 18.2. Cortex Vulnerability Risk Score 

➡️n…

Posture Management → Vulnerability
Management → Vulnerability Issues

18.3. Vulnerability policies 

🧭 18.3.1. Create a vulnerability policy 

➡️n…

Posture Management → Rules & Policies →
Policies → Vulnerability Management

18.3.2. Update the Ignored CVEs, Asset Groups,
and Assets policy



🧭 18.3.3. Modify a vulnerability policy 

➡️n…

Posture Management → Rules & Policies →
Vulnerability Management

🧭 18.3.4. Configure a block grace period 

➡️n…

Posture Management → Rules & Policies →
Vulnerability Management → Vulnerability
Policies - Prevention

18.3.5. Enable or disable a vulnerability policy 

18.4. Investigate and remediate vulnerabilities 

🧭 18.4.1. View all Vulnerabilities 

➡️n…

Posture Management → Vulnerability
Management → Vulnerabilities

18.4.2. View vulnerability issues 

🧭 18.4.3. View All Vulnerability Findings 

➡️n…

Posture Management → Vulnerability
Management → Vulnerability Issues and click
the All Vulnerability Findings button

🧭 18.4.4. View vulnerable assets 

➡️n…

Posture Management → Vulnerability
Management → Vulnerable Assets

🧭 18.5. Vulnerability Intelligence 

➡️n…

Posture Management → Vulnerability
Management → Vulnerability Intelligence

18.6. Recast CVSS scores and CVSS severities 

19. Cloud Security Rules and Policies 

19.1. Overview 

19.2. Cloud Security Rules 

19.3. Cloud Security Policies 

📝

🟢note

If you have the following Scope Based Access
Control (SBAC) settings in place, User Settings
> Cases and Issues Scope > Select domains >
Posture, you may encounter a Case mismatch in
Issues/Cases/Findings counts. This is because
the Case count on the Rules page captures
Cases belonging to the Posture domain.
Whereas Platform pages, capture Issues...

19.3.1. Issues 

19.4. Create Rules 

19.4.1. Create a configuration rule 

19.4.1.1. Guidelines for creating cloud security
rules 📝 
🟢note

XQL queries are supported for cloud security
configuration rules only. XQL queries are not
yet supported for other types of cloud security
rules.

19.4.1.2. Cloud security rule status for custom
configuration rules 

19.4.2. Create a Data Rule 

19.4.3. Create a Network Exposure Rule 

19.4.4. Edit a Rule 

19.5. Create Policies 

📝

🟢note

When creating Policies, note that Rules with the
Informational severity level are excluded.

19.5.1. Edit a Policy 

19.5.2. Use an Existing Policy to Create a New
One 

20. Cloud Workload Policies and

Rules 

☑️

🟣pr…

Users need View/Edit RBAC permissions (under
Policies → Compute Policies) or the Instance
Administrator role to view, edit, and modify
Cloud Workload Policies policies.

👀

🟠i…

Users with SBAC granular scoping (in addition
to the RBAC permissions required for Cloud
Workload Policies) can only view Cloud
Workload Policies, when their access is scoped
to any of the available options: All assets, No
assets, or Select asset groups. For more
information on granular scoping, see Manage
user scope. When no SBAC restriction is...

20.1. How policies and rules work together 

20.2. Cloud Workload Policies 

📝

🟢note

Issues are automatically resolved when the
finding is no longer applicable to the asset or
when the affected asset is removed from the
inventory.
For more details on Prevent and create issues,
see Cloud Workload Preventive Action.

20.2.1.1. Trusted image cloud workload policies  20.2.1. Types of Cloud Workload Policies 

📝

🟢note

Registry scanning is for finding problems that
are objectively issues regardless of context,
organization, or scope. Trust, however, is
subjective. Depending on the scope and other
factors, an issue may or may not be
problematic, and can change over time
depending on the context.

📝

🟢note

Trusted image policies do not override other
policies and their rules, such as malware
policies, misconfiguration policies, and secrets
policies.

⚠️

🟡c…

The default behavior for the prevent action
when trust criteria are unavailable is Create an
Issue. Be aware that changing this default will
mean an incomplete deployment file can block
a workload.

20.2.2. Cloud Workload Policies page 

📝

🟢note

Keep the following caveats in my mind when
working with Policies:
Instance Administrators are able to view all
facets of policies without restrictions, even if
Scope Based Access Control (SBAC) roles are
in effect. Learn more about SBAC.Manage user
scope

If you've been assigned a custom role with
View/Edit permissions limited by SBAC, you may
not be...

20.2.2.1.1. Show or hide the widget panel  20.2.2.1. Widgets panel 

20.2.2.2. Change the layout of the policies table 

20.2.2.3. Policy Details Panel 

20.2.3. Enable or disable a Cloud Workload

Policy 🧭 
➡️n…

Posture Management → Rules & Policies →
Policies → Cloud Workload

20.2.4. Create a Cloud Workload Policy 

📝

🟢note

The Evaluation stage for Misconfiguration
policies is supported only in the Runtime SDLC
stage and is enforceable through the
Kubernetes Admission Controller for clusters
on-boarded using the Posture Management
(KSPM) Connector.

📝

🟢note

Use Create a new Custom Detection Rule to
define and add new custom rules as required.

📝

🟢note

Each rule may support different actions. While
some include both Create an Issue and Prevent
and Create an Issue, others provide only the
Create an Issue option.

📝

🟢note

All non-relevant (non-compute) assets are
automatically excluded from the included asset
list.

📝

🟢note

Currently, dynamic asset groups for Kubernetes
Prevent Policies support only the following
attributes:
Kubernetes Resource Cluster: The cloud
identifier for the cluster.
Kubernetes Resource Namespace: The
namespace where the resource is located.
Kubernetes Resource Labels: The labels
assigned to the resource.
Kubernetes Resource Category: The category...

📝

🟢note

See Cloud Workload Preventive Action, to learn
more about the Prevent action behavior and
prerequisites.
If the Prevent and create an issue action is
selected, the preventive actions in the CI
pipeline will trigger a Fail Pipeline by returning
an exit code of 2 in the CI tool.

📝

🟢note

Choosing this option results in the automatic
selection of all other asset scopes in the list.

📝

🟢note

See Cloud Workload Preventive Action, to learn
more about the Prevent action behavior and
prerequisites.

📝

🟢note

See Cloud Workload Preventive Action, to learn
more about the Prevent action behavior and
prerequisites.

📝

🟢note

Choosing this option results in the automatic
selection of all other asset scopes in the list.

📝

🟢note

See Cloud Workload Preventive Action, to learn
more about the Prevent action behavior and
prerequisites.

20.2.5. Use an existing policy to create a new
Cloud Workload policy



20.2.6. Edit a Cloud Workload Policy 

20.2.7. Delete a Cloud Workload Policy 

20.2.8. Cloud Workload Preventive Action 

20.3. Cloud Workload Rules 

20.3.1. Default (pre-defined) Rules 

20.3.2. Custom (user-defined) Rules 

20.3.3. Cloud Workload Rules page 

📝

🟢note

Keep the following caveats in my mind when
working with Rules:
Instance Administrators are able to view all
facets of Rules without restrictions, even if
Scope Based Access Control (SBAC) roles are
in effect. Learn more about SBAC.Manage user
scope

If you've been assigned a custom role with
View/Edit permissions limited by SBAC, you may
not be able...

20.3.3.1. Filter page results 

20.3.3.2. Change the layout of the rules table 

20.3.3.3. Rule details panel 

20.3.4. Create a new Custom Detection Rule 

📝

🟢note

The custom rule must use the match term (not
deny or others) to function properly.

📝

🟢note

Only users with the following roles can enable
or disable Custom Code Execution:
Account Admin
Instance Administrator
Deployment Admin
Privileged Security Admin

📝

🟢note

Only Custom Detection Rules (not built-in
rules) can be assigned to custom controls.

👀

🟠i…

The custom Python scripts are intended to be
executed exclusively for compliance checks
and validations. To ensure the scripts are used
properly and no security risks or unintended
changes occur, the system implements the
following restrictions and safeguards:
Only a predefined set of Python libraries and
functions required for compliance checks are...

20.3.5. Use an existing rule to create a new
Custom Detection Rule 🧭 
➡️n…

Posture Management → Rules & Policies →
Rules → Cloud Workload

20.3.6. Edit a Custom Detection Rule 

20.3.7. Delete a Custom Detection Rule 

21. Monitor and track compliance
adherence



21.1. Cortex compliance flow 

21.2. Choose compliance standards from the
compliance catalog 

🧭

➡️n…

Cortex provides lists of available standards and
controls in the Standards and Controls
catalogs under Posture Management →
Compliance → Catalogs

21.2.1.1. Use a built-in or custom standard  21.2.1. Standards Catalog 

21.2.2.1. Use a built-in or custom control  21.2.2. Controls catalog 

21.2.2.1.1. Add a custom detection rule to a

custom control 📝 
🟢note

Only custom detection rules (not built-in) can
be assigned to custom controls.

21.2.2.1.2. Create a new Custom Detection Rule 

📝

🟢note

The custom rule must use the match term (not
deny or others) to function properly.

📝

🟢note

Only users with the following roles can enable
or disable Custom Code Execution:
Account Admin
Instance Administrator
Deployment Admin
Privileged Security Admin

📝

🟢note

Only Custom Detection Rules (not built-in
rules) can be assigned to custom controls.

👀

🟠i…

The custom Python scripts are intended to be
executed exclusively for compliance checks
and validations. To ensure the scripts are used
properly and no security risks or unintended
changes occur, the system implements the
following restrictions and safeguards:
Only a predefined set of Python libraries and
functions required for compliance checks are...

21.3. Use an assessment profile to run
compliance checks on your assets 🧭 
➡️n…

Under Posture Management → Compliance →
Assessment Profiles, click Create New
Assessment

21.4. View and manage compliance assessments
and reports 

📝

🟢note

Compliance assessment results may take up to
six hours to be generated.

📝 21.4.1. Assessments 

🟢note

If there are no rules associated with the control,
the control will be assigned a severity of low.

🧭 21.4.2. Reports 

➡️n…

The Posture Management → Compliance →
Results → Reports page shows a table listing
compliance assessment report files

21.5. View the compliance assessment of an
individual asset 

22. Data management 

22.1. Broker VM 

📝 22.1.1. What is the Broker VM? 

🟢note

The Broker VM is a closed, hardened appliance.
To maintain its security integrity and
performance standards, third-party agents
cannot be installed on the Broker VM.

22.1.2. Set up and configure Broker VM 

📝

🟢note

The Broker VM comes with a 512 GB disk.
Therefore, deploy the Broker VM with thin
provisioning, meaning the hard disk can grow up
to 512 GB but will do so only if needed.

📝

🟢note

When DHCP is not enabled in your network and
there isn't an IP address for your Broker VM,
configure the Broker VM with a static IP using
the serial console menu.

📝

🟢note

When configuring a Local Agent Settings
applet with installer and content caching, you
need to configure an SSL certificate for the
Broker VM as explained in the task below. For
more information on specific requirements for
the Local Agent Settings applet, see Activate
Local Agent Settings.
Keep in mind that several Broker VM applets,
such as the...

📝

🟢note

When a Broker VM is disconnected for more
than 30 days, it will have to go through a re-
registration process.

📝

🟢note

When DHCP is not enabled in your network and
there isn't an IP address for your Broker VM,
configure the Broker VM with a static IP using
the serial console menu.

📝

🟢note

You can configure another Broker VM as a
proxy server for this Broker VM by selecting the
HTTP type. When selecting HTTP to route
Broker VM communication, you need to add the
IP Address and Port number (set when
activating the Agent Proxy) for another Broker
VM registered in your tenant. This designates
the other Broker VM as a proxy for this Broker
VM.

📝

🟢note

The Palo Alto Networks Broker VM supports
only strong cipher SHA256-based certificates.
MD5/SHA1-based certificates are not
supported.

📝

🟢note

If adding a CA certificate to the Broker VM is
not possible, ensure that you've added the
Broker Service FQDNs to the SSL Decryption
Exclusion list on your firewalls. See Enable
Access to Cortex XDR.
If you need to enable legacy SSL renegotiation,
use the Broker VM Serial console.

📝

🟢note

If you need to enable legacy SSL renegotiation,
use the Broker VM Serial console.

👀

🟠i…

There can be instances in which the Broker VM
requires up to 50% of the incoming bandwidth
as outgoing. Such instances can be, network
instability between the Broker VM and Cortex
Cloud, or data that is being collected, but not
well compressed.

👀

🟠i…

The internal network for the Broker VM must be
unique and reserved. Other devices should not
use the same IP as the Broker VM internal
network as it can lead to communication issues
with the Broker VM.

👀

🟠i…

If you use SSL decryption in your firewalls and
proxies, see the Understanding CA certificate
functionality in Broker VM deployments section
below. In addition, verify that the proxies used
support HTTP/2, gRPC-specific headers, and
HTTP/2 trailers, and the inspection policies
support gRPC traffic. Any devices that you use
with this configuration...

👀

🟠i…

When configuring more than one network
interface, ensure that only one Default
Gateway is defined. The rest must be set to
0.0.0.0, which configures them as undefined. In
addition, we recommend assigning each
network interface to a different subnet, as
oppose to configuring two interfaces on the
same subnet which can potentially cause
unexpected...

👀

🟠i…

Internal IP must be:
Formatted as prefix/mask, for example
192.0.2.1/24.
Must be within /8 to /24 range.
Cannot be configured to end with a zero.
For Broker VM version 9.0 and earlier, Cortex
Cloud will only accept 172.17.0.0/16.

👀

🟠i…

We strongly recommend disabling SSH
connectivity when it's not being used.
Therefore, activate SSH connectivity when it's
needed and disable it right afterwards.
When generating a new SSH key ensure to avoid
embedding the domain-style username, by not
using any backslashes (\) in the comment field,
to ensure the SSH key passes validation.

22.1.2.1. Broker VM image installations 

22.1.2.1.1. Set up Broker VM on Alibaba Cloud 

☑️

🟣pr…

Download a Cortex Cloud Broker VM QCOW2
image. For more information, see the virtual
machine compatibility requirements in Set up
and configure Broker VM.

📝

🟢note

For Linux and Windows uploads, you can use
Alibaba Cloud's graphical management tool
called ossbrowser.

📝

🟢note

The Object Storage Service must be created in
the same Region as the image of the virtual
machine.

🧭

➡️n…

Hamburger menu → Elastic Compute Service
→ Instances & Images → Instances

22.1.2.1.2. Set up Broker VM on Amazon Web
Services 

☑️

🟣pr…

Download a Cortex Cloud Broker VM VMDK
image. For more information, see the virtual
machine compatibility requirements in Set up
and configure Broker VM.
You need to set up an AWS VM Import role
(vmimport) before you continue with the steps
to convert the image as it is required for the
import-snapshot CLI command. You can use a
different role, if the...

📝

🟢note

Assigning security groups can take up to 15
minutes.

👀

🟠i…

A t3.xlarge (16 GB RAM) is the lowest machine
type that can be used as an instance type to
meet the mandatory 4 vCPU requirement.

🧭

➡️n…

Log in to the AWS IAM Console, and in the
navigation pane, select Access Management →
Users → Add Users

In the Cortex Cloud Web Console , select
Settings → Configurations → Data Broker →
Broker VMs → Add Broker → VMDK

Services → Storage → S3 → Buckets

Services → EC2 → IMAGES → AMIs and locate
your AMI image using the task ID

Obtain a registration token from the Cortex
Cloud Web Console by selecting Settings →
Configurations → Data Broker → Broker VMs →
Add Broker → Generate Token

22.1.2.1.3. Set up Broker VM on Google Cloud
Platform (GCP) 

☑️

🟣pr…

Download a Cortex Cloud Broker VM VMDK
image. For more information, see the virtual
machine compatibility requirements in Set up
and configure Broker VM.
To complete the set up, you must have G Cloud
installed and have an authenticated user
account.

☑️

🟣pr…

Before importing a GCP image using the gcloud
CLI, ensure that you update the Google Cloud
components to version 371.0.0 and above using
the following command:

📝

🟢note

The import tool uses Cloud Build API, which
must be enabled in your project. For the import
to work, Cloud Build service account must have
compute.admin and iam.serviceAccountUser
roles. When using the Google Cloud console to
import the image, you will be prompted to add
these permissions automatically.

22.1.2.1.4. Set up Broker VM on KVM using
Ubuntu ☑️ 

🟣pr…

Download a Cortex Cloud Broker VM QCOW2
image. For more information, see the virtual
machine compatibility requirements in Set up
and configure Broker VM.

22.1.2.1.5. Set up Broker VM on Microsoft Azure 

☑️

🟣pr…

Download a Cortex Cloud Broker VM VHD
(Azure) image. For more information, see the
virtual machine compatibility requirements in
Set up and configure Broker VM.

📝

🟢note

Extraction can take up to a few hours.

📝

🟢note

Upload can take up to a few hours.

📝

🟢note

For more information, see the Azure
Documentation.

📝

🟢note

Creating the VM can take up to 15 minutes. The
Broker VM Web UI is not accessible during this
time.

📝

🟢note

For more information on creating a rule in an
Azure VM, see Create a Security Rule in the
Azure Documentation.

22.1.2.1.6. Set up Broker VM on Microsoft Hyper-
V ☑️ 

🟣pr…

Download a Cortex Cloud Broker VM VHD
image. For more information, see the virtual
machine compatibility requirements in Set up
and configure Broker VM.

22.1.2.1.7. Set up Broker VM on Nutanix
Hypervisor 

☑️

🟣pr…

Download a Cortex Cloud Broker VM QCOW2
image. For more information, see the virtual
machine compatibility requirements in Set up
and configure Broker VM.

📝

🟢note

Saving the image to Nutanix hypervisor can
take time as it's a large file. We recommend
verifying periodically that the connection is
alive for the upload process to finish
successfully.

📝

🟢note

Creating the VM can take up to 15 minutes. The
Broker VM Web user interface is not accessible
during this time.

22.1.2.1.8. Set up Broker VM on VMware ESXi
using vSphere Client ☑️ 
🟣pr…

Ensure you have a virtualization platform
installed that is compatible with an OVA image,
and have an authenticated user account.
Download a Cortex Cloud Broker VM OVA
image. For more information, see the virtual
machine compatibility requirements in Set up
and configure Broker VM.

🔒 22.1.2.2. Broker VM data collector applets 

🟣c…

Some data collector applets require the Data
Collection add-on.

22.1.3. Manage Broker VM 

📝

🟢note

For more information on troubleshooting errors
and warnings for these broker applets, see
Troubleshoot Broker VM applet errors.

📝

🟢note

Certain fields are also exposed in the Clusters
tab, when a Broker VM node is added to a High
Availability (HA) cluster, and each cluster node
is expanded to view the Broker VM nodes table.
An asterisk (*) is beside every field that is also
included in the Broker VM nodes table for each
HA cluster.

🧭

➡️n…

Settings → Configurations → Data Broker →
Broker VMs

22.1.3.1. Edit Broker VM Configuration 

📝

🟢note

For all Broker VM nodes added to a HA cluster,
you can also Configure the Broker VM nodes
from the Clusters tab.

📝

🟢note

Internal IP must be:
Formatted as prefix/mask, for example
192.0.2.1/24.
Must be within /8 to /24 range.
Cannot be configured to end with a zero.
For Broker VM version 9.0 and lower, Cortex
Cloud accepts only 172.17.0.0/16.

📝

🟢note

Make sure you save the password before
closing the window. The only way to re-
generate a password is to disable ssh and re-
enable.

22.1.3.2. Increase Broker VM storage allocated
for data caching 

📝

🟢note

Follow your hypervisor documentation to
understand how to add a persistent disk
storage to your VM.

📝

🟢note

If your disk is not listed and you didn't
shutdown your Broker VM in your hypervisor
before manually adding a disk to the VM, you'll
need to reboot the Broker VM before the disk
details are detected by the Broker VM. This can
be performed either in the hypervisor or
directly in the Broker VMs page.

⛔

🔴w…

This operation is irreversible, and will make the
disk become an integral part of the broker,
where disconnecting the disk will result in
errors and data loss.

22.1.3.3. Monitor Broker VM using Prometheus 

📝

🟢note

For all Broker VM nodes added to a HA cluster,
you can also Configure the Broker VM nodes
from the Clusters tab.

📝

🟢note

For more information on Docker, see the Docker
website.

📝 22.1.3.4. Collect Broker VM Logs 

🟢note

The Download Logs (<TIMESTAMP>) is only
displayed when you've downloaded your logs
previously using Generate New Logs.

👀 22.1.3.5. Upgrade Broker VM 

🟠i…

After a Broker VM upgrade, your broker may
require a reboot to finish installing important
updates. A notification about this will be sent
to your Cortex Cloud console Notification
Center.

22.1.3.6. Import Broker VM Configuration 

👀

🟠i…

This option can only be used on Broker VMs
with version 20.0 and later, and is only suitable
for importing a configuration of brokers in the
same version, or from a broker in an older
version to a broker in a newer version.

👀

🟠i…

If your source Broker VM configuration includes
a WEC applet, you'll need to ensure that you
update the DNS record of this Broker VM's
FQDN to point to the target Broker VM IP
address.

📝 22.1.3.7. Open Live Terminal 

🟢note

You can either restart_routes or reboot the
Broker VM for the changes in the
/etc/network/routes file to take affect.

22.1.3.8. Add Broker VM to cluster 

22.1.3.9. Switchover Primary Node in Cluster 

📝 22.1.3.10. Remove from Cluster 

🟢note

If you've enabled a Load Balancer Health-
Check on the cluster, you need to exclude this
Broker VM from your Load Balancer settings.

22.1.4. Manage Broker VM data collector
applets 

📝

🟢note

For more information on the Broker VM applet
connectivity status, see Manage Broker VM.

22.1.4.1. Activate DSPM Fileshare 

☑️

🟣pr…

Set up and configure Broker VM
Know the complete path to the files and
folders that you want Cortex Cloud to monitor.
Necessary user permissions to access the
network shares. For the SMB connection type,
you need the user name and password.

📝

🟢note

The applet list displays only the applets for
which you have permissions.

📝

🟢note

By default, all configured connections are
saved.

📝

🟢note

For details regarding the connection fields, see
the table above under File Share Connection.

📝

🟢note

You can add multiple connections under a
single instance of the DSPM Fileshare applet by
returning to the File Share Connection screen
and clicking Add Connection. Each new
connection can be of either the NFS or SMB
connection type.

22.1.5. Broker VM High Availability Cluster 

📝

🟢note

For Load Balancing, you must install a Load
Balancer in your network, which will distribute
the incoming data between the nodes.

📝

🟢note

The following applets aren't supported when
configuring Broker VMs in HA clusters: DSPM
Fileshare, Registry Scanner, and Transporter.

22.1.5.1. Configure High Availability Cluster 

☑️

🟣pr…

Be sure you do the following task before
creating a cluster from an existing Broker VM:
If the Broker VM is explicitly specified in some
Agent Settings profile, which means Cortex
Cloud agents retrieve release upgrades and
content updates from this Broker VM, you must
change the Broker VM's current designated
role. To do this, you need to modify the...

📝

🟢note

The Broker VM Load Balancer Health Check
requires HTTP/1.1 or higher. Legacy HTTP/1.0 is
no longer supported. Ensure your external load
balancer is configured to use HTTP/1.1 or above
when performing health checks.

👀

🟠i…

You can only create a new cluster from an
existing Broker VM, when the Broker VM version
is 19.0 and later, and the STATUS is Connected.
The Create a Cluster from this Broker option is
only listed if the Broker VM is not already
added to a cluster.

👀

🟠i…

When the Protocol is set to HTTPS, you may
need to perform a few follow-up steps to
establish a validated secure SSL connection
with the Broker VM.
If you're using your own Certificate Authority
(CA) to sign the certificates, you'll need to
place the CA in the client, such as the Load
Balancer, and upload the certificates to the
Broker VM.
If you're...

22.1.5.2. Manage Broker VM clusters 

🧭

➡️n…

After you've configured a cluster, you can
manage all your Broker VM clusters from the
Clusters tab on the Broker VMs page (Settings
→ Configurations → Data Broker → Broker VMs
→ Clusters

22.1.5.2.1. View cluster details 

📝

🟢note

For more information on troubleshooting errors
and warnings for these applets, see
Troubleshoot Broker VM applet errors.

🧭

➡️n…

The Clusters tab of the Broker VMs page
(Settings → Configurations → Data Broker →
Broker VMs) enables you

🧭 22.1.5.2.2. Edit cluster 

➡️n…

Settings → Configurations → Data Broker →
Broker VMs, and select the Clusters tab

22.1.5.2.3. Add applet to cluster 

📝

🟢note

With Cortex XDR Prevent, it's only relevant to
configure a HA cluster with a Local Agent
Settings applet as this is the only applet
supported for this product license. The other
applets are collector applets, which are only
available in Cortex XDR Pro or Cortex XSIAM.

📝

🟢note

For more information on troubleshooting errors
and warnings for these applets, see
Troubleshoot Broker VM applet errors.

22.1.5.2.4. Add Broker VM to cluster 

22.1.5.2.5. Remove cluster 

22.1.6. Broker VM notifications 

📝 22.1.7. Monitor Broker VM activity 

🟢note

Certain fields are exposed and hidden by
default. An asterisk (*) is beside every field that
is exposed by default.

22.1.8. Troubleshoot Broker VM applet errors 

22.2. Dataset management 

🔒

🟣c…

Managing Roles requires an Account Admin or
Instance Administrator role. For more
information, see Predefined user roles.

☑️

🟣pr…

Dataset Management requires View/Edit RBAC
permissions for Data Management (under
Configurations → Data Management), which
are the same permissions required for Parsing
Rules, Data Model Rules, and Event Forwarding.

📝

🟢note

Cold storage, in addition to a cold storage
license, requires compute units (CU) to run cold
storage queries. For more information on CU,
see Manage compute units.

📝

🟢note

Only datasets that are already handled as part
of the GB license are supported for this license.
In addition, the retention configuration is only
available in Cortex Cloud, as opposed to the
public APIs.

📝

🟢note

Certain fields are exposed and hidden by
default. An asterisk (*) is beside every field that
is exposed by default.
Datasets include dataset permission
enforcements in the Cortex Query
Language(XQL), Query Center, and XQL
Widgets. For example, to view or access any of
the endpoints and host_inventory datasets, you
need role-based access control (RBAC)...

📝

🟢note

You'll only be able to save the dataset view if
the query contains no errors; otherwise, the
Save button is disabled.

📝

🟢note

For more information on user roles, see Manage
user roles.

👀

🟠i…

Cortex Cloud enforces retention on all log-
type datasets excluding Host Inventory,
Vulnerability Assessment, Metrics, and Users.

👀

🟠i…

This column is updated once a day. Therefore,
if the dataset was created or updated by the
target or lookup flows, it's possible that the
Last Updated value is a day behind when the
queries or reports were run as it was before this
column was updated.

22.2.1. What are datasets? 

☑️

🟣pr…

Dataset Management requires View/Edit RBAC
permissions for Data Management (under
Configurations → Data Management), which
are the same permissions required for Parsing
Rules, Data Model Rules, and Event Forwarding.

📝

🟢note

For more information on tasks specific to
lookup datasets, see Lookup datasets.

📝

🟢note

Schema changes to datasets may not be
reflected in the autocomplete suggestions and
definitions as you type in real time the XQL
query and can appear with a slight delay.

👀

🟠i…

By default, forensic datasets are not included
in XQL query results, unless the dataset query
is explicitly defined to use a forensic dataset.

22.2.2. Lookup datasets 

☑️

🟣pr…

Dataset Management requires View/Edit RBAC
permissions for Data Management (under
Configurations → Data Management), which
are the same permissions required for Parsing
Rules, Data Model Rules, and Event Forwarding.

📝

🟢note

A lookup dataset can only be deleted if there
are no other dependencies. For example, if a
Correlation Rule is based on a lookup dataset,
you wouldn't be able to delete the lookup
dataset until you removed the dataset from the
XQL query of the Correlation Rule.

👀

🟠i…

When you create or add data to a lookup
dataset using the target stage, the _time field
won't be included by default unless you
explicitly add it with the fields stage.

22.2.2.1. Import a lookup dataset 

☑️

🟣pr…

Dataset Management requires View/Edit RBAC
permissions for Data Management (under
Configurations → Data Management), which
are the same permissions required for Parsing
Rules, Data Model Rules, and Event Forwarding.

☑️

🟣pr…

When uploading a CSV, TSV, or JSON file,
ensure that the file meets the following
requirements:
The maximum size for the total data to be
imported into a lookup dataset is 30 MB from
the Dataset Management page. Otherwise, the
limit is 50 MB using Cortex Query Language
(XQL) or APIs.
Field names can contain characters from
different languages, special...

👀

🟠i…

The name of a dataset created from a TSV file
must always include the extension. For example,
if the original file name is mrkdptusrsnov23.tsv,
you can save the dataset with the name
marketing_dept_users_Nov_2023.tsv.

22.2.2.2. Download JSON file of lookup dataset 

☑️

🟣pr…

Dataset Management requires View/Edit RBAC
permissions for Data Management (under
Configurations → Data Management), which
are the same permissions required for Parsing
Rules, Data Model Rules, and Event Forwarding.

🧭

➡️n…

Open the Settings → Configurations → Data
Management → Dataset Management page

☑️ 22.2.2.3. Set time to live for lookup datasets 

🟣pr…

Dataset Management requires View/Edit RBAC
permissions for Data Management (under
Configurations → Data Management), which
are the same permissions required for Parsing
Rules, Data Model Rules, and Event Forwarding.

22.2.3. Monitor datasets and dataset views

activity 

☑️

🟣pr…

Dataset Management requires View/Edit RBAC
permissions for Data Management (under
Configurations → Data Management), which
are the same permissions required for Parsing
Rules, Data Model Rules, and Event Forwarding.

📝

🟢note

Certain fields are exposed and hidden by
default. An asterisk (*) is beside every field that
is exposed by default.

22.3. Manage Event Forwarding 

☑️

🟣pr…

Event Forwarding requires View/Edit RBAC
permissions for Data Management (under
Configurations → Data Management), which
are the same permissions required for Parsing
Rules, Data Model Rules, and Dataset
Management.

📝

🟢note

After exporting logs, you can download them
from GCP for up to 14 days. The Pub/Sub
subscription messages are available for 7 days.

📝

🟢note

Requires the Endpoints Event Forwarding add-
on.

📝

🟢note

Requires the GB Event Forwarding add-on

👀

🟠i…

Since data is aggregated and compressed, it
may take up to two hours until the data is
available in the forwarding bucket.

🧭

➡️n…

Under Settings → Configurations → Data
Management → Event Forwarding, activate one
or more of the following: Enable GB Event
Forwarding Enable Endpoints Event Forwarding

Under Settings → Configurations → Data
Management → Event Forwarding, activate one
or more of the following

22.3.1. Endpoints Event Forwarding -
included/excluded fields by event type ☑️ 
🟣pr…

Event Forwarding requires View/Edit RBAC
permissions for Data Management (under
Configurations → Data Management), which
are the same permissions required for Parsing
Rules, Data Model Rules, and Dataset
Management.

22.4. Manage compute units 

📝

🟢note

For more information, see Import historical data
into cold storage.Import historical data into
cold storage

📝

🟢note

The CU consumption of cold storage queries is
based on the number of days in the query time
frame. For example, when querying 1 hour of a
specific day, the CU of querying this entire day
is consumed. When querying 1 hour that
extends past 2 days, such as from 23�50 to
00�50 of the following day, the CU of querying
these two days is consumed.

22.4.1. Compute units usage 

📝

🟢note

To enable the add-on, select Settings →
Configurations → Cortex Cloud License →
Addons tile, and select the Compute Unit tile
and Enable.

🧭

➡️n…

Settings → Configurations → Data
Management → Compute Unit Usage

23. Cortex Cloud Data Sources 

23.1. What are Cortex Cloud data sources? 

📝

🟢note

Some content packs provide parsing rules and
data model rules for data sources ingested
using a Syslog Collector applet of the Broker
VM or for standard data sources, and won't be
listed in the Data Sources & Integrations page.

👀

🟠i…

We are actively working to evolve the UI to
create a single, unified point of configuration
for all data ingestion and integration points.
This documentation is structured to help you
navigate the current segmented process until
that UI update is available.

23.2. Complete data source catalog 

23.3. Vendor-specific data sources 

👀

🟠i…

In some cases, the same vendor is available
through one or more of these options: standard
data source, CSP onboarding data source, or
content pack integration. Check the available
descriptions for each data source in both the
user interface and documentation to decide
which option is more suitable for your needs.

23.3.1. Amazon S3 

23.3.1.1. Ingest audit logs from AWS Cloud Trail 

📝

🟢note

For more information on configuring data
collection from Amazon S3 using AWS
CloudTrail, see the AWS CloudTrail
Documentation.

📝

🟢note

For more information on creating an AWS
CloudTrail trail, see Create a trail.
If you already have an Amazon S3 bucket
configured with AWS CloudTrail audit logs,
skip this step and go to Configure an Amazon
Simple Queue Service (SQS).

📝

🟢note

It is your organization's responsibility to define
a retention policy for your Amazon S3 bucket
by creating a Lifecycle rule in the Management
tab. We recommend setting the retention policy
to at least 7 days to ensure that the data is
retrieved under all circumstances.

📝

🟢note

The CloudWatch Logs - optional settings are
not supported and should be left disabled.

📝

🟢note

Ensure that you create your Amazon S3 bucket
and Amazon SQS queue in the same region.

📝

🟢note

For more information on granting permissions
to publish messages to an SQS queue, see
Granting permissions to publish event
notification messages to a destination.

📝

🟢note

If your receive an error when trying to save your
changes, you should ensure that the
permissions are set up correctly.

📝

🟢note

It your organization's responsibility to ensure
that the user who performs this task of creating
the access key is designated with the relevant
permissions. Otherwise, this can cause the
process to fail with errors.
Skip this step if you are using an Assumed Role
for Cortex Cloud.

📝

🟢note

For more information, see Managing access
keys for IAM users.

📝

🟢note

Skip this step if you are using an Assumed Role
for Cortex Cloud.

📝

🟢note

For more information on granting permissions
to publish messages to an SQS queue, see
Granting permissions to publish event
notification messages to a destination.

23.3.1.2. Ingest network flow logs from Amazon
S3 

📝

🟢note

For more information on configuring data
collection from Amazon S3, see the Amazon S3
Documentation.

📝

🟢note

For the Amazon S3 bucket created using
CloudFormation, it is the customer's
responsibility to define a retention policy by
creating a Lifecycle rule in the Management
tab. We recommend setting the retention policy
to at least 7 days to ensure that the data is
retrieved under all circumstances.

📝

🟢note

To create a new VPC, click Launch VPC Wizard.
For more information, see AWS VPC Flow Logs.

📝

🟢note

It is the responsibility of the customer's
organization to ensure that the user who
performs this task of creating the access key is
designated with the relevant permissions.
Otherwise, this can cause the process to fail
with errors.
Skip this step if you are using an Assumed Role
for Cortex Cloud.

📝

🟢note

For more information, see Managing access
keys for IAM users.

📝

🟢note

Skip this step if you are using an Access Key to
provide access to Cortex Cloud.

23.3.1.3. Ingest generic logs from Amazon S3 

📝

🟢note

For more information on configuring data
collection from Amazon S3, see the Amazon S3
Documentation.

📝

🟢note

You need to set up an Amazon S3 data
collector to receive generic logs when
collecting logs from BeyondTrust Privilege
Management Cloud. For more information, see
Ingest logs from BeyondTrust Privilege
Management Cloud.

📝

🟢note

It is the customer's responsibility to define a
retention policy for your Amazon S3 bucket by
creating a Lifecycle rule in the Management
tab. We recommend setting the retention policy
to at least 7 days to ensure that the data is
retrieved under all circumstances.

📝

🟢note

Ensure that you create your Amazon S3 bucket
and Amazon SQS queue in the same region.

📝

🟢note

For more information on granting permissions
to publish messages to an SQS queue, see
Granting permissions to publish event
notification messages to a destination.

📝

🟢note

If your receive an error when trying to save your
changes, you should ensure that the
permissions are set up correctly.

📝

🟢note

It is the responsibility of your organization to
ensure that the user who performs this task of
creating the access key is assigned the relevant
permissions. Otherwise, this can cause the
process to fail with errors.
Skip this step if you are using an Assumed Role
for Cortex Cloud.

📝

🟢note

For more information, see Managing access
keys for IAM users.

📝

🟢note

Skip this step if you are using an Assumed Role
for Cortex Cloud.

📝

🟢note

For more information on granting permissions
to publish messages to an SQS queue, see
Granting permissions to publish event
notification messages to a destination.

📝

🟢note

-The Vendor and Product defaults to Auto-
Detect when the Log Format is set to CEF or
LEEF.
-For a Log Format set to CEF or LEEF, Cortex
Cloud reads events row by row to look for the
Vendor and Product configured in the logs.
When the values are populated in the event log
row, Cortex Cloud uses these values even if you
specified a value in the Vendor...

23.3.1.4. Ingest network Route 53 logs from
Amazon S3 

📝

🟢note

For more information on configuring data
collection from Amazon S3 for Route 53 DNS
logs, see the AWS Documentation.

📝

🟢note

For more information on creating a Stack, see
Creating a stack on the AWS CloudFormation
console.

📝

🟢note

For the Amazon S3 bucket created using
CloudFormation, it is the customer's
responsibility to define a retention policy by
creating a Lifecycle rule in the Management
tab. We recommend setting the retention policy
to at least 7 days to ensure that the data is
retrieved under all circumstances.

📝

🟢note

It is the responsibility of the customer's
organization to ensure that the user who
performs this task of creating the access key is
designated with the relevant permissions.
Otherwise, this can cause the process to fail
with errors.
Skip this step if you are using an Assumed Role
for Cortex Cloud.

📝

🟢note

For more information, see Managing access
keys for IAM users.

📝

🟢note

Skip this step if you are using an Access Key to
provide access to Cortex Cloud.

23.3.1.5. Create an assumed role 

☑️

🟣pr…

You need ensure you have an Amazon S3
bucket and Amazon Simple Queue Service
(SQS) already configured as it's needed to
configure an IAM policy. The S3 bucket and
SQS required depends on how you plan to
configure your Amazon S3 data source:
When using a CloudFormation script provided
by Cortex Cloud to configure Amazon S3 with
SQS notifications, you'll...

📝

🟢note

You can retrieve your bucket's ARN by opening
the Amazon S3 Console in a browser window. In
the Buckets section, select the bucket, click
Copy ARN, and paste the ARN in the field.
You can retrieve the SQS queue ARN by
opening another instance of the AWS
Management Console in a browser window, and
opening the Amazon SQS Console, and
selecting the...

📝

🟢note

For more information, see the AWS instructions.

23.3.1.6. Configure data collection from
Amazon S3 manually 

📝

🟢note

If you already have an Amazon S3 bucket
configured with VPC flow logs that you want
to use for this configuration, you do not need
to perform the prerequisite steps detailed in
the first two bullets.

📝

🟢note

It is your responsibility to define a retention
policy for your Amazon S3 bucket by creating a
Lifecycle rule in the Management tab. We
recommend setting the retention policy to at
least 7 days to ensure that the data is retrieved
under all circumstances.

📝

🟢note

For resources such as Amazon S3, SQS, VPC
flow logs, and event notifications, you may use
the default configurations; yet, these settings
remain fully customizable to meet your specific
environment requirements.

📝

🟢note

If you already have an Amazon S3 bucket
configured with VPC flow logs, skip this step
and go to Configure an Amazon Simple Queue
Service (SQS).

📝

🟢note

Ensure that you create your Amazon S3 bucket
and Amazon SQS queue in the same region.

📝

🟢note

It is the responsibility of the customer's
organization to ensure that the user who
performs this task of creating the access key is
designated with the relevant permissions.
Otherwise, this can cause the process to fail
with errors.
Skip this step if you are using an Assumed Role
for Cortex Cloud.

📝

🟢note

Skip this step if you are using an Assumed Role
for Cortex Cloud.

23.3.2. Amazon Web Services 

23.3.3.1. Ingest data for API security  23.3.3. API Security 

23.3.3.1.1. Ingest AWS API Gateway 

📝

🟢note

You must generate a new key if you did not save.

📝

🟢note

If no logs are showing, send some API requests
on Postman or CURL.

👀

🟠i…

Leave '–'filter-pattern empty as displayed
above.

23.3.3.1.2. Ingest Azure APIM 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

Microsoft Azure APIM service must be running
before starting to configure the integration.

📝

🟢note

From the URL, save the UUID and the resource
group -

/resource/subscriptions/<UUID>/resourceGroup
s/<ResourceGroup>.
The UUID is the Azure account/subscription ID
and the resource group, which is the group
where the APIM Service is defined.

📝

🟢note

Use the search to navigate to the specific
section.

📝

🟢note

131072 bytes = 128 KB. This value determines
the size (in bytes) of request and response
bodies to send to Cortex. Any bytes beyond this
limit are truncated.

📝

🟢note

If any other inbound policies should be added,
they must be added after these elements.

👀

🟠i…

If you want to add additional data to the
<outbound>, add it at the start of the
<outbound> code.

23.3.3.1.3. Ingest Apigee Proxy 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

The steps include the prerequisites that run the
automated script that deploys files and
configurations to Apigee. For manual
configuration, refer to the section Manual
deployment.

📝

🟢note

If you have an existing hookand would like to
integrate with the shared flow, run the
deploy.sh script, and select n' and exit at the
prompt to create a new hook. Refer to the
section Connect to existing hook.

📝

🟢note

Start with the hook in pre-proxy.

📝

🟢note

Archive deployments are currently in preview
and are subject to change.

👀

🟠i…

This should only be done when there are
already existing hooks, and API security shared
flows can't be hooked as a standalone. Run the
deployment script, but skip step 9 by passing n.
This step publishes API security shared flows to
the desired Apigee environment without setting
them to flow hooks.

23.3.3.1.4. Ingest Kong 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

Contact support to obtain the custom plugin
file.

📝

🟢note

The queue parameters can be updated to
change when the plugin mirrors data to Cortex.

📝

🟢note

Use commas to separate tags.

👀

🟠i…

The size of the buffer must be equal or larger
than the max body size setting in the plugin
configuration, on every data plane node.

23.3.3.1.5. Ingest-F5 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

Extract the files from the F5 plugin to a folder
before selecting to upload to F5.

📝

🟢note

The provider for "context_provider" should
always be uppercase.
Supported providers: AWS, GCP, Azure, On-
prem.

📝

🟢note

Select panw_apisec_set_ssl_data only if your
client SSL profile is enabled.

23.3.4. Azure Event Hub 

23.3.4.1. Ingest logs from Microsoft Azure Event
Hub 

🔒

🟣c…

Requires the Data Collection add-on.

☑️

🟣pr…

Ensure that you do the following tasks before
you begin configuring data collection from
Azure Event Hub.
Before you set up an Azure Event Hub,
calculate the quantity of data that you expect
to send to Cortex Cloud, taking into account
potential data spikes and potential increases in
data ingestion, because partitions cannot be
modified after creation....

📝

🟢note

For more information on Azure Event Hub audit
logs, see Overview of Azure platform logs.

📝

🟢note

These logs are from the management plane.

📝

🟢note

Even though you can collect Azure AD Activity
logs and Azure Sign-in logs using the Azure
Event Hub data collector, we recommend using
the Microsoft Office 365 data collector,
because it is easier to configure. In addition,
ensure that you do not configure both
collectors to collect the same types of logs,
because if you do so, you will be creating...

📝

🟢note

These logs are from the data plane.

📝

🟢note

For every resource that you want to confiure,
you'll have to repeat this step, or use Azure
policy for a general configuration.

📝

🟢note

There are additional log categories displayed.
We recommend selecting all the available
options.

📝

🟢note

When you Normalize and enrich audit logs, the
log format is automatically configured. As a
result, the Log Format option is removed and is
no longer available to configure (default).

📝

🟢note

For a Log Format set to CEF or LEEF, Cortex
Cloud reads events row by row to look for the
Vendor and Product configured in the logs.
When the values are populated in the event log
row, Cortex Cloud uses these values even if you
specified a value in the Vendor and Product
fields in the Azure Event Hub data collector
settings. Yet, when the values are...

📝

🟢note

When you Normalize and enrich audit logs, the
Vendor and Product fields are automatically
configured, so these fields are removed as
available options (default).

⛔

🔴w…

Misconfiguration of Event Hub resources could
cause ingestion delays.
In an existing Event Hub integration, do not
change the mapping to a different Event Hub.
Do not use the same Event Hub for more than
two purposes.

23.3.5. BeyondTrust Privilege Management
Cloud 

23.3.5.1. Ingest logs from BeyondTrust Privilege
Management Cloud 📝 
🟢note

For a Log Format set to Beyondtrust Cloud
ECS, the following fields are automatically set
and not configurable.
Vendor: Beyondtrust
Product: Privilege Management
Compression: Uncompressed

23.3.6.1. Ingest logs and data from Box  23.3.6. Box 

🔒

🟣c…

Requires the Data Collection and Identity
Threat Detection & Response add-ons.

☑️

🟣pr…

Set up an Enterprise Box plan.
Important
To collect Box Shield Alerts, you must purchase
Box Shield and it must be enabled on Box
enterprise.
Set up an Enterprise Box plan.
Create a valid Box account that is assigned to a
role with sufficient permissions for the data you
want to collect. For example, create an account
assigned to an Admin role to...

📝

🟢note

The Fetch Intervals are non-configurable.

📝

🟢note

Collecting Box Shield Alerts requires
implementing Box Shield,

📝

🟢note

There is a current bug with the Groups API from
Box. If you don't configure the Box app with the
proper permissions for managing groups data,
the Groups API from Box won't return an error
message to Cortex Cloud indicating that the
API failed to receive the data, and the Groups
data will not be collected.

📝

🟢note

For any future change that you make to your
Box app, ensure that you send the changes for
approval to the administrator, who will need to
approve them as explained above.

📝

🟢note

Inventory data snapshots are collected every 10
minutes.

23.3.7. Check Point FW1/VPN1 

23.3.8. Cisco ASA firewalls and AnyConnect 

23.3.9. Corelight Zeek 

23.3.10.1. Ingest data from Cribl  23.3.10. Cribl 

☑️

🟣pr…

Ensure that you have the credentials and IDs for
each data source, such as Tenant ID, App ID and
Client secret.

☑️

🟣pr…

Ensure that you have the copies of the Cortex
Cloud authorization token and API URL
obtained in Task 2.

📝

🟢note

We do not support email data collection via
Cribl.

📝

🟢note

Only one Cribl data collector instance can be
configured in Cortex Cloud.

📝

🟢note

When the logs are in CEF format, Cortex Cloud
reads events row by row to look for the Vendor
and Product configured in the logs. When the
values are populated in the event log row,
Cortex Cloud uses these values even if you
specified a value in the Vendor and Product
fields in the Cribl collector settings. However,
when the values are blank in the...

👀

🟠i…

Raw data must be collected by Cribl and
streamed as-is from the passed through source,
because any changes made by Cribl might
affect the way that Cortex Cloud handles the
data.

📝 23.3.10.1.1. Data source UUIDs 

🟢note

Do not use this generic UUID when your data
source is listed in this table.

📝 23.3.11.1. How to onboard Databricks  23.3.11. Databricks 

🟢note

If you turn on this feature, both the cloud and
region will be used for scanning, possibly
incurring cost and requiring adherence to
certain compliance policies.

23.3.12.1. Ingest logs and data from Dropbox  23.3.12. Dropbox 

🔒

🟣c…

Requires the ITDR Module add-on and Data
Collection add-on.

☑️

🟣pr…

Set up an Advanced Dropbox plan.
Create a Dropbox Business admin account with
Security admin permissions, which is required to
authorize Cortex Cloud to access the Dropbox
Business account and generate the OAuth 2.0
access token.

📝

🟢note

The Fetch Interval is non-configurable.

📝

🟢note

When the App Key field is empty, the here link
in the tooltip is disabled. When an incorrect
App Key is entered, clicking the link results in a
404 error.

📝

🟢note

Whenever you change the permissions of the
Dropbox app, we recommend that you
generate a new Access Code for the Dropbox
data collector instance so that the permissions
match the updates.

📝

🟢note

Event data is collected every 60 seconds with a
10 minute lag time.

📝

🟢note

Inventory data snapshots are collected every 10
minutes.

23.3.13. Elasticsearch Filebeat 

📝

🟢note

You can configure collecting container logs
from Google Kubernetes Engine using
Elasticsearch Filebeat with a Custom - Filebeat
based Collector or with a content pack
Integration. For more information, see Google
Kubernetes Engine.

🔒 23.3.13.1. Ingest logs from Elasticsearch Filebeat 

🟣c…

Requires the Data Collection add-on.

23.3.14. Forcepoint DLP 

23.3.15. Fortinet Fortigate 

23.3.16. Google Cloud Platform 

23.3.16.1. Ingest logs and data from a GCP
Pub/Sub 

📝

🟢note

When collecting flow logs, we recommend that
you include GKE annotations in your logs, which
enable you to view the names of the containers
that communicated with each other. GKE
annotations are only included in logs if
appended manually using the custom metadata
configuration in GCP. For more information, see
VPC Flow Logs Overview. In addition, to...

📝

🟢note

For a Log Format set to CEF or LEEF, Cortex
Cloud reads events row by row to look for the
Vendor and Product configured in the logs.
When the values are populated in the event log
row, Cortex Cloud uses these values even if you
specified a value in the Vendor and Product
fields in the GCP data collector settings. Yet,
when the values are blank in the...

📝

🟢note

For a Log Format set to CEF or LEEF, Cortex
Cloud reads events row by row to look for the
Vendor and Product configured in the logs.
When the values are populated in the event log
row, Cortex Cloud uses these values even if you
specified a value in the Vendor and Product
fields in the GCP data collector settings. Yet,
when the values are blank in the...

23.3.17. Google Workspace 

23.3.17.1. Ingest logs and data from Google
Workspace



🔒

🟣c…

Requires the ITDR Module add-on and Data
Collection add-on.

📝

🟢note

For all types of data collected via the Admin
Reports API, except alerts and emails, the log
events are collected with a preset lag time as
reported by Google Workspace. For more
information on these lag times for the different
types of data, see Google Workspace Data
retention and lag times.

📝

🟢note

Spam emails from the compliance email
account, and from all other monitored email
accounts, are not collected.
Any draft emails written in the compliance
email account are collected by the Google
Workspace data collector, and are then
deleted even if the email was never sent.

📝

🟢note

For more information on the entire process, see
Perform Google Workspace Domain-Wide
Delegation of Authority.

📝

🟢note

You don't need to add permissions to the GCP
Project-org service account.

📝

🟢note

When collecting Google Workspace alerts with
other types of data, except emails, you need to
configure a service account in Google with the
applicable permissions to collect events from
the Google Reports API and alerts from the
Alert Center API. If you prefer to use different
service accounts to collect events and alerts
separately, you'll need to...

📝

🟢note

For more information on the Admin Reports API
scopes, see OAuth 2.0 Scopes for Google APIs.

📝

🟢note

For more information on the Gmail API scopes,
see OAuth 2.0 Scopes for Google APIs.

📝

🟢note

For more information about the events
collected from the various Google Reports, see
Google Workspace Reports API Documentation.

23.3.18. Google Kubernetes Engine 

📝

🟢note

It's also possible to use a Custom - Filebeat
based Collector to ingest logs related to file
activity on your endpoints and servers without
using the Cortex XDR agent. For more
information, see Elasticsearch Filebeat.

23.3.18.1. Ingest logs from Google Kubernetes
Engine 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

Cortex Cloud supports logs in single line
format or multiline format. For more
information on handling messages that span
multiple lines of text in Elasticsearch Filebeat,
see Manage Multiline Messages.

23.3.19. HTTP log collector 

23.3.19.1. Set up an HTTP log collector to
receive logs 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

-The Vendor and Product defaults to Auto-
Detect when the Log Format is set to CEF or
LEEF.
-For a Log Format set to CEF or LEEF, Cortex
Cloud reads events row by row to look for the
Vendor and Product configured in the logs.
When the values are populated in the event log
row, Cortex Cloud uses these values even if you
specified a value in the Vendor...

📝

🟢note

Each record cannot exceed 5 MB in size.

23.3.20. Kubernetes 

23.3.20.1. Onboard the Kubernetes Connector 

📝

🟢note

This option is not supported for Fargate.

📝

🟢note

Enabling Realtime Protection installs the agent
on your Kubernetes clusters as a DaemonSet.

📝

🟢note

For Fargate, you must provide the cluster
resource identifier.
The format of the identifier is arn:aws:eks:
<region>:<account-id>:cluster/<cluster-name>.

📝

🟢note

Basic authentication is only supported in
Posture Management. If deploying Realtime
Protection, select None .

📝

🟢note

If you install the latest version but select the N-
1 strategy, this policy will take effect starting
from the next upgrade cycle (it will not
immediately downgrade your current
installation).

23.3.20.2. What's new in Kubernetes Connector? 

📝 23.3.20.3. Supported Kubernetes distributions 

🟢note

In Google Container-Optimized OS release 100
and earlier, where the FANOTIFY EXEC flag is
not supported, the Kernel configuration may be
partial for the user mode agent to properly
function. In such cases, the agent will fallback
to asynchronous mode.
In RHCOS version 4.12 and earlier, the Kernel
configuration may be partial for the user mode
agent to...

23.3.21. Microsoft Azure 

23.3.22.1. Ingest logs from Microsoft Office 365  23.3.22. Microsoft Office 365 

🔒

🟣c…

Requires the ITDR Module add-on and Data
Collection add-on. For viewing alerts and Azure
AD Activity Logs only the Data Collection add-
on is required.

📝

🟢note

Ingesting Microsoft Entra ID (formerly known as
Azure AD) authentication and audit events from
Microsoft Graph API requires a Microsoft Azure
Premium 1 or Premium 2 license. Alternatively, if
the directory type is Azure AD B2C, the sign-in
reports are accessible through the API without
any additional license requirement.
To ingest email logs and data...

📝

🟢note

When auditing is turned off from the default
setting, you need to first turn on auditing for
your organization to collect Microsoft Office
365 audit events from the Management Activity
API. Log duplication of up to 5% in Microsoft
products is considered normal. In some cases,
such as login to a portal using MFA, two log
entries are recorded by design.

📝

🟢note

To address Azure reporting latency, there is a
10-minute latency period for Cortex Cloud to
receive Azure AD logs.

📝

🟢note

For more information, see the Office 365
Management Activity API schema.

📝

🟢note

In FedRAMP environments, Azure sign-in logs
are not supported, due to vendor technical
constraints.

📝

🟢note

Use this option when you don't want to grant
permissions for Azure AD Authentication and
Azure AD Audit.

23.3.23. Microsoft Office 365 (email) 

23.3.23.1. Ingest logs and data from Microsoft
365 

🔒

🟣c…

Email content visibility and licensing: Email
subjects and bodies are stored in an encrypted
format to ensure data privacy. To view this
content or generate alerts for it, an Email
Security module license is required.
Without the license: Sensitive email content
(subject, body, and attachments) remains
encrypted and is not accessible for viewing or...

☑️

🟣pr…

A user account with the Microsoft Azure
Account Administrator role is required to set up
a new Microsoft 365 email collector.
The following Microsoft Graph API permissions
are required:
Mailbox access (read-write)
Read and write mail in all mailboxes
Read contacts in all mailboxes
Read all user mailbox settings
User information, groups,...

📝

🟢note

For other logs from Microsoft Office 365, use
the Office 365 data collector. For more
information, see Ingest logs from Microsoft
Office 365.

☑️ 23.3.24.1. How to onboard Microsoft 365  23.3.24. Microsoft 365 (Posture) 

🟣pr…

You have generated a Globally Unique Identifier
(GUID), also known as a Universally Unique
Identifier (UUID). You will need this ID for the
tenant you want to use for the Microsoft 365
instance.
In order to use Microsoft 365, you must be
registered with Microsoft Azure.

☑️ 23.3.25.1. Ingest logs and data from Okta  23.3.25. Okta 

🟣pr…

Administrator privileges: Your Okta user must
have a role capable of creating API tokens,
such as Read-only Administrator, Super
Administrator, or Organization Administrator.
For more information, see the Okta
Administrators Documentation.

23.3.26.1. Ingest logs and data from OneLogin  23.3.26. OneLogin 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

Event data is collected every 30 seconds.

📝

🟢note

Inventory data snapshots are collected every 10
minutes.

23.3.27. Oracle Cloud Infrastructure 

23.3.28. PingFederate 

23.3.29. PingOne 

23.3.29.1. Ingest authentication logs and data
from PingOne 

🔒

🟣c…

Requires the Data Collection add-on.

🧭

➡️n…

Reporting → Subscriptions → Add Subscription

23.3.30.1. How to onboard Snowflake  23.3.30. Snowflake 

☑️

🟣pr…

In order to use Snowflake, you must be
registered with one of these cloud providers:
Amazon AWS, Microsoft Azure, or Google
Cloud Platform (GCP).
Ensure you have the necessary account
permissions to onboard. It is recommended to
use Account Admin as the role for the
onboarding.

📝

🟢note

The account identifier can be found using the
user information at the bottom left. Hover over
the account you wish to onboard and select
the copy option at the top right. The account
identifier is usually of the format:
(organization).[account]

23.3.31. Windows DHCP via Elasticsearch

Filebeat 

23.3.31.1. Ingest logs from Windows DHCP using
Elasticsearch Filebeat 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

To avoid formatting issues in filebeat.yml, we
recommend that you edit the text file inside the
user interface, instead of copying it and editing
it elsewhere. Validate the syntax of the YML file
before you finish creating the profile.

📝

🟢note

For more information on configuring the
filebeat.yml file, see the Elastic Filebeat
Documentation.

📝

🟢note

To avoid formatting issues in your filebeat.yml,
we recommend that you use the download
example file to make your customizations. Do
not copy and paste the code syntax examples
provided later in this procedure into your file.

📝

🟢note

The tokenizer definition is dependent on the
Windows server version that you are using as
the log format differs.
-For platforms earlier than Windows Server
2008, use "%{id},%{date},%{time},%
{description},%{ipAddress},%{hostName},%
{macAddress}"
-For Windows Server 2008 and 2008 R2, use "%
{id},%{date},%{time},%{description},%
{ipAddress},%{hostName},%{mac...

23.3.32.1. Ingest report data from Workday  23.3.32. Workday 

🔒

🟣c…

Requires the Data Collection and Identity
Threat Detection & Response add-ons.

☑️

🟣pr…

Create an Integration System User that is
designated to access the custom report from
Workday for data collection in Cortex Cloud.
Create an Integration System Security Group
for the Integration System User created in Step
1 for accessing the report. When setting this
group ensure to define the following:
Type of Tenanted Security Group: Select...

📝

🟢note

Any user with permissions to view issues and
cases can view the Workday data.

📝

🟢note

For more information on completing any of the
prerequisite steps, see the Workday
documentation.

📝

🟢note

The Business Object changes depending on the
Data Source selected.

📝

🟢note

Cortex Cloud uses a structured schema when
integrating Workday data. To get the best
Analytics results, specify all the fields marked
with an asterisk from the recommended schema.
workday_user_id*
full_name*
workday_manager_user_id*
manager*
worker_type*
position_title*
department*
private_email_address*
business_email_address*
employment_start_date*
em...

23.3.33. Zscaler Internet Access 

23.3.34. Zscaler Private Access 

23.4. Cloud service provider (CSP) onboarding 

📝 23.4.1. Ingest cloud assets 

🟢note

The cloud accounts being onboarded must be
owned by the customer performing the
onboarding process.

23.4.2. Onboard Amazon Web Services 

☑️

🟣pr…

Ensure you have access to AWS Management
Console.
Ensure you have the Required AWS permissions.

📝

🟢note

Scanning with an outpost may require
additional CSP permissions and may incur
additional CSP costs.

📝

🟢note

When onboarding an AWS organization or
organizational unit (OU), Cortex Cloud creates
IAM resources in every account within that
organization or OU. This occurs even if you
choose to exclude specific accounts from
being scanned. While excluded accounts will
not be scanned and will not appear in the asset
inventory, the IAM resources may still be...

23.4.2.1. Onboard Amazon Web Services with

basic configuration 

🔒

🟣c…

Onboarding Amazon Web Services (AWS) using
the basic configuration is included with Cortex
Cloud NG SIEM, Cortex Cloud Enterprise, and
Cortex Cloud Enterprise+ licenses. For
onboarding AWS with a Cortex Cloud Premium
license, see Onboard Amazon Web Services.

☑️

🟣pr…

Ensure you have access to AWS Management
Console.
Ensure you have the Required AWS permissions.

📝

🟢note

When onboarding an AWS organization or
organizational unit (OU), Cortex Cloud creates
IAM resources in every account within that
organization or OU. This occurs even if you
choose to exclude specific accounts from
being scanned. While excluded accounts will
not be scanned and will not appear in the asset
inventory, the IAM resources may still be...

23.4.2.2. Manually upload template to AWS 

☑️

🟣pr…

Before you begin, ensure you have:
An AWS account
Access to AWS Management Console
Permission to create a stack and its resources in
AWS CloudFormation

📝

🟢note

You must ensure that the KMS key region and
the SNS topic region are the exact same as the
AWS region where you are deploying the
CloudFormation stack.

23.4.2.3. Configure AWS integration instances
and monitor integration instance health 

📝

🟢note

If you have not yet onboarded your cloud
integration, see Ingest cloud assets.

📝

🟢note

Currently, automation permission errors or
missing automation permissions do not affect
the Automation health status. You can view any
permission errors or missing permissions in the
the Permissions health status.

23.4.3. Onboard Google Cloud Platform 

☑️

🟣pr…

Ensure you have cccess to Google Cloud
Console.
Ensure you have an admin user with the
required admin GCP permissions.
Ensure you have the following APIs in the GCP
project you are onboarding:
Cloud Resource Manager API
Identity and Access Management (IAM) API
Cloud Pub/Sub API (if audit logs are enabled)
If you plan on enabling Automation as...

📝

🟢note

Scanning with an outpost may require
additional CSP permissions and may incur
additional CSP costs.

📝

🟢note

If you want to connect to your GCP Workspace,
you must first complete onboarding with the
option disabled. Once the GCP cloud instance
is created, perform the steps detailed in
Connect Google Workspace with your GCP
cloud instance.

23.4.3.1. Onboard Google Cloud Platform with
basic configuration ☑️ 
🟣pr…

Ensure you have cccess to Google Cloud
Console.
Ensure you have an admin user with the
required admin GCP permissions.
Ensure you have the following APIs in the GCP
project you are onboarding:
Cloud Resource Manager API
Identity and Access Management (IAM) API
Cloud Pub/Sub API (if audit logs are enabled)
If you plan on enabling Automation as...

23.4.3.2. Manually upload template to GCP 

☑️

🟣pr…

Before you begin, ensure you have:
A GCP account.
Permission to create the required resources in
Google Cloud Deployment Manager.
Installed Terraform on your local machine. You
can download Terraform from the official
Terraform website and follow the installation
instructions for your operating system.
Installed the GCP gcloud CLI tool.
Reviewed the...

📝

🟢note

The directory you create must be a
subdirectory of the home directory.

👀

🟠i…

You must not delete or move the Terraform files
from this folder. It will prevent you from being
able to edit your cloud instance in the future.

23.4.3.3. Connect Google Workspace with your
GCP cloud instance 

☑️

🟣pr…

Ensure you have the Super Admin role in
Google Workspace.

☑️

🟣pr…

Ensure you have the organization ID of the
Google Workspace you want to connect:
Log in to your Google Admin Console. and
navigate to Account → Account settings →
Profile. Next to Customer ID is your
organization ID.
Ensure you have the organization ID of the
Google Workspace you want to connect:
Log in to your Google Admin Console. and
navigate...

23.4.3.4. Configure GCP integration instances
and monitor integration instance health 

📝

🟢note

If you have not yet onboarded your cloud
integration, see Ingest cloud assets.

📝

🟢note

Currently, automation permission errors or
missing automation permissions do not affect
the Automation health status. You can view any
permission errors or missing permissions in the
the Permissions health status.

23.4.3.5. Monitor GCP resources inside service

perimeters 📝 
🟢note

Each GCP cloud instance is assigned a scope
within GCP. If the scope, whether it be
organization, folder, or project, includes any
projects with a service perimeter, this
procedure must be performed for that cloud
instance to authorize Cortex Cloud to scan the
resources in the perimeter.

23.4.4. Onboard Microsoft Azure 

☑️

🟣pr…

Ensure you have a Microsoft Azure subscription.
Ensure you have the admin permissions required
to onboard Microsoft Azure or the built-in
Security Administrator role.
Obtain the tenant ID and subscription ID. You
can view these in the Microsoft Azure Portal in
Management groups.

23.4.4.1. Onboard Microsoft Azure with basic

configuration 

🔒

🟣c…

Onboarding Microsoft Azure using the basic
configuration is included with Cortex Cloud NG
SIEM, Cortex Cloud Enterprise, and Cortex
Cloud Enterprise+ licenses. For onboarding
Microsoft Azure with a Cortex Cloud Premium
license, see Onboard Microsoft Azure.

☑️

🟣pr…

Ensure you have a Microsoft Azure subscription.
Ensure you have the admin permissions required
to onboard Microsoft Azure or the built-in
Security Administrator role.
Obtain the tenant ID and subscription ID. You
can view these in the Microsoft Azure Portal in
Management groups.

23.4.4.2. Finalize Microsoft Azure onboarding by
executing the authentication template 

☑️

🟣pr…

Before you begin, ensure you have:
An Azure subscription.
A user with the required permissions for the
relevant scope (subscription, management
group, tenant). We recommend you create a
dedicated role.
Tenant ID and subscription ID. You can view
these in Microsoft Azure Portal in Management
groups.

Installed Terraform on your local machine. You
can...

☑️

🟣pr…

Before you begin, ensure you have:
An Azure subscription.
A user with the required permissions for the
relevant scope (subscription, management
group, tenant). We recommend you create a
dedicated role.
Tenant ID and subscription ID. You can view
these in Microsoft Azure Portal in Management
groups.
Installed the Azure CLI tool.
Authorization to create...

☑️

🟣pr…

Before you begin, ensure you have:
An Azure subscription.
A user with the required permissions for the
relevant scope (subscription, management
group, tenant). We recommend you create a
dedicated role.
Tenant ID and subscription ID. You can view
these in Microsoft Azure Portal in Management
groups.
Installed the Azure CLI tool.
Authorization to create...

👀

🟠i…

Do not delete or move the Terraform files from
this folder. It will prevent you from being able
to edit your cloud instance in the future.

23.4.4.3. Configure Azure integration instances
and monitor integration instance health 

📝

🟢note

If you have not yet onboarded your cloud
integration, see Ingest cloud assets.

📝

🟢note

Currently, automation permission errors or
missing automation permissions do not affect
the Automation health status. You can view any
permission errors or missing permissions in the
the Permissions health status.

23.4.5. Onboard Oracle Cloud Infrastructure 

☑️

🟣pr…

Ensure you have access to the Oracle Cloud
Infrastructure console
Permissions for all of the following are required:
Creation of identity groups (for more
information, refer to Managing Groups)
Policies (for more information, refer to How
Policies Work)
Tag namespaces in the root compartment (for
more information, refer to Tags and Tag
Namespace...

23.4.5.1. Onboard Oracle Cloud Infrastructure

with basic configuration 

🔒

🟣c…

Onboarding Oracle Cloud Infrastructure (OCI)
using the basic configuration is included with
Cortex Cloud NG SIEM, Cortex Cloud
Enterprise, and Cortex Cloud Enterprise+
licenses. For onboarding OCI with a Cortex
Cloud Premium license, see Onboard Oracle
Cloud Infrastructure.

☑️

🟣pr…

Ensure you have access to the Oracle Cloud
Infrastructure console
Permissions for all of the following are required:
Creation of identity groups (for more
information, refer to Managing Groups)
Policies (for more information, refer to How
Policies Work)
Tag namespaces in the root compartment (for
more information, refer to Tags and Tag
Namespace...

☑️ 23.4.5.2. Manually upload template to OCI 

🟣pr…

Before you begin, ensure you have:
An Oracle Cloud Infrastructure account and
the tenancy OCID.
Permission to deploy a custom template and
create its resources in OCI.
Installed Terraform on your local machine. You
can download Terraform from the official
Terraform website and follow the installation
instructions for your operating system.
Installed...

23.4.5.3. Configure the OCI connector for log
collection 

23.4.6. Manually connect a cloud instance 

👀 23.4.7. Edit your onboarded CSP configuration 

🟠i…

When using Terraform authentication
templates, you must execute the updated
Terraform template from the same folder where
the original Terraform template was executed.

23.4.8. Outposts 

👀

🟠i…

Outpost scan is an alternative to the
recommended standard cloud scan. Cloud scan
is recommended because it is fully managed by
Palo Alto Networks and incurs no compute
costs for your organization. Outpost scan is an
advanced deployment model reserved for
specific data residency or architectural
requirements.

23.4.8.1. Outpost fundamentals and planning 

📝

🟢note

Before you create your outpost, verify that
your internet connection is active. An active
internet connection is necessary for the
notification to be sent to Cortex Cloud to
create the new outpost.

👀

🟠i…

While outposts provide maximum control over
the scanning environment, cloud scan mode is
the recommended default for most
organizations.

🧭

➡️n…

Before onboarding your Cortex Cloud with the
cloud service provider (CSP) onboarding
wizard, create an outpost by navigating to
Settings → Data Sources & Integrations →
Outposts

Settings → Data Sources & Integrations → Add
New

View and manage existing outposts by
navigating to Settings → Data Sources &
Integrations → Outposts

23.4.8.2. Create an outpost 

☑️

🟣pr…

Before you begin, ensure you have:
An AWS account
Permission to create a stack and its resources in
AWS
Installed Terraform on your local machine. You
can download Terraform from the official
Terraform website and follow the installation
instructions for your operating system.
Installed the AWS CLI tool and configured your
profile with the aws...

☑️

🟣pr…

Before you begin, ensure you have:
A GCP account
Permission to create the required resources in
Google Cloud Deployment Manager
Installed Terraform on your local machine. You
can download Terraform from the official
Terraform website and follow the installation
instructions for your operating system.
Installed the GCP gcloud CLI tool

☑️

🟣pr…

Before you begin, ensure you have:
An active Azure subscription.
Installed the Azure CLI tool.
Permission to deploy a custom template and
create its resources in Microsoft Azure
("Owner" or "Contributor" on the designated
outpost subscription scope, and Active
Directory "Cloud Application Administrator" or
"Application Administrator" privileged...

📝

🟢note

Verify that your internet connection is active.
An active internet connection is necessary for
notifications to be sent to Cortex Cloud for
creating the new outpost. If you are unable to
establish an internet connection, contact
customer support for a manual workaround.

📝

🟢note

When creating an outpost for a specific Azure
subscription, the outpost account must be in
the same Azure organization as the monitored
subscriptions.

👀

🟠i…

While outposts provide maximum control over
the scanning environment, cloud scan mode is
the recommended default for most
organizations. For details, see When to choose
outpost scan.

🧭

➡️n…

Start the outpost creation wizard by navigating
to Settings → Data Sources & Integrations →
Outposts and clicking New Outpost

A new outpost is created in pending status and
can be viewed in the Outpost page at Settings
→ Data Sources & Integrations → Outposts

23.4.9. Introduction to Terraform for Cloud

service provider (CSP) onboarding 

👀

🟠i…

Always run these commands in the same folder
where the original .tf files and .terraform folder
live—this is where the state is stored.

👀

🟠i…

This .tfstate state file is critical because it is
needed by the terraform destroy command to
clean up created resources. It is critical that
you never delete this file.

23.4.10. Container Registry Scanning 

23.4.10.1. Overview of container registry
scanning 

23.4.10.1.1. Registry Components 

23.4.10.1.2. How Container Registry Scanning
Works



23.4.10.2. Configure registry scanning for cloud
accounts ☑️ 

🟣pr…

Ensure that you have performed the all steps
till Additional Security Capabilities as listed in
the onboarding wizard for the required CSP:
Onboard Amazon Web ServicesOnboard
Amazon Web Services
Onboard Google Cloud PlatformOnboard
Google Cloud Platform
Onboard Microsoft AzureOnboard Microsoft
Azure
Onboard Oracle Cloud InfrastructureOnboard
Oracle...

23.4.10.3. Modify the container registry scanning
scope



23.4.10.4. Scan re-evaluation process 

23.4.10.5. Connect Docker Hub registry 

☑️

🟣pr…

Ensure an Outpost is connected to your tenant.

☑️

🟣pr…

Set up and configure Broker VM
Configure High Availability Cluster

📝

🟢note

If you choose Azure as the Cloud Provider, you
must also select the Tenant Id. The Tenant Id is
required to approve Cortex as an enterprise
application in your Azure tenant.

📝

🟢note

If you choose Azure as the cloud provider, only
Outposts associated with the selected tenant
ID are displayed.

📝

🟢note

The list of Broker VMs displays only VMs that
support registry scanning.
The list of high-availability Clusters displays
only clusters that contain at least one VM
supporting registry scanning.
The registry scanning status for each VM
appears in brackets if it was previously
activated for that specific VM.

📝 23.4.10.5.1. Manage a Docker Hub connector 

🟢note

If you selected Scan with Broker VM mode, you
can't change to a different scan mode (such as
Cloud Discovery or Scan with Outpost) when
you edit the instance.
When editing an instance configured for Scan
with Broker VM, you must re-enter your
authentication credentials, including Username,
Password, and CA certificate.

23.4.10.6. Connect Docker V2 compliant
container registry 

☑️

🟣pr…

Ensure an Outpost is connected to your tenant.
Outposts

☑️

🟣pr…

Ensure one of the following is configured:
Set up and configure Broker VM.
Configure High Availability Cluster.

📝

🟢note

If you choose Azure as the Cloud Provider, you
must also select the Tenant Id. The Tenant Id is
required to approve Cortex as an enterprise
application in your Azure tenant.

📝

🟢note

If you choose Azure as the cloud provider, only
Outposts associated with the selected tenant
ID are displayed.

📝

🟢note

The list of Broker VMs displays only VMs that
support registry scanning.
The list of high-availability Clusters displays
only clusters that contain at least one VM
supporting registry scanning.
The registry scanning status for each VM
appears in brackets if it was previously
activated for that specific VM.

📝 23.4.10.6.1. Manage a Docker V2 connector 

🟢note

If you selected Scan with Broker VM mode, you
can't change to a different scan mode (such as
Cloud Discovery or Scan with Outpost) when
you edit the instance.
When editing an instance configured for Scan
with Broker VM, you must re-enter your
authentication credentials, including Username,
Password, and CA certificate.

23.4.10.7. Connect GitLab container registry 

☑️

🟣pr…

Ensure an Outpost is connected to your tenant.

☑️

🟣pr…

Set up and configure Broker VM
Configure High Availability Cluster

📝

🟢note

When both the group ID and project ID are
provided, the system retrieves container images
from all projects within the specified group as
well as from the specified project.
If neither the group ID nor the project ID is
provided, the system retrieves container images
from all registries (across all groups and
projects) accessible to the authenticated...

📝

🟢note

When both the group ID and project ID are
provided, the system retrieves container images
from all projects within the specified group as
well as from the specified project.
If neither the group ID nor the project ID is
provided, the system retrieves container images
from all registries (across all groups and
projects) accessible to the authenticated...

📝

🟢note

If you choose Azure as the Cloud Provider, you
must also select the Tenant Id. The Tenant Id is
required to approve Cortex as an enterprise
application in your Azure tenant.

📝

🟢note

If you choose Azure as the cloud provider, only
Outposts associated with the selected tenant
ID are displayed.

📝

🟢note

When both the group ID and project ID are
provided, the system retrieves container images
from all projects within the specified group as
well as from the specified project.
If neither the group ID nor the project ID is
provided, the system retrieves container images
from all registries (across all groups and
projects) accessible to the authenticated...

📝

🟢note

When both the group ID and project ID are
provided, the system retrieves container images
from all projects within the specified group as
well as from the specified project.
If neither the group ID nor the project ID is
provided, the system retrieves container images
from all registries (across all groups and
projects) accessible to the authenticated...

📝

🟢note

The list of Broker VMs displays only VMs that
support registry scanning.
The list of high-availability Clusters displays
only clusters that contain at least one VM
supporting registry scanning.
The registry scanning status for each VM
appears in brackets if it was previously
activated for that specific VM.

📝

🟢note

When both the group ID and project ID are
provided, the system retrieves container images
from all projects within the specified group as
well as from the specified project.
If neither the group ID nor the project ID is
provided, the system retrieves container images
from all registries (across all groups and
projects) accessible to the authenticated...

📝

🟢note

When both the group ID and project ID are
provided, the system retrieves container images
from all projects within the specified group as
well as from the specified project.
If neither the group ID nor the project ID is
provided, the system retrieves container images
from all registries (across all groups and
projects) accessible to the authenticated...

23.4.10.7.1. Manage a Gitlab Container Registry
connector 📝 
🟢note

If you selected Scan with Broker VM mode, you
can't change to a different scan mode (such as
Cloud Discovery or Scan with Outpost) when
you edit the instance.
When editing an instance configured for Scan
with Broker VM, you must re-enter your
authentication credentials, including Username,
Password, and CA certificate.

23.4.10.8. Connect Harbor registry 

☑️

🟣pr…

Ensure an Outpost is connected to your tenant.

☑️

🟣pr…

Ensure one of the following is configured:
Set up and configure Broker VM.
Configure High Availability Cluster.

📝

🟢note

If you choose Azure as the Cloud Provider, you
must also select the Tenant Id. The Tenant Id is
required to approve Cortex as an enterprise
application in your Azure tenant.

📝

🟢note

If you choose Azure as the cloud provider, only
Outposts associated with the selected tenant
ID are displayed.

📝

🟢note

The list of Broker VMs displays only VMs that
support registry scanning.
The list of high-availability Clusters displays
only clusters that contain at least one VM
supporting registry scanning.
The registry scanning status for each VM
appears in brackets if it was previously
activated for that specific VM.

📝 23.4.10.8.1. Manage a Harbor connector 

🟢note

If you selected Scan with Broker VM mode, you
can't change to a different scan mode (such as
Cloud Discovery or Scan with Outpost) when
you edit the instance.
When editing an instance configured for Scan
with Broker VM, you must re-enter your
authentication credentials, including Username,
Password, and CA certificate.

23.4.10.9. Connect JFrog container registry 

☑️

🟣pr…

Ensure an Outpost is connected to your tenant.

☑️

🟣pr…

Set up and configure Broker VM
Configure High Availability Cluster

📝

🟢note

If you choose Azure as the Cloud Provider, you
must also select the Tenant Id. The Tenant Id is
required to approve Cortex as an enterprise
application in your Azure tenant.

📝

🟢note

If you choose Azure as the cloud provider, only
Outposts associated with the selected tenant
ID are displayed.

📝

🟢note

The list of Broker VMs displays only VMs that
support registry scanning.
The list of high-availability Clusters displays
only clusters that contain at least one VM
supporting registry scanning.
The registry scanning status for each VM
appears in brackets if it was previously
activated for that specific VM.

📝 23.4.10.9.1. Manage a JFrog connector 

🟢note

If you selected Scan with Broker VM mode, you
can't change to a different scan mode (such as
Cloud Discovery or Scan with Outpost) when
you edit the instance.
When editing an instance configured for Scan
with Broker VM, you must re-enter your
authentication credentials, including Username,
Password, and CA certificate.

23.4.10.10. Connect Sonatype Nexus registry 

☑️

🟣pr…

Ensure an Outpost is connected to your tenant.
Outposts

☑️

🟣pr…

Ensure one of the following is configured:
Set up and configure Broker VM.
Configure High Availability Cluster.

📝

🟢note

If you are using a CA certificate, enter the
server IP address instead of the registry url.

📝

🟢note

If you choose Azure as the Cloud Provider, you
must also select the Tenant Id. The Tenant Id is
required to approve Cortex as an enterprise
application in your Azure tenant.

📝

🟢note

If you choose Azure as the cloud provider, only
Outposts associated with the selected tenant
ID are displayed.

📝

🟢note

If you are using a CA certificate, enter the
server IP address instead of the registry URL.

📝

🟢note

The list of Broker VMs displays only VMs that
support registry scanning.
The list of high-availability Clusters displays
only clusters that contain at least one VM
supporting registry scanning.
The registry scanning status for each VM
appears in brackets if it was previously
activated for that specific VM.

📝

🟢note

If you are using a CA certificate, enter the
server IP address instead of the registry URL.

📝 23.4.10.10.1. Manage a Sonatype connector 

🟢note

If you had selected Scan with Broker VM mode,
you can't change to a different scan mode
(such as Cloud Discovery or Scan with
Outpost) when you edit the instance.
When editing an instance configured for Scan
with Broker VM, you must re-enter your
authentication credentials, including Username,
Password, and CA certificate.

23.4.11. Cloud service provider permissions 

23.4.11.1. Amazon Web Services provider
permissions 

23.4.11.2. Google Cloud Platform provider
permissions 

23.4.11.3. Microsoft Azure provider permissions 

23.4.11.4. Oracle Cloud Infrastructure provider
permissions 

23.5. Generic on-premise data collectors 

23.5.1. Broker VM data collector applets 

23.5.1.1. Activate Apache Kafka Collector 

🔒

🟣c…

Requires the Data Collection add-on.
This data source is only available in your tenant
if the tenant was activated before October 1,
2025 with an active Data Collection add-on.

☑️

🟣pr…

Apache Kafka version 2.5.1 and above.
Kafka cluster set up on premises, from which
the data will be ingested.
Privileges to manage Broker Service
configuration, such as Instance Administrator
privileges.
Create a user in the Kafka cluster with the
necessary permissions and the following
authentication details:
Broker Certificate and Private Key for...

📝

🟢note

For CEF and LEEF logs, Cortex Cloud takes the
vendor and product names from the log itself,
regardless of what you configure on this page.

23.5.1.2. Activate CSV Collector 

🔒

🟣c…

Requires the Data Collection add-on.
This data source is only available in your tenant
if the tenant was activated before October 1,
2025 with an active Data Collection add-on.

☑️

🟣pr…

Set up and configure Broker VM.
Ensure that you share the applicable CSV files.
Know the complete file path for the Windows
directory.

📝

🟢note

When you implement a wildcard file search,
ensure that the CSV files share the same
columns and header rows as all other logs that
are collected from the CSV files to create a
single dataset.

📝

🟢note

The CSV Collector checks for new CSV files
every 10 minutes.

23.5.1.3. Activate Database Collector 

🔒

🟣c…

Requires the Data Collection add-on.
This data source is only available in your tenant
if the tenant was activated before October 1,
2025 with an active Data Collection add-on.

☑️

🟣pr…

Set up and configure Broker VM

📝

🟢note

The DB Collector does not support passwords
containing semicolons (;).

📝

🟢note

The reference data ingested using the DB
Collector is counted towards license utilization.

👀

🟠i…

When configuring the DB collector to work
with an Oracle database and enabling this
option, ensure the following steps are
completed for a successful connection:
Ensure you have the certificate of the Database
server.

Upload certificate to Broker CA Trust Store
(Conditional):
This step is only required if the Oracle server's
certificate is self-signed...

👀

🟠i…

The Database Collector applet supports a
single row size of up to 65 KB. Ensure your
source data does not exceed this limit per
record to avoid ingestion errors.

23.5.1.4. Activate DSPM Fileshare 

☑️

🟣pr…

Set up and configure Broker VM
Know the complete path to the files and
folders that you want Cortex Cloud to monitor.
Necessary user permissions to access the
network shares. For the SMB connection type,
you need the user name and password.

📝

🟢note

The applet list displays only the applets for
which you have permissions.

📝

🟢note

By default, all configured connections are
saved.

📝

🟢note

For details regarding the connection fields, see
the table above under File Share Connection.

📝

🟢note

You can add multiple connections under a
single instance of the DSPM Fileshare applet by
returning to the File Share Connection screen
and clicking Add Connection. Each new
connection can be of either the NFS or SMB
connection type.

23.5.1.5. Activate Files and Folders Collector 

🔒

🟣c…

Requires the Data Collection add-on.
This data source is only available in your tenant
if the tenant was activated before October 1,
2025 with an active Data Collection add-on.

☑️

🟣pr…

Set up and configure Broker VM.
Know the complete path to the files and
folders that you want Cortex Cloud to monitor.
Ensure that the user permissions for the
network share include the ability to rename and
delete files in the folder that you want to
configure collection.

📝

🟢note

Cortex Cloud only supports ingestion of files
encoded in UTF-8 format.

📝

🟢note

When using the Linux file share, including the
Linux share with NFS, a Username and Password
are not required, so these fields are grayed out
in the screen.

📝

🟢note

In Batch mode, the Files and Folders Collector
supports collecting logs from a network share
for a maximum file size of 500 MB.

📝

🟢note

When uploading JSON files, Cortex Cloud only
parses the first level of nesting and only
supports single line JSON format, such that
every new line means a separate entry.

📝

🟢note

Use this option only in cases where your files
contain some sort of "header" lines, such as a
general description, an introduction, a
disclaimer, or similar, and you want to skip
ingesting them. The Lines to Skip are not part of
the file format. For example, in CSV files, there
is no need to skip lines.

📝

🟢note

When the storing method is Replace, the
maximum size for the total data to be imported
into a lookup dataset is 30 MB each time the
data is fetched.
The inventory data ingested using the Files and
Folders collector is counted towards license
utilization.
When you use a JOINT function with a lookup
table in a query or correlation rule, make sure
you...

📝

🟢note

You can't specify a file name that's the same as
a system file name.
The name of a dataset created from a tsv file
must always include the extension. If the
original file name is mrkdptusrsnov23.tsv, you
can name save the dataset with the name
marketing_dept_users_Nov_2023.tsv.

📝

🟢note

The Vendor and Product defaults to Auto-
Detect when the Log Format is set to CEF or
LEEF.

23.5.1.6. Activate FTP Collector 

🔒

🟣c…

Requires the Data Collection add-on.
This data source is only available in your tenant
if the tenant was activated before October 1,
2025 with an active Data Collection add-on.

☑️

🟣pr…

Set up and configure Broker VM.
Ensure that the user permissions for the FTP,
SFTP, or FTPS include the ability to rename and
delete files in the folder that you want to
configure collection.
When setting up an FTPS Collector with a
server using a Self-signed certificate, you must
upload the certificate first to the Broker VM as
a Trusted CA...

📝

🟢note

When configuring an SFTP connection, Cortex
Cloud expects the private key to be in the RSA
format that is included in the -----BEGIN RSA
PRIVATE KEY----- tag. Cortex Cloud does not
support providing the private key in the
OpenSSH format from the -----BEGIN
OPENSSH PRIVATE KEY----- tag.
When using ssh-keygen using a Mac, you get
the OpenSSH format by...

📝

🟢note

When uploading JSON files, Cortex Cloud only
parses the first level of nesting and only
supports single line JSON format, such that
every new line means a separate entry.

📝

🟢note

Use this option only in cases where your files
contain some sort of "header" lines, such as a
general description, an introduction, a
disclaimer, or similar, and you want to skip
ingesting them. The Lines to Skip are not part of
the file format. For example, in CSV files, there
is no need to skip lines.

📝

🟢note

The Vendor and Product defaults to Auto-
Detect when the Log Format is set to CEF or
LEEF.

23.5.1.7. Activate Local Agent Settings 

📝

🟢note

When routing traffic through a Broker VM
proxy that sits behind a firewall, you may
experience intermittent agent disconnections if
the firewall is configured to drop challenge
ACK reset (RST) packets. For environments
using a Palo Alto Networks Next-Generation
Firewall (NGFW), see this Knowledge Base
article for details on managing this behavior
via...

📝

🟢note

This is assuming a standard hardware setup
with 2vCPU 8 GB memory.

📝

🟢note

Broker VMs can act as as a proxy for routing
XDR Collector traffic to the Cortex Cloud
tenant. The Broker VM does not cache XDR
Collector installers.

📝

🟢note

When you install your XDR agents, you need to
configure the IP address of the Broker VM and
a port number during the installation. You can
use the default 8888 port or set a custom port.
You are not permitted to configure port
numbers between 0-1024 and 63000-65000, or
port numbers 4369, 5671, 5672, 5986, 6379,
8000, 9100, 15672, 25672. Additionally,...

👀

🟠i…

A FQDN must be configured for Agent Installer
and Content Caching to function properly.

👀

🟠i…

You can only enable Agent Installer and
Content Caching, when in the Broker VM
Configuration, you've uploaded your signed
SSL Server Certificate and key and set the
FQDN. For more information, see the Agent
Installer and Content Caching requirements
explained above.

23.5.1.8. Activate NetFlow Collector 

🔒

🟣c…

Requires the Data Collection add-on.

☑️

🟣pr…

Set up and configure Broker VM

📝

🟢note

Since multiple network devices can send data
to a single NetFlow Collector, we recommend
that you configure a maximum of 50 NetFlow
Collectors per Broker VM applet, with a
maximum aggregated rate of approximately
50K flows per second (FPS) to maintain system
performance.

📝

🟢note

Since Cortex Cloud reserves some port
numbers, it is best to select a port number that
is not in the range of 0-1024 (except for 514), in
the range of 63000-65000 or has one of the
following values: 4369, 5671, 5672, 5986, 6379,
8000, 8888, 9100, 15672, or 28672.

📝

🟢note

The options available change, according to the
set Source Network value.

23.5.1.9. Activate Network Mapper 

☑️

🟣pr…

After you have configured and registered your
Broker VM, you can choose to activate the
Network Mapper application.

📝

🟢note

Each IP address range can receive multiple scan
requests based on it's availability.

📝

🟢note

IP address ranges are displayed according to
what you defined as your Network Parameters.

23.5.1.10. Activate Registry Scanner 

☑️

🟣pr…

To initialize registry scanning on your Broker
VM, you must first add the necessary data
connectors. For details, see:
Connect Docker Hub registry
Connect Docker V2 compliant container
registry
Connect GitLab container registry
Connect Harbor registry
Connect JFrog container registry
Connect Sonatype Nexus registry
To initialize registry...

📝

🟢note

You cannot activate the Registry Scanner
directly on a new or existing Broker VM. You
can only activate or deactivate existing
Registry Scanner applets. To activate or
deactivate existing applets, see Step 4 under
Verify Registry Scanner connection section.

23.5.1.11. Syslog Collector applet 

23.5.1.11.1. Activate Syslog Collector 

🔒

🟣c…

Requires the Data Collection add-on.

☑️

🟣pr…

Set up and configure Broker VM

📝

🟢note

Once configured, you cannot change the
Port/PROTOCOL. If you don't want to use a
data source, ensure to remove the data source
from the list as explained in Task 5.

📝

🟢note

The Vendor and Product defaults to Auto-
Detect when the Log Format is set to CEF or
LEEF.
For a Log Format set to CEF or LEEF, Cortex
Cloud reads events row by row to look for the
Vendor and Product configured in the logs.
When the values are populated in the event log
row, Cortex Cloud uses these values even if you
specified a value in the Vendor and...

📝

🟢note

The server certificate and private key pair is
expected in a PEM format.
Cortex Cloud will notify you when your
certificates are about to expire.

📝

🟢note

Because some port numbers are reserved by
Cortex Cloud , you must choose a port number
that is not:
In the range of 0-1024 (except for 514)
In the range of 63000-65000
Values of 4052, 4369, 5671, 5672, 5986, 6379,
8000, 8888, 9100, 15672, or 28672

📝

🟢note

CORELIGHT is not available for a UDP protocol.

🔒 23.5.1.11.2. Ingest logs from a Syslog receiver 

🟣c…

Requires the Data Collection add-on.

23.5.1.11.3. Check Point FW1/VPN1 

23.5.1.11.3.1. Ingest logs from Check Point
firewalls 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

Logs with sessionid = 0 are dropped.
Destination Port data is available only in the
raw logs.

23.5.1.11.4. Cisco ASA firewalls and AnyConnect 

23.5.1.11.4.1. Ingest logs from Cisco ASA firewalls
and AnyConnect 🔒 
🟣c…

Requires the Data Collection add-on.

🔒 23.5.1.11.5.1. Ingest logs from Corelight Zeek  23.5.1.11.5. Corelight Zeek 

🟣c…

Requires the Data Collection add-on.

🔒 23.5.1.11.6.1. Ingest logs from Forcepoint DLP  23.5.1.11.6. Forcepoint DLP 

🟣c…

Requires the Data Collection add-on.

23.5.1.11.7. Fortinet Fortigate 

23.5.1.11.7.1. Ingest logs from Fortinet Fortigate
firewalls 🔒 
🟣c…

Requires the Data Collection add-on.

23.5.1.11.8. Next-Generation Firewall 

23.5.1.11.8.1. Ingest Next-Generation Firewall logs
using the Syslog collector 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

In the following procedure, general information
is provided for NGFW and Panorama. For
detailed instructions, consult the
documentation for your specific devices and
Panorama version, to ensure that you have
configured log forwarding correctly for all the
log types that you would like to forward to
Cortex Cloud. The following steps only cover...

📝

🟢note

To avoid the possible effects of line formatting,
do not copy/paste the message formats
directly into the PAN-OS web interface. Instead,
paste into a text editor, remove any carriage
return or line feed characters, and then copy
and paste into the web interface.

📝

🟢note

From version 10.0 and later, the log format
documented for log types (Traffic, Threat, and
URL) exceeds the maximum supported 2048
characters in the Custom Log Format tab on
the firewall and Panorama. Select the CEF keys
and values to limit the number of characters to
2048, as per your requirements.

23.5.1.11.9. PingFederate 

23.5.1.11.9.1. Ingest authentication logs from
PingFederate 🔒 
🟣c…

Requires the Data Collection add-on.

23.5.1.11.10. Zscaler Internet Access 

23.5.1.11.10.1. Ingest logs from Zscaler Internet
Access 🔒 
🟣c…

Requires the Data Collection add-on.

23.5.1.11.11. Zscaler Private Access 

23.5.1.11.11.1. Ingest logs from Zscaler Private
Access



🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

For more information on configuring the
parameters on the screen, see the Zscaler
Private Access (ZPA) documentation for
Configuring a Log Receiver.

📝

🟢note

You can only configure a ZPA log receiver to
collect one type of log with your Syslog
Collector in Cortex Cloud. To configure more
that one log type, you'll need to add another
log receiver.

23.5.1.12. Activate Transporter 

☑️

🟣pr…

Permissions: To configure and manage
Transporter applet settings, you must have
permissions to manage Broker Service
configurations (such as an Instance
Administrator)
Set up and configure Broker VMSet up and
configure Broker VM
Confirm that your Broker is v 28 or above
Whitelist IP addresses to enable access to
Cortex Cloud resources. The IP addresses...

⛔

🔴w…

The Transporter applet is not supported for
FedRAMP customers.

23.5.1.13. Activate Windows Event Collector 

🔒

🟣c…

Requires the Data Collection add-on.

☑️

🟣pr…

Set up and configure Broker VM
Broker VM version 8.0 and later
You have knowledge of Windows Active
Directory and Domain Controllers.
You must configure different settings related to
the FQDN where the instructions differ
depending on whether you are configuring a
standalone Broker VM or High Availability (HA)
cluster.
Standalone broker
A...

📝

🟢note

You can also activate the Windows Event
Collector on Windows Core. For more
information, see Activate Windows Event
Collector on Windows Core.

📝

🟢note

By default, Cortex Cloud collects Palo Alto
Networks predefined Security events that are
used by the Cortex Cloud detectors. Removing
the Security collector interferes with the
Cortex Cloud detection functionality. Restore
to Default to reinstate the Security event
collection.

📝

🟢note

You must install the WEF certificate on every
Windows Server, whether DC or not, for the
WEFs that are supposed to forward logs to the
Windows Event Collector applet on the Broker
VM.

📝

🟢note

You must install the WEF certificate on every
Windows Server, whether DC or not, for the
WEFs that are supposed to forward logs to the
Windows Event Collector applet on the Broker
VM.

📝

🟢note

This step overrides any local policy settings.

📝

🟢note

You must type out the name, do not select the
name from the browse button.

📝

🟢note

If Windows Firewall is enabled on your event
forwarders, you will have to define an
outbound rule to enable the WEF to reach port
5986 on the WEC.

23.5.1.13.1. Activate Windows Event Collector on
Windows Core



🔒

🟣c…

Requires the Data Collection add-on.

☑️

🟣pr…

Set up and configure Broker VM
Broker VM version 8.0 and later
You have knowledge of Windows Active
Directory and Domain Controllers.
You must configure different settings related to
the FQDN where the instructions differ
depending on whether you are configuring a
standalone Broker VM or High Availability (HA)
cluster.
Standalone broker
A...

📝

🟢note

By default, Cortex Cloud collects Palo Alto
Networks predefined Security events that are
used by the Cortex Cloud detectors. Removing
the Security collector interferes with the
Cortex Cloud detection functionality. Restore
to Default to reinstate the Security event
collection.

📝

🟢note

You must install the WEF certificate on every
Windows Server, whether DC or not, for the
WEFs that are supposed to forward logs to the
Windows Event Collector applet on the Broker
VM.

📝

🟢note

This step overrides any local policy settings.

📝

🟢note

You must type out the name, do not select the
name from the browse button.

📝

🟢note

If Windows Firewall is enabled on your event
forwarders, you will have to define an
outbound rule to enable the WEF to reach port
5986 on the WEC.

23.5.1.13.2. Renew WEC certificates 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

You must install the WEF certificate on every
Windows Server, whether DC or not, for the
WEFs that are supposed to forward logs to the
Windows Event Collector applet on the Broker
VM.

📝

🟢note

You can see more than one
ca.wec.paloaltonetworks.com and
forwarder.wec.paloaltonetworks.com file from a
previous installation in the directory, so select
the file with the most extended Expiration Date.
You can verify that you are using the correct
certificate:
To verify the client certificate in the Personal →
Certificates folder is related to the...

📝

🟢note

Only perform this step under the following
conditions:
You have completed the WEF certification
renewal process for ALL clients in your
environment. Otherwise, events from the WEFs
that you did not install the new client
certificate will not be collected by the WEC.
You are approaching the WEC server CA
certificate expiration date, which is 2 years...

📝

🟢note

If this query does not display results with a
timestamp from after the renewal process, it
could indicate that the renewal process is not
complete, so wait a few minutes before running
another query. If you are still having a problem,
contact Technical Support.

👀

🟠i…

After you receive a notification for renewing
your WEC CA certificate, we recommend that
you do not add any new WEF clients until the
WEC certification renewal process is complete.
Events from these WEF clients that are added
afterwards will not be collected by the server
until the WEC certificates are renewed.

23.5.2. XDR Collectors 

📝

🟢note

Ingestion of log events larger than 5 MB is not
supported.

🔒 23.5.2.1. XDR Collector audit logs 

🟣c…

Requires the Data Collection add-on.

23.5.2.2. XDR Collector machine requirements
and supported operating systems 🔒 
🟣c…

Requires the Data Collection add-on.

23.5.2.3. Resources required to enable access to
XDR Collectors



🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

Some of the IP addresses required for access
are registered in the United States. As a result,
some GeoIP databases do not correctly
pinpoint the location in which IP addresses are
used. All customer data is stored in your
deployment region, regardless of the IP address
registration and restricts data transmission
through any infrastructure to that...

📝

🟢note

Throughout this topic, refers to the chosen
subdomain of your Cortex Cloud tenant and
<region> is the region in which your Strata
Logging Service is deployed.

23.5.2.4. Manage XDR Collectors 

🔒

🟣c…

Requires the Data Collection add-on.

23.5.2.4.1. XDR Collectors installation resource

for Windows and Linux 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

In some circumstances, such as after an XDR
Collectors upgrade, the configured settings in
the XML configuration file can be erased. Yet,
this won't affect the saved settings in the XDR
Collectors database.

23.5.2.4.2. Create an XDR Collector installation
package



🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

XDR Collectors cannot be moved between
Cortex Cloud managing servers. In this
situation, you need to uninstall the existing
collector, and then install a new collector using
an installation package from the new managing
server. For more information on uninstalling, see
Uninstall the XDR Collector.

📝

🟢note

Since Cortex Cloud relies on the installation
package ID to approve XDR Collector
registration during install, it is not
recommended to delete the installation
package for any active on-premise collector
machines. Hiding the installation package will
remove it from the default list of available
installation packages, and can be useful to
eliminate...

23.5.2.4.3. Install the XDR Collector installation
package for Windows



🔒

🟣c…

Requires the Data Collection add-on.

23.5.2.4.3.1. Install the XDR Collector on

Windows using the MSI 

📝

🟢note

When the package is executed using the MSI, an
installation log is generated in
%TEMP%\MSI<Random characters>.log by
default.

📝

🟢note

If the XDR Collector does not connect to
Cortex Cloud, verify your internet connection
on the collector machine. If the XDR Collector
still does not connect, verify that the
installation package has not been removed
from the Cortex Cloud tenant.

23.5.2.4.3.2. Install the XDR Collector on
Windows using Msiexec



23.5.2.4.4. Install the XDR Collector installation

package for Linux 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

We recommend that you perform a Linux RPM
or Linux DEB installation.

📝

🟢note

If the XDR Collector does not connect to
Cortex Cloud, verify your Internet connection
on the collector machine. If the XDR Collector
still does not connect, verify the installation
package has not been removed from the Cortex
Cloud management console.

📝

🟢note

The XDR Collector does not support proxy
communication in environments where proxy
authentication is required.

👀

🟠i…

When performing a XDR Collector installation
or upgrade in Linux using a shell installer, the
/tmp folder cannot be marked as noexec.
Otherwise, the installation or upgrade fails. As a
workaround, before the installation or upgrade,
use the following command:

23.5.2.4.5. Configure the XDR Collector
upgrade scheduler 

🔒

🟣c…

Requires the Data Collection add-on.

🧭

➡️n…

In Cortex Cloud, select Settings →
Configurations → XDR Collectors →
Configuration

23.5.2.4.6. Set an application proxy for XDR
Collectors



🔒

🟣c…

Requires the Data Collection add-on.

🧭

➡️n…

In Cortex Cloud, select Settings →
Configurations → XDR Collectors →
Administration

23.5.2.4.7. Set an alias for an XDR Collector

machine 

🔒

🟣c…

Requires the Data Collection add-on.

🧭

➡️n…

Settings → Configurations → XDR Collectors
→ Administration

23.5.2.4.8. Upgrade XDR Collectors 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

The XDR Collector keeps the name of the
original installation package after every
upgrade.

🔒 23.5.2.4.9. Uninstall the XDR Collector 

🟣c…

Requires the Data Collection add-on.

23.5.2.5. Define XDR Collector machine groups 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

XDR Collectors only support IPv4 addresses.

📝

🟢note

Disconnecting Directory Sync in your Cortex
Cloud deployment can affect existing collector
machine groups and policy rules based on
Active Directory properties.

23.5.2.6. About Cortex XDR Collector content

updates 🔒 
🟣c…

Requires the Data Collection add-on.

23.5.2.7. XDR Collector profiles 

🔒

🟣c…

Requires the Data Collection add-on.

23.5.2.7.1. Add an XDR Collector profile for
Windows



📝

🟢note

Ingestion of log events larger than 5 MB is not
supported.

📝

🟢note

Fileset validation is enforced. You must enable
at least one fileset in the module, because
filesets are disabled by default.
Cortex Cloud collects all logs in either an
uncompressed JSON or text format.
Compressed files, such as the gzip format, are
not supported.
Cortex Cloud supports logs in single line
format or multiline format. For more...

23.5.2.7.1.1. Ingest logs from Windows DHCP
using Elasticsearch Filebeat 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

Although this enrichment is available when
configuring a Windows DHCP collector for a
cloud data collection integration, we
recommend configuring Cortex Cloud to
receive Windows DHCP logs with an XDR
Collector Windows Filebeat profile, because it
is simpler to set up.

📝

🟢note

Configuration activities include editing the
filebeat.yml file. To avoid formatting issues in
this file, use the template provided by Cortex
Cloud to make your customizations. We
recommend that you edit the file inside the user
interface, instead of copying it and editing it
elsewhere. Validate the syntax of the YML file
before you finish creating your...

📝

🟢note

Configuration activities include editing the
filebeat.yml file. To avoid formatting issues in
this file, use the example file provided by
Cortex Cloud to make your customizations. Do
not copy and paste the code syntax examples
provided later in this procedure into your
filebeat.yml file. Validate the syntax of the YML
file before you finish creating...

📝

🟢note

The tokenizer definition is dependent on the
Windows server version that you are using,
because the log format differs.
For platforms earlier than Windows Server
2008, use "%{id},%{date},%{time},%
{description},%{ipAddress},%{hostName},%
{macAddress}"
For Windows Server 2008 and 2008 R2, use "%
{id},%{date},%{time},%{description},%
{ipAddress},%{hostName},%...

23.5.2.7.1.2. Ingest Windows DNS debug logs
using Elasticsearch Filebeat 

📝

🟢note

Detailed logs are significantly larger, because
more information is added to the logs.

📝

🟢note

To avoid formatting issues in your filebeat.yml
file, we recommend that you validate the
syntax of the file.

23.5.2.7.2. Add an XDR Collector profile for
Linux 

📝

🟢note

Ingestion of log events larger than 5 MB is not
supported.

📝

🟢note

Fileset validation is enforced. You must enable
at least one fileset in the module, because
filesets are disabled by default.
Cortex Cloud collects all logs in either an
uncompressed JSON or text format.
Compressed files, such as the gzip format, are
not supported.
Cortex Cloud supports logs in single line
format or multiline format. For more...

23.5.2.8. Apply profiles to collection machine
policies 

🔒

🟣c…

Requires the Data Collection add-on.

🧭

➡️n…

Settings → Configurations → XDR Collectors
→ Policies → +Add Policy

🔒 23.5.2.9. XDR Collector datasets 

🟣c…

Requires the Data Collection add-on.

23.5.2.10. Troubleshoot XDR Collectors errors 

23.6. Palo Alto Networks integrations 

🔒

🟣c…

Data collection may require an add-on.

📝

🟢note

Roll-back of Strata Logging Service integration
migration is not supported.

23.6.1. About Palo Alto Networks integrations 

📝

🟢note

For Palo Alto Networks Integrations there is an
option to turn on or off the collection of URL
and File log types. For more information, see
Collecting URL and File log types.

📝

🟢note

Roll-back of Strata Logging Service integration
migration is not supported.

23.6.2. Next-Generation Firewall 

23.6.2.1. Ingest data from Next-Generation
Firewall 

🔒

🟣c…

Requires the Data Collection add-on.

☑️

🟣pr…

Ensure that you have completed the following
on the NGFW or Panorama side:
For Panorama only, ensure that the Panorama
Cloud Services plugin is installed.
Enable log forwarding profiles on firewall rules.
On the Cortex Cloud side, ensure that you have
user role permissions for Data Collection >
Data Sources & Integrations.
Configuration of data...

📝

🟢note

Roll-back of Strata Logging Service integration
migration is not supported.

📝

🟢note

If your firewalls are located in a different
region, or bandwidth issues are encountered
due to large log size, you can ingest NGFW
logs in CEF format, using the Syslog collector.
However, the Syslog solution is not as powerful
nor as comprehensive as this data collector,
and should only be used when this data
collector cannot be used. For more...

📝

🟢note

In the following procedure, general information
is provided for NGFW and Panorama. For
detailed instructions, consult the
documentation for your specific devices and
Panorama version.

📝

🟢note

Cortex Cloud does not validate your firewall
credentials, you must ensure the certificates or
PSK details have been updated in your firewalls
in order for data to stream.

📝

🟢note

It can take an hour or longer after connecting
the firewall in Cortex Cloud until you start
seeing notifications that the certificate has
been approved, and that the logging service
license has appeared on the firewall.

📝

🟢note

IOC and BIOC issues are applicable on stitched
data only, and are not available on raw data.

👀

🟠i…

This cross-account support is limited to the
same SFDC hierarchy; it does not extend to
MSSP scenarios where the customer and
provider own separate ends of the solution.

💡

🔵tip

You can see an overview of ingestion status for
all log types, and a breakdown of each log type
and its daily consumption quota on the NGFW
Ingestion Dashboard.

23.6.2.2. Ingest Next-Generation Firewall logs
using the Syslog collector 

🔒

🟣c…

Requires the Data Collection add-on.

📝

🟢note

In the following procedure, general information
is provided for NGFW and Panorama. For
detailed instructions, consult the
documentation for your specific devices and
Panorama version, to ensure that you have
configured log forwarding correctly for all the
log types that you would like to forward to
Cortex Cloud. The following steps only cover...

📝

🟢note

To avoid the possible effects of line formatting,
do not copy/paste the message formats
directly into the PAN-OS web interface. Instead,
paste into a text editor, remove any carriage
return or line feed characters, and then copy
and paste into the web interface.

📝

🟢note

From version 10.0 and later, the log format
documented for log types (Traffic, Threat, and
URL) exceeds the maximum supported 2048
characters in the Custom Log Format tab on
the firewall and Panorama. Select the CEF keys
and values to limit the number of characters to
2048, as per your requirements.

23.6.3. Ingest data from Prisma Access 

🔒

🟣c…

Requires the Data Collection add-on.

☑️

🟣pr…

Configuration of data ingestion from multiple
accounts requires Super User permissions in
both Cortex Cloud tenant and Prisma Access
accounts.

📝

🟢note

Roll-back of Strata Logging Service integration
migration is not supported.

📝

🟢note

Cortex Cloud does not validate your Prisma
Access account credentials. You must ensure
the account has been deployed in order for
data to stream.

🔒 23.6.4. Ingest logs from Prisma Access Browser 

🟣c…

Requires the Data Collection add-on.

23.6.5. Ingest detection data from Strata
Logging Service 📝 
🟢note

IOC and BIOC issues are applicable on stitched
data only and are not available on raw data.

23.6.6. IoT Security 

23.6.6.1. Ingest alerts and assets from IoT
Security 🔒 
🟣c…

Requires the Data Collection add-on.

23.6.7. Collecting URL and File log types 

23.7. Cloud Posture and Runtime Security data
sources



📝 23.7.1. How to onboard Databricks 

🟢note

If you turn on this feature, both the cloud and
region will be used for scanning, possibly
incurring cost and requiring adherence to
certain compliance policies.

23.7.2. How to onboard on-premise file shares
to Cortex Cloud Data Security 

📝

🟢note

Your data is scanned on the Broker VM itself,
and only the metadata and classification
results are transmitted from the on-premise
environment to Cortex Cloud.

📝

🟢note

For information about activating the DSPM
Fileshare applet, see Activate DSPM Fileshare.

23.7.2.1. Activate DSPM Fileshare 

☑️

🟣pr…

Set up and configure Broker VM
Know the complete path to the files and
folders that you want Cortex Cloud to monitor.
Necessary user permissions to access the
network shares. For the SMB connection type,
you need the user name and password.

📝

🟢note

The applet list displays only the applets for
which you have permissions.

📝

🟢note

By default, all configured connections are
saved.

📝

🟢note

For details regarding the connection fields, see
the table above under File Share Connection.

📝

🟢note

You can add multiple connections under a
single instance of the DSPM Fileshare applet by
returning to the File Share Connection screen
and clicking Add Connection. Each new
connection can be of either the NFS or SMB
connection type.

☑️ 23.7.3. How to onboard Microsoft 365 

🟣pr…

You have generated a Globally Unique Identifier
(GUID), also known as a Universally Unique
Identifier (UUID). You will need this ID for the
tenant you want to use for the Microsoft 365
instance.

In order to use Microsoft 365, you must be
registered with Microsoft Azure.

☑️ 23.7.4. Ingest logs and data from Okta 

🟣pr…

Administrator privileges: Your Okta user must
have a role capable of creating API tokens,
such as Read-only Administrator, Super
Administrator, or Organization Administrator.
For more information, see the Okta
Administrators Documentation.

23.7.5. Activate Registry Scanner 

☑️

🟣pr…

To initialize registry scanning on your Broker
VM, you must first add the necessary data
connectors. For details, see:
Connect Docker Hub registry
Connect Docker V2 compliant container
registry
Connect GitLab container registry
Connect Harbor registry
Connect JFrog container registry
Connect Sonatype Nexus registry
To initialize registry...

📝

🟢note

You cannot activate the Registry Scanner
directly on a new or existing Broker VM. You
can only activate or deactivate existing
Registry Scanner applets. To activate or
deactivate existing applets, see Step 4 under
Verify Registry Scanner connection section.

23.7.6. How to onboard Snowflake 

☑️

🟣pr…

In order to use Snowflake, you must be
registered with one of these cloud providers:
Amazon AWS, Microsoft Azure, or Google
Cloud Platform (GCP).
Ensure you have the necessary account
permissions to onboard. It is recommended to
use Account Admin as the role for the
onboarding.

📝

🟢note

The account identifier can be found using the
user information at the bottom left. Hover over
the account you wish to onboard and select
the copy option at the top right. The account
identifier is usually of the format:
(organization).[account]

23.7.7. Activate Transporter 

☑️

🟣pr…

Permissions: To configure and manage
Transporter applet settings, you must have
permissions to manage Broker Service
configurations (such as an Instance
Administrator)
Set up and configure Broker VMSet up and
configure Broker VM
Confirm that your Broker is v 28 or above
Whitelist IP addresses to enable access to
Cortex Cloud resources. The IP addresses...

⛔

🔴w…

The Transporter applet is not supported for
FedRAMP customers.

🧭

➡️n…

Settings → Configurations → Broker VMs
(under Data Broker

the Brokers tab → locate your Broker VM →
hover and click + Add under the Apps column
→ AppSec Transporter

the Appsec Transporter under the App column
→ Deactivate → Confirm when prompted All
existing connections are deleted but their
configurations are saved in the database

the Appsec Transporter under the App column
→ Deactivate → Confirm when prompted

23.8. Administration and troubleshooting 

23.8.1. Manage instances 

23.8.1.1. Add a new data source or instance 

📝

🟢note

If a data source contains multiple integrations,
the integration configured as the default
integration will used by the Data Onboarder.
The default integration of the content pack is
indicated in each content pack's
documentation. The other integrations are
available for configuration in the Data Sources
& Integrations page after installing the content...

📝

🟢note

If you are adding a new instance to an existing
data source, these options are View only.
You can adjust the view-only options on the
relevant page in the system, for example
Correlations, Playbooks, or Scripts.
If you are adding a new instance to an existing
data source, these options are View only.
You can adjust the view-only options on the
relevant...

23.8.1.2. How to configure the scanning settings
for supported services 📝 
🟢note

If you do not select a scanning cadence, the
default setting is applied. For more information,
contact your Customer Support team.

📝 23.8.1.3. Manage cloud instances 

🟢note

Currently, automation permission errors or
missing automation permissions do not affect
the Automation health status. You can view any
permission errors or missing permissions in the
the Permissions health status.

23.8.1.4. Update cloud permissions after Cortex
release updates 

☑️

🟣pr…

Ensure that the user account used to modify
permissions has the necessary privileges within
both the Cortex platform and your cloud
environment, for example, AWS or Azure.
You received a notification regarding a new
version available that requires permission
updates, or viewed a Needs Update status in
the Data Sources & Integrations page.

📝

🟢note

Instances requiring updates will not change
their connection status, for example,
Connected, Warning, Error, Disabled, due to the
pending permission update.

23.8.1.5. Pending cloud instances 

📝 23.8.1.6. Troubleshoot errors on cloud instances 

🟢note

Errors related to collection of audit logs in the
cloud instance are recorded in the
collection_auditing dataset. For more
information, see Audit logs fields and query
examples.

23.8.1.7. Manage Kubernetes Connector
instances 📝 
🟢note

After uninstalling the Kubernetes connector,
the connector status updates to Not connected
48 hours after the uninstall process is initiated.

23.8.2. Integrations 

📝

🟢note

In addition to content packs that you install
from Marketplace, related content packs are
automatically downloaded when you adopt
playbooks or edit tasks that require content
items such as scripts or integrations.

📝

🟢note

If the integration was installed from a content
pack, you need to duplicate the integration
before editing.

📝

🟢note

Integration commands are only available when
the integration instance is enabled. Some
commands depend on a successful connection
between Cortex Cloud and third-party
integrations.

💡

🔵tip

In the Playground, you can clear the context
data, if needed, which deletes everything in the
Playground context data, but does not affect
the actual issue or case. To clear the context,
run !DeleteContext all=yes' from the CLI or
click Clear Context Data while viewing the
context data.

🧭

➡️n…

Settings → Configurations → Marketplace

23.8.2.1. Integration use cases 

23.8.2.2. Add an integration instance 

🧭 23.8.2.3. Configure integration permissions 

➡️n…

Settings → Configurations → Data Collection
→ Integration Permissions

23.8.2.4. Fetch issues from an integration
instance 

📝

🟢note

In some integrations, the Issue Fetch interval is
called Feed Fetch Interval.
If the integration instance does not have the
Issue Fetch Interval field, you need to add this
field by editing the integration settings. If the
integration is from a content pack, you need to
create a copy of the integration. Any future
updates to this integration will not...

📝

🟢note

Some content packs include preconfigured
correlation rules, but you should review them to
see if they suit your use case and duplicate
them if required. Go to Threat Management →
Detection Rules → Correlations, search for the
relevant rule, right-click, and select Preview
Rule. For example, the ServiceNow v2 Alerts
(automatically generated)...

23.8.2.4.1. Map fields to issue types 

📝

🟢note

In the Classification & Mapping page, the
mapping does not indicate for which issue
types they are configured. Therefore, when
creating a mapper, it is best practice to add to
the mapper name, the issue types the mapper is
for. For example, Mail Listener - Phishing.

📝

🟢note

When mapping a list, we recommend you map
to a multi select field. Short text fields do not
support lists. If you do need to map a list to a
short text field, add a transformer in the
relevant playbook task, to split the data back
into a list.

🧭

➡️n…

Settings → Configurations → Object Setup →
Issues → Classification & Mapping

23.8.2.4.2. Classify events using a classifier for
issue types 

23.8.2.5. Troubleshoot Integrations 

23.8.2.6. Forward requests to long-running
integrations 

📝

🟢note

This integration replaces the External Dynamic
list integration, which is deprecated. For more
information about how to set up the
integration, see Manage external dynamic
lists.Manage external dynamic lists

📝

🟢note

When running on the tenant, you can only use
long-running integrations provided by Cortex
Cloud, you cannot create custom ones. Custom
long-running integrations are supported only
on engines at this time.
Configuring custom certificates or private API
Keys in the long-running integration instance is
supported only on engines, not on the Cortex
Cloud...

📝

🟢note

For the TAXII Server and TAXII2 Server
integrations, the xdr string is automatically
replaced by crtx. For the Microsoft Teams
integration, you can use the microsoft-teams-
create-messaging-endpoint command to get
the correct messaging endpoint based on the
server URL, the server version, and the instance
configurations. For more information, see...

📝 23.8.3. Verify collector connectivity 

🟢note

For more information on troubleshooting data
collector applet errors, see Troubleshoot
Broker VM applet connectivity.

23.8.4. Overview of data ingestion metrics 

☑️

🟣pr…

For Cortex Cloud to monitor data ingestion
health and create health issues, you must
enable Cortex - Analytics. Go to Configurations
→ Cortex - Analytics. For more information, see
Enable the Analytics Engine and Identity
Analytics.Enable the Analytics Engine and
Identity Analytics

23.8.4.1. Creating correlation rules to monitor
data ingestion health 

📝 23.8.4.2. Measuring data freshness 

🟢note

The max_delay metric is taken from the
maximum bucket value with a restricted limit;
therefore, metrics show whole numbers.
The median and ninetieth_percentile metrics are
statistical calculations that give an
approximation of the real value; therefore,
metrics show decimal numbers.
Time slots with a zero log count or zero byte
count display records...

23.8.5. About health issues 

☑️

🟣pr…

For Cortex Cloud to monitor data ingestion
health and create health issues, you must
enable the following settings under
Configurations:
Cortex - Analytics: Go to Configurations →
Cortex - Analytics. For more information, see
Enable the Analytics Engine and Identity
Analytics.Enable the Analytics Engine and
Identity Analytics

📝

🟢note

The Health Issues page displays issues that
were triggered after July 2024. To see health
issues that were triggered before this date,
click Legacy Health Issues.

📝

🟢note

Cortex Cloud enforces the dedup logic to
health issues. This logic reduces the likelihood
of identical health issues from flooding the
issues dataset.

23.8.5.1. Investigate and resolve health issues 

23.8.5.2. Monitor data ingestion health 

📝 23.8.5.3. Monitor correlation rules 

🟢note

In the dataset, the Query start time and Query
end time indicate the timeframe of the data
that was queried. The actual start and end
times of the correlation rule execution are
recorded in the _time field for the Initiated and
Completed entries.

24. Marketplace 

🧭 24.1. Cortex Marketplace 

➡️n…

Settings → Configurations → Marketplace) or
at Cortex Developer Docs Marketplace

📝 24.2. Content Pack Support Types 

🟢note

Palo Alto Networks is not liable for and does
not warrant or support any content pack
produced by a third-party publisher.
Palo Alto Networks does not support content
packs that do not have official available
documentation.

24.3. Cortex Cloud content 

24.4. Manage content packs 

📝

🟢note

Optional content packs that are already
installed are treated like they are required
content packs to preserve content integrity.

📝

🟢note

Cortex Cloud includes a built-in default mail
sender. You also have the option of installing a
different mail sender content pack, such as
Microsoft Exchange Online.

📝

🟢note

In addition to content packs that you install
from Marketplace, related content packs are
automatically downloaded when you adopt
playbooks or edit tasks that require content
items such as scripts or integrations.

📝

🟢note

Third-party product integrations are developed
and tested against a specific product version.
For products that are on-prem or cloud-based
with specific API versions, the version
developed and tested against will be included
in the integration's documentation. Newer
versions of the product are not always
immediately tested, and it is expected that...

⚠️

🟡c…

If you want to downgrade, any content that
depends on the content pack including any
customizations may be deleted if it does not
exist in the target content pack version.

⚠️

🟡c…

If another content pack is dependent on the
content pack you want to delete, it may break
the other content pack. You can reinstall the
content pack, but you cannot restore detached
and customized content.

💡

🔵tip

You can also find content packs that require
updates by going to Settings → Data Sources &
Integrations and filtering by Pack Version =
Update Available. If you click on an integration
in the filtered list, there is a link to the content
pack in Marketplace for updates.

24.5. Marketplace FAQs 

24.6. Content changes when upgrading Cortex
Cloud versions 

25. Serverless function posture
security 

25.1. Onboard cloud providers for serverless
functions 

📝

🟢note

When scanning serverless functions with layers,
those layers need to be from the same cloud
account.

📝

🟢note

Only functions containing zip files are
supported.

👀

🟠i…

Cortex supports Google Cloud Functions: 1st
gen and 2nd gen Cloud Functions API.

25.2. Serverless function posture policies 

25.2.1. Manage serverless function policies 

📝

🟢note

You can select multiple types to view all your
serverless function rules across your cloud
providers.

📝

🟢note

Refer to Create serverless function policies for
more information on how to define the steps of
a policy in the wizard.

25.2.2. Create serverless function policies 

📝

🟢note

You can select multiple options.

📝

🟢note

For more information about rules, refer to
Manage serverless function rules.

25.3. Serverless function posture rules 

25.3.1. Manage serverless function rules 

📝

🟢note

You can select multiple types to view all your
serverless function policies across your cloud
providers.

📝

🟢note

Refer to Create serverless function rules for
more information on how to define the steps of
a rule in the wizard.

25.3.2. Create serverless function rules 

25.3.3. Create an attack path rule for serverless
functions 🧭 
➡️n…

Under Posture Management, select Rules &
Policies → Cloud Security (under Rules) →
click Create Rule

in as the operator → enter the CVE ID → Search

> or >= as the operator → Severity level (such
as High, Low) → Search

> or >= as the operator → enter a score →
Search

25.3.4. Create a configuration rule for
serverless functions 

25.3.5. Create a network exposure rule for
serverless functions 

🧭 25.4. Serverless function usage 

➡️n…

To access serverless function assets, under
Inventory, select All Assets → Compute →
Serverless Functions

Posture Management → Vulnerability
Management) → Vulnerability Issues

Add Filters → Asset Category → Serverless
Function

Posture Management → Vulnerability
Management) → Vulnerable Assets

Add Filters → Asset Category → Serverless
Functions

26. Serverless function runtime

security 

26.1. Overview 

26.2. Set up serverless function protection 

📝 26.3. Serverless runtime issues 

🟢note

Issues triggered within 24 hours, sharing the
same name and description, will be aggregated
into cases along with issues from the same
function per execution.

27. Cortex CLI 

📝

🟢note

The CLI Read Only Role is not supported for
CWP as the system does not support offline
mode.

27.1. Connect Cortex CLI 

☑️

🟣pr…

System requirements:
macOS (Intel Core i7, such as Sequoia): To
ensure all functionalities work correctly, you
must install the vectorscan dependency via
Homebrew, using this command: brew install
vectorscan

RHEL 8.10 and Red Hat UBI9. The following
prerequisites must be met:
Install patchelf
Install zstd
Ubuntu 20 requires the prefetch...

📝

🟢note

The Cortex CLI requires an API key with the
Standard security level.

📝

🟢note

On macOS arm 64 architecture you must
unpack the downloaded file to retrieve the
executable.

📝

🟢note

To add an additional CLI instance, navigate to
Settings → Data Sources → select the menu
for your connected CLI instance → + New
Instance, and repeat the onboarding steps.

📝

🟢note

For more information about CLI usage for CWP,
refer to Cortex CLI for Cloud Workload
Protection
For more information about CLI usage for API
Security, refer to Cortex CLI for API Security
For more information about CLI usage for
Cortex Cloud Application Security, refer to
Cortex CLI usage for Cortex Cloud Application
Security

27.2. Self-service API keys for CLI scans 

27.3. Cortex CLI common command line

reference guide 📝 
🟢note

For soft fails, a failed check matches the
defined severity threshold. If multiple soft fail
severities are specified, the highest severity acts
as the threshold for determining a soft fail.
However, a successful scan will always return
an exit code of 0, even if block-level findings
(which might trigger soft fails based on
severity) are present.

27.4. Cortex CLI for API Security 

☑️

🟣pr…

Ensure you have the required user permissions.
Refer to Cortex CLI for more information
Onboard and install the Cortex CLI. Refer to
Connect Cortex CLI for more information
Ensure your application exposes APIs and
provides a corresponding OpenAPI
Specification file
Ensure that you have installed Java v 11 and
above

27.4.1. Cortex CLI API Security command line
reference guide 

27.5. Cortex CLI for Cloud Workload Protection 

☑️

🟣pr…

Ensure you have the required user permissions.
Refer to Cortex CLI for more information
Onboard and install the Cortex CLI. Refer to
Connect Cortex CLI for more information
Verify that Java version 11 and above is
installed: Run java -version in your terminal. If
not, refer to Java SE Development Kit 11.0.25 for
information about installing Java

☑️

🟣pr…

Before you begin, ensure you have sudo
privileges to execute the image scan.

📝

🟢note

CWP does not support container image secret
scanning for systems running on ARM
architecture.

📝

🟢note

For available CWP commands, refer to Cloud
Workload Protection command line reference.

📝

🟢note

--archive: When used with image scan, sets the
scan source to an archive file. When used with
image sbom, indicates the SBOM should be
exported from an archive file
The --archive flag can also be explicitly set as -
-archive=true
--archive-format <value>: The image archive
format (such as docker-archive or oci-archive).
Default: docker-archive.

27.5.1. Cloud Workload Protection command
line reference



27.6. Cortex CLI for Code Security 

☑️

🟣pr…

For the Cortex CLI binary:
Ensure you have Node.js v22 installed on your
host machine before running any scans with the
Cortex CLI. This is crucial to prevent runtime
errors, as the CLI depends on Node.js for
executing JavaScript analysis
Note

To check your version of Node.js, run node -v
To download Node.js, refer to the official
Node.js...

27.6.1. Cortex CLI usage for Cortex Cloud
Application Security 

27.6.2. Cortex CLI Cortex Cloud Application
Security command line reference



📝

🟢note

The repo-id flag must not end with .config, .log
or .ini. -config is acceptable.
-- repo-id foo.config will be blocked
-- repo-id foo-config will pass

👀

🟠i…

The Cortex CLI Cortex Cloud Application
Security only supports single occurrences of
each flag. If the same flag is passed multiple
times, only the last provided value will be used.
For example, in the following command, only TF
CloudFormation will be the scanned framework.
./cortexcli --api-base-url <YOUR_API_URL> --
api-key <YOUR_API_KEY> --auth-id...

28. Cortex Cloud XQL 

28.1. Get started with XQL 

28.1.1. XQL language features 

💡

🔵tip

When creating XQL queries, you can:
Use the up and down arrow keys to navigate
through the auto-suggestion commands and
definitions.
Select an auto-suggestion command by
pressing either the Enter or Tab key.
Press Shift+Enter to add a new line, and easily
ignore the auto-suggestion output.
Close the auto-suggestion output by pressing
the Esc key.

🧭

➡️n…

You submit XQL queries to Cortex Cloud using
the Investigation & Response → Search →
Query Builder user interface

28.1.2. XQL Language Structure 

📝

🟢note

Specifying a dataset is not required because
Cortex Cloud uses xdr_data as the default
dataset. If you have more than one dataset or
lookup, you can change your default dataset
by navigating to Settings → Configurations →
Data Management → Dataset Management,
right-click on the appropriate dataset, and
select Set as default.

📝

🟢note

You can also build a query that investigates
data in both a cold dataset and hot dataset in
the same query. In addition, as the hot storage
dataset format is the default option and
represents the fully searchable storage, this
format is used throughout this guide. For more
information on hot and cold storage, see
Dataset management.

28.1.2.1. Adding comments in queries 

📝 28.1.3. Supported operators 

🟢note

In some cases, using an IN or NOT IN operator
combined with a dataset and filterfilter stage
can be a better alternative to using a join stage.

28.1.4. Datasets and presets 

📝

🟢note

To set up this Cloud Identity Engine (previously
called Directory Sync Service (DSS)) dataset,
you need to set up a Cloud Identity Engine.
Otherwise, you will not have a pan_dss_raw
dataset. For more information, see Set up Cloud
Identity Engine.Set up Cloud Identity Engine

📝

🟢note

INFO issues are not included in this dataset.
The issue fields included in this dataset are
limited to certain fields available in the API. For
the full list, see Get Alerts Multi-Events v2 API.

📝

🟢note

The fields contained in this dataset are a subset
of the fields in the xdr_data dataset.

📝

🟢note

If the vendor and product are not specified in
the Winlogbeat profile's configuration file,
Cortex Cloud creates a default dataset called
microsoft_windows_raw.

📝

🟢note

To ensure GlobalProtect access authentication
logs are sent to Cortex Cloud, verify that your
PANW firewall's Log Settings for GlobalProtect
has the Cortex Data Lake checkbox selected.

📝

🟢note

The fields contained in this dataset are a subset
of the fields in the xdr_data dataset.

📝

🟢note

Prisma Access firewalls do not send
configuration logs to the Structured Log
Storage (SLS).

📝

🟢note

You can query in XQL for this data and build
widgets based on the xdr_data dataset or using
the preset device_control.
To view in an XQL query these events, the
Device Configuration of the endpoint profile
must be set to Block. Otherwise, the USB events
are not captured. The events are also captured
when a group of device types are blocked on
the...

📝

🟢note

The fields contained in this dataset are a subset
of the fields in the xdr_data dataset.

🧭 28.1.5. About examples 

➡️n…

Investigation & Response → Search → Query
Builder → XQL → Query Library

👀 28.1.6. JSON functions 

🟠i…

JSON field names are case sensitive, so the key
to field pairing must be identical in an XQL
query for results to be found. For example, if a
field value is "TIMESTAMP" and your query is
defined to look for "timestamp", no results will
be found.

28.1.7. How to filter for empty values in the
results table 

28.1.8. Understanding string manipulation in XQL 

28.2. Build XQL queries 

📝

🟢note

If you have the Cortex Agentic Assistant, you
can use natural language prompts to create
and run XQL queries within the chat interface.
For more information, see Create and run XQL
queries with Agentic Assistant chat.Cortex
Agentic Assistant

28.2.1. About the Query Builder 

📝

🟢note

Schema changes to datasets may not be
reflected in the autocomplete suggestions and
definitions as you type in real time the XQL
query, and can appear with a slight delay.

💡

🔵tip

When creating XQL queries, you can:
Use the up and down arrow keys to navigate
through the auto-suggestion commands and
definitions.
Select an auto-suggestion command by
pressing either the Enter or Tab key.
Press Shift+Enter to add a new line, and easily
ignore the auto-suggestion output.
Close the auto-suggestion output by pressing
the Esc key.

28.2.2. How to build XQL queries 

📝

🟢note

Users with different dataset permissions can
receive different results for the same XQL query.
An administrator or a user with a predefined
user role can create and view queries built with
an unknown dataset that currently does not
exist in Cortex Cloud. All other users can only
create and view queries built with an existing
dataset.
When you have more...

📝

🟢note

You can build a query that investigates data in
both a cold dataset and a hot dataset in the
same query. In addition, as the hot storage
dataset format is the default option and
represents the fully searchable storage, this
format is used throughout this guide for
investigation and threat hunting. For more
information on hot and cold storage, see...

👀

🟠i…

Forensic datasets are not inlcuded by default in
XQL query results, unless the dataset query is
explicitly defined to use a forensic dataset.

28.2.2.1. Get started with XQL queries 

28.2.2.2. Useful XQL user interface features 

📝

🟢note

Schema changes to datasets may not be
reflected in the autocomplete suggestions and
definitions as you type in real time the XQL
query and can appear with a slight delay.

💡

🔵tip

When creating XQL queries, you can:
Use the up and down arrow keys to navigate
through the auto-suggestion command
suggestions and definitions.
Select an auto-suggestion command by
pressing either the Enter or Tab key.
Press Shift+Enter to add a new line, and easily
ignore the auto-suggestion output.
Close the auto-suggestion output by pressing
the Esc...

28.2.2.3. XQL Query best practices 

28.2.2.4. Expected results when querying fields 

28.2.2.5. Create XQL query 

📝

🟢note

Whenever the time period is changed in the
query window, the config timeframe is
automatically set to the time period defined for
the entire query, including queries that are part
of the join stage. Yet, this won't be visible as
part of the query. Only if you manually type in
the config timeframe will this be seen in the
query.

These time picker...

💡

🔵tip

When creating XQL queries, you can:
Use the up and down arrow keys to navigate
through the auto-suggestion command
suggestions and definitions.
Select an auto-suggestion command by
pressing either the Enter or Tab key.
Press Shift+Enter to add a new line, and easily
ignore the auto-suggestion output.
Close the auto-suggestion output by pressing
the Esc...

💡

🔵tip

While the query is running, you can navigate
away from the page. A notification is sent when
the query has finished. You can also Cancel the
query or run a new query, where you have the
option to Run only new query (cancel previous)
or Run both queries.

28.2.2.6. Review XQL query results 

📝

🟢note

It's also possible to graph the results displayed.
For more information, see Graph query results.

📝

🟢note

Real time query results are available only in the
Query Builder and in free text query.
Real time results are displayed only for queries
run on hot datasets.
The Sort option is available only after all the
data is retrieved.
When you formulate complex queries, the
results will be displayed when the query has
finished running completely, and not in real...

📝

🟢note

Results are received incrementally for the first
100K records, or up to 100MB worth of records,
whichever comes first. After that, the next
update is when the query has finished running
completely.

📝

🟢note

In order for Cortex Cloud to provide a
histogram for a field, the field must not contain
an array or a JSON object.

👀 28.2.2.7. Translate to XQL 

🟠i…

This feature is still in a Beta state and you will
find that not all Splunk queries can be
converted to XQL. This feature will be improved
upon in the upcoming releases to support
greater Splunk query translations to XQL.

28.2.2.8. Graph query results 

🔒

🟣c…

Building Cortex Query Language (XQL) queries
in the Query Builder requires a Data Collection
add-on.

📝

🟢note

To display the result of as a time duration,
choose the graph type Single Value and enable
Show as Time. You can then select the Time
Unit (millisecond, second, minute, or hour) and
the Display format.

28.2.3. XQL query entities 

🧭 28.2.3.1. Create authentication query 

➡️n…

Investigation & Response → Search → Query
Builder

28.2.3.2. Create event log query 

28.2.3.3. Create file query 

28.2.3.4. Create image load query 

28.2.3.5. Create network connections query 

📝 28.2.3.6. Create network query 

🟢note

When you run the query, depending on the
outcome of the results, the value specified in
this field might be displayed in the dst_ip field
in the query results. This occurs if an RDP event
is recorded whereby a user connected from the
source IP to the destination IP.

28.2.3.7. Create process query 

👀 28.2.3.8. Create registry query 

🟠i…

Ensure the KEY NAME is entered as a real
registry key name, and not as a symbolic link.
Otherwise, the query will not retrieve results.
Instead of
HKEY_LOCAL_MACHINE\System\CurrentControl
Set, which is a symbolic link, use
KEY_LOCAL_MACHINE\System\ControlSet001.
Instead of HKEY_CURRENT_USER, use
HKEY_USERS\<SID>, where SID is either a SID of
the current...

28.2.3.9. Query across all entities 

28.2.4. Overview of the Query Center 

📝

🟢note

Very short queries might not be listed.
You cannot cancel correlation queries.
The default retention period for historic queries
is aligned with issue retention.

28.2.4.1. Edit and run queries in Query Center 

📝

🟢note

If query limits are applied to your tenant, the
number of concurrent running queries is limited
per user. If query usage is reaching the defined
limit, a system message warns you that a high
query load is impacting performance. If you
exceed the limit, new queries are blocked until
query usage drops. You can view all active
queries under Query Center...

📝

🟢note

You can cancel your own queries. To cancel
queries run by other users, you must have
View/Edit permissions for Configurations →
Query Management. By default, Instance
administrators have View/Edit permission.

📝

🟢note

Cancelled queries show a Canceled status. You
can see details of all canceled queries in the
Query History tab.
You cannot cancel correlation rule queries.
If you cancel a scheduled query, only the
current query is cancelled. Future recurrences
of the scheduled query are not affected.

📝 28.2.4.1.1. Query Center reference information 

🟢note

Certain fields are exposed and hidden by
default. An asterisk (*) is beside every field that
is exposed by default.

28.2.5. Manage scheduled queries 

🧭

➡️n…

Investigation & Response → Search →
Scheduled Queries

28.2.5.1. Scheduled Queries reference

information 📝 
🟢note

Certain fields are exposed and hidden by
default. An asterisk (*) is beside every field that
is exposed by default.

🧭 28.2.6. Manage your personal query library 

➡️n…

From the Query Builder Select Investigation &
Response → Search → Query Builder → XQL

Investigation & Response → Search → Query
Builder → XQL

28.3. Stages 

28.4. Functions 

29. Graph Search 

☑️ 29.1. What is Graph Search? 

🟣pr…

Graph Search requires View or View/Edit RBAC
permissions for Graph Search under
Investigation & Response → Search.

29.2. Get started with Graph Search queries 

🔒

🟣c…

This feature is included with a Cortex XSIAM
Premium license. It is also included with any
other Cortex XSIAM license that has the Cloud
Posture Security or Cloud Runtime Security
add-on.

☑️

🟣pr…

Graph Search requires View or View/Edit RBAC
permissions for Graph Search under
Investigation & Response → Search.

🧭

➡️n…

Understand your assets and findings data:
Graph Search queries are based on the current
data that has been collected for assets and
findings from the data sources configured and
then sent to the Unified Asset Inventory (UAI),
which is displayed in the All Assets page
(Inventory → Assets → All Assets

29.3. How to build Graph Search queries? 

☑️

🟣pr…

Graph Search requires View or View/Edit RBAC
permissions for Graph Search under
Investigation & Response → Search.

🧭

➡️n…

You submit Graph Search queries using the
Investigation & Response → Search → Query
Builder → Graph Search built-in query interface

☑️ 29.4. Understand Graph Search query results 

🟣pr…

Graph Search requires View or View/Edit RBAC
permissions for Graph Search under
Investigation & Response → Search.

29.5. Create Graph Search query 

☑️

🟣pr…

Graph Search requires View or View/Edit RBAC
permissions for Graph Search under
Investigation & Response → Search.

📝

🟢note

For more information about the Query Library,
see Manage the Graph Search Query Library.

💡

🔵tip

After running the query, you can view the
complete query by hovering over the last
THAT... in the Graph Search query interface, and
the query is displayed in a tooltip.
If your query doesn't find any results or you
want to change your query for any reason, you
can always click anywhere in the Graph Search
query interface, where your existing query is...

29.6. Graph Search examples 

🔒

🟣c…

This feature is included with a Cortex XSIAM
Premium license. It is also included with any
other Cortex XSIAM license that has the Cloud
Posture Security or Cloud Runtime Security
add-on.

☑️

🟣pr…

Graph Search requires View or View/Edit RBAC
permissions for Graph Search under
Investigation & Response → Search.

29.7. Manage the Graph Search Query Library 

☑️

🟣pr…

Graph Search requires View or View/Edit RBAC
permissions for Graph Search under
Investigation & Response → Search.

🧭

➡️n…

From Graph Search in the Query Builder Select
Investigation & Response → Search → Query
Builder → Graph Search

Investigation & Response → Search → Query
Builder → Graph Search

Investigation & Response → Search → Query
Center

29.8. Edit and run queries in Query Center 

📝

🟢note

If query limits are applied to your tenant, the
number of concurrent running queries is limited
per user. If query usage is reaching the defined
limit, a system message warns you that a high
query load is impacting performance. If you
exceed the limit, new queries are blocked until
query usage drops. You can view all active
queries under Query Center...

📝

🟢note

You can cancel your own queries. To cancel
queries run by other users, you must have
View/Edit permissions for Configurations →
Query Management. By default, Instance
administrators have View/Edit permission.

📝

🟢note

Cancelled queries show a Canceled status. You
can see details of all canceled queries in the
Query History tab.
You cannot cancel correlation rule queries.
If you cancel a scheduled query, only the
current query is cancelled. Future recurrences
of the scheduled query are not affected.

🧭

➡️n…

Investigation & Response → Search → Query
Center → Query History

📝 29.8.1. Query Center reference information 

🟢note

Certain fields are exposed and hidden by
default. An asterisk (*) is beside every field that
is exposed by default.

☑️ 29.9. Supported assets and findings 

🟣pr…

Graph Search requires View or View/Edit RBAC
permissions for Graph Search under
Investigation & Response → Search.

☑️ 29.10. FAQ on Graph Search 

🟣pr…

Graph Search requires View or View/Edit RBAC
permissions for Graph Search under
Investigation & Response → Search.